

Microlocal Analysis

Lecture Notes

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Preface

These notes present the basic ideas of microlocal analysis, beginning with tempered distributions and the Euclidean pseudodifferential calculus and continuing through elliptic regularity, wave front sets, and propagation of singularities. Standard references include Hörmander [Hör03; Hör07] and Shubin [Shu01]; complementary introductions are provided by Hintz [Hin25], Wunsch [Wun08], and Lefeuvre [Lef25].

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Part I

Foundations

Chapter 1

Overview and Motivating Examples

Microlocal analysis can be approached from two complementary viewpoints:

- (1) **The theory of pseudodifferential operators.** One enlarges the algebra of differential operators to a larger algebra $\Psi^\bullet$ in which one can invert elliptic operators, take square roots, construct parametrices, and so on.
- (2) **A paradigm for studying singularities of distributions.** Rather than asking *where* a distribution is singular (the singular support, a subset of the base $\mathbb{R}^n$), one asks *where and in which direction* it is singular. The answer lives on the cotangent bundle: the **wave front set**.

The term *microlocal* refers to simultaneous localization in the spatial variable x and in the frequency direction ξ .

1.1 From the resolvent of $\Delta + 1$ to symbols

Notation Throughout,

$$D_{x_k} = \frac{1}{i} \partial_{x_k}, \quad \Delta = \sum_{k=1}^n D_{x_k}^2 = - \sum_{k=1}^n \partial_{x_k}^2.$$

The factor $1/i$ is a convention chosen so that the Fourier transform turns D^α into multiplication by ξ^α **without** extra powers of i ; with this sign convention Δ is the *nonnegative* Laplacian.

Fourier convention For $u \in \mathcal{S}(\mathbb{R}^n)$, we use

$$\widehat{u}(\xi) = \int_{\mathbb{R}^n} e^{-ix \cdot \xi} u(x) dx, \quad u(x) = \int_{\mathbb{R}^n} e^{ix \cdot \xi} \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n}.$$

A model problem Consider

$$L = \Delta + 1 : \mathcal{S}(\mathbb{R}^n) \longrightarrow \mathcal{S}(\mathbb{R}^n).$$

Claim. L is an isomorphism of $\mathcal{S}(\mathbb{R}^n)$ onto itself.

Question. What kind of object is L^{-1} ?

Computing L^{-1} by Fourier transform For $u, f \in \mathcal{S}$,

$$Lu = f \iff (1 + |\xi|^2) \widehat{u}(\xi) = \widehat{f}(\xi)$$

because $\widehat{D^\alpha u} = \xi^\alpha \widehat{u}$, hence $\widehat{\Delta u} = |\xi|^2 \widehat{u}$. Since $1 + |\xi|^2 \geq 1 > 0$ never vanishes, we may divide and invert:

$$u(x) = \int_{\mathbb{R}^n} e^{ix \cdot \xi} \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n} = \int_{\mathbb{R}^n} e^{ix \cdot \xi} \underbrace{(1 + |\xi|^2)^{-1}}_{\text{symbol of } L^{-1}} \widehat{f}(\xi) \frac{d\xi}{(2\pi)^n}.$$

So L^{-1} is again a "Fourier multiplier", but with multiplier $(1 + |\xi|^2)^{-1}$ — a function that **decays** like $|\xi|^{-2}$ rather than growing like a polynomial. It is not a differential operator (a polynomial in ξ), but it is built by exactly the same recipe.

Two features are worth isolating:

- The nonvanishing of $1 + |\xi|^2$ (i.e. **ellipticity**) is what made inversion possible.
- The inverse is still " $e^{ix \cdot \xi} \times$ (a function of ξ) integrated against $\widehat{f}$ "; only the class of allowed multipliers changed.

The general recipe Allow a more general class of **symbols** $a(x, \xi)$, functions on $\mathbb{R}_x^n \times \mathbb{R}_\xi^n$ subject to suitable growth conditions in ξ (and boundedness of all derivatives in x), and use the quantization formula

$$(\text{Op}(a)u)(x) = \int_{\mathbb{R}^n} e^{ix \cdot \xi} a(x, \xi) \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n}.$$

The typical growth condition, giving the class S^m , is

$$|\partial_x^\beta \partial_\xi^\alpha a(x, \xi)| \leq C_{\alpha\beta} \langle \xi \rangle^{m - |\alpha|}, \quad \langle \xi \rangle := (1 + |\xi|^2)^{1/2},$$

i.e. a grows at most like $|\xi|^m$, and **each ξ -derivative gains one power of decay**. This last clause is the whole engine of the symbolic calculus.

This produces the algebras

$$\Psi^m(\mathbb{R}^n), \quad m \in \mathbb{R},$$

where m records the growth of the symbol, i.e. the "order" of the operator. Basic structural facts:

- **(*) Differential operators sit inside.** For $m \in \mathbb{N}_0$,

$$\text{Diff}^m(\mathbb{R}^n) \subset \Psi^m(\mathbb{R}^n),$$

Indeed, the differential operator and its polynomial symbol are

$$P = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha, \quad a(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha.$$

- **(*) Composition is graded.**

$$\Psi^m(\mathbb{R}^n) \circ \Psi^{m'}(\mathbb{R}^n) \subset \Psi^{m+m'}(\mathbb{R}^n),$$

so $\Psi^\bullet = \bigcup_m \Psi^m$ is a **graded algebra** (filtered, strictly speaking) — the orders add, exactly as for differential operators.

Example 1.1 (An elliptic inverse). $L = \Delta + 1 \in \text{Diff}^2(\mathbb{R}^n)$, and $L^{-1} \in \Psi^{-2}(\mathbb{R}^n)$. The order flips sign, as the composition rule demands: $-2 + 2 = 0$, and Ψ^0 contains the identity.

Remark 1.2. Slogan. $\Psi^\bullet$ is the smallest natural enlargement of $\text{Diff}^\bullet$ in which elliptic operators become invertible modulo smoothing errors.

1.2 Analysis of singularities: the wave front set

For $u \in \mathcal{S}'(\mathbb{R}^n)$, the wave front set will be a subset

$$\text{WF}(u) \subset \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\}) = T^*\mathbb{R}^n \setminus 0,$$

characterized by

$$(x_0, \xi_0) \in \text{WF}(u) \iff u \text{ is singular at } x_0, \text{ and } \xi_0 \text{ is a direction of that singularity.}$$

Two remarks on the definition to come:

- The zero section is removed because "direction" only makes sense for $\xi \neq 0$; indeed $\text{WF}(u)$ is **conic**: $(x_0, \xi_0) \in \text{WF}(u) \Rightarrow (x_0, \lambda \xi_0) \in \text{WF}(u)$ for all $\lambda > 0$.
- Projecting to the base recovers the classical notion: $\pi_x(\text{WF}(u)) = \text{sing supp}(u)$.

Example 1.3 (A characteristic function). Let $\Omega \subset \mathbb{R}^n$ be a domain with smooth boundary and $u = \mathbf{1}_\Omega$ its characteristic function. Then

$$\text{WF}(\mathbf{1}_\Omega) = N^*\partial\Omega \setminus \{0\},$$

the **conormal bundle** of the boundary with the zero section removed. Concretely, the jump occurs along $\partial\Omega$; at a boundary point the function is smooth in tangent directions and singular only in the normal direction, as illustrated in figure 1.1.

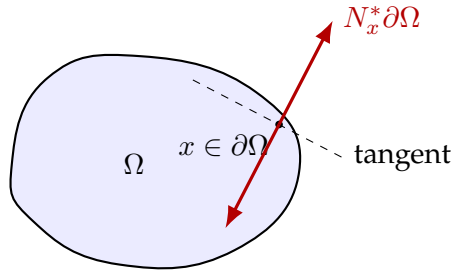


Figure 1.1: The wave front directions of $\mathbf{1}_\Omega$ are conormal to $\partial\Omega$.

1.3 The two fundamental microlocal theorems

Setup: let $P \in \text{Diff}^m(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$ with

$$Pu = 0.$$

Question. What can be said about $\text{WF}(u)$?

Definition 1.4 (Principal symbol and characteristic set). Write the full symbol of P as

$$a(x, \xi) = \sigma^m(P)(x, \xi) + (\text{lower order terms}),$$

where the **principal symbol**

$$\sigma^m(P) : T^*\mathbb{R}^n \longrightarrow \mathbb{C}$$

is the top-degree part, i.e. $\sigma^m(P)(x, \xi) = \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha$. It is the piece that is invariantly defined (independent of coordinates), which is why it lives on $T^*\mathbb{R}^n$.

Example 1.5 (A lower-order perturbation). Let $P = \Delta + V$ with $V \in C^\infty$. Its full symbol is $a(x, \xi) = |\xi|^2 + V(x)$, while $\sigma^2(\Delta + V) = |\xi|^2$. Thus the potential is lower order and invisible to the principal symbol.

Definition 1.6 (Characteristic set and ellipticity).

$$\text{Char}(P) = \{(x, \xi) \in T^*\mathbb{R}^n \setminus 0 : \sigma^m(P)(x, \xi) = 0\}.$$

(The condition $\xi \neq 0$ is essential: for $\xi = 0$ the principal symbol vanishes for trivial reasons whenever $m \geq 1$.) P is called **elliptic** iff $\text{Char}(P) = \emptyset$.

Microlocal elliptic regularity

Theorem 1.7 (microlocal elliptic regularity).

$$Pu = 0 \implies \text{WF}(u) \subset \text{Char}(P).$$

Interpretation: a solution can only be singular in the directions where the principal symbol degenerates. Everywhere else the equation can be inverted microlocally.

Example 1.8 (Elliptic equation). For $P = \Delta + V$ one has $\sigma^2(P) = |\xi|^2 \neq 0$ when $\xi \neq 0$. Hence $\text{Char}(P) = \emptyset$, so $\text{WF}(u) = \emptyset$ and every solution is smooth. This recovers classical elliptic regularity.

Example 1.9 (The wave operator). On $\mathbb{R}_t \times \mathbb{R}_x^n$ take

$$\square = D_t^2 - \Delta.$$

With dual variables (τ, ξ) to (t, x) ,

$$\sigma^2(\square)(t, x, \tau, \xi) = \tau^2 - |\xi|^2,$$

so the characteristic set is the **light cone**

$$\text{Char}(\square) \cap T_{(t,x)}^*\mathbb{R}^{n+1} = \{(\tau, \xi) \in \mathbb{R}^{n+1} \setminus \{0\} : \tau^2 = |\xi|^2\}.$$

The corresponding characteristic cone is shown in figure 1.2. This is nonempty, so solutions **may** be singular. Indeed, in one space dimension

$$u(t, x) = H(t - x), \quad H = \mathbf{1}_{[0, \infty)},$$

is a distributional solution ($\square$ applied to a function of $t - x$ vanishes). Its singular support is the line $\{t = x\}$ and its wave front set is the conormal to that line:

$$\text{WF}(u) = \{(t, x, \tau, \xi) : t = x, (\tau, \xi) = \lambda(1, -1), \lambda \neq 0\}.$$

One checks $\tau^2 = \xi^2$: the microlocal elliptic regularity statement is satisfied, with equality of the two sets in this example. The jump set and a conormal direction are shown in figure 1.3.

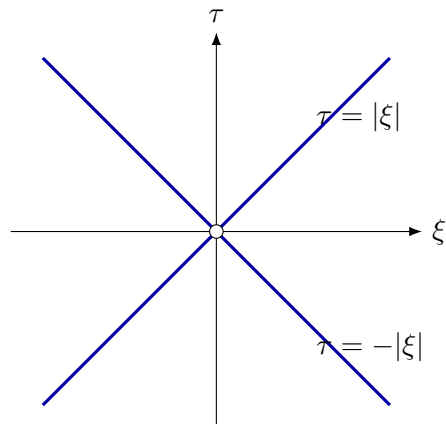


Figure 1.2: A two-dimensional slice of the characteristic cone of $\square$; the origin is removed.

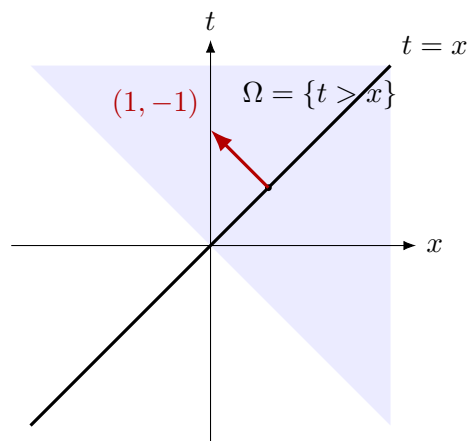


Figure 1.3: The jump set of $H(t - x)$ and one of its conormal directions.

Propagation of singularities The principal symbol does more than cut out $\text{Char}(P)$: since $\sigma^m(P)$ is a function on the symplectic manifold $T^*\mathbb{R}^n$, it generates a **Hamiltonian flow**

$$\varphi_s = \exp(sH_{\sigma^m(P)}), \quad H_a = \sum_j \left(\frac{\partial a}{\partial \xi_j} \partial_{x_j} - \frac{\partial a}{\partial x_j} \partial_{\xi_j} \right),$$

and (for real principal symbol) this flow preserves $\text{Char}(P) = \{\sigma^m(P) = 0\}$. Its integral curves inside $\text{Char}(P)$ are the **bicharacteristics**.

Computation for the wave operator. With $a = \tau^2 - |\xi|^2$ on $T^*(\mathbb{R}_t \times \mathbb{R}_x^n)$: $\partial_\tau a = 2\tau$, $\partial_\xi a = -2\xi$, and a is independent of (t, x) , so (τ, ξ) is conserved and

$$\varphi_s(t, x, \tau, \xi) = (t - 2\tau s, x + 2\xi s, \tau, \xi).$$

Up to reparametrization, these are the light rays $\dot{x}/\dot{t} = -\xi/\tau$, that is, straight lines traveling at unit speed on $\text{Char}(\square)$.

Theorem 1.10 (propagation of singularities, Hörmander).

$$Pu = 0 \implies \text{WF}(u) \text{ is invariant under the flow } (\varphi_s).$$

Slogan: **singularities of solutions to a wave-type equation travel along light rays.** Combined with microlocal elliptic regularity, this is a rather complete description: singularities live on $\text{Char}(P)$ and move along the bicharacteristic flow inside it.

The rest of Part I builds the vocabulary needed to make all of this precise.

Chapter 2

Distributional Foundations

2.1 Definitions and continuity

Definition 2.1 (Schwartz space and tempered distributions). The **Schwartz space** $\mathcal{S}(\mathbb{R}^n)$ consists of the smooth functions ϕ all of whose derivatives decay faster than any polynomial. It is topologized by the seminorms

$$\|\phi\|_k := \sup_{|\alpha|, |\beta| \leq k} \sup_{x \in \mathbb{R}^n} |x^\beta \partial^\alpha \phi(x)|, \quad k \in \mathbb{N}_0.$$

The space of **tempered distributions** is the continuous dual

$$\mathcal{S}'(\mathbb{R}^n) = \mathcal{L}(\mathcal{S}(\mathbb{R}^n), \mathbb{C}).$$

Notation. $u(\phi) = \langle u, \phi \rangle$.

Continuity criterion. A linear functional u on $\mathcal{S}$ is continuous iff

$$\exists k \in \mathbb{N}_0, C > 0 \quad \text{s.t.} \quad |\langle u, \phi \rangle| \leq C \|\phi\|_k \quad \forall \phi \in \mathcal{S}(\mathbb{R}^n).$$

This is the usual characterization of continuity for linear maps out of a Fréchet space whose topology is given by an increasing family of seminorms: continuity at 0 suffices, and it is controlled by a *single* seminorm.

Definition 2.2 (Weak-* topology on $\mathcal{S}'$).

$$u_j \rightarrow u \iff \langle u_j, \phi \rangle \rightarrow \langle u, \phi \rangle \quad \forall \phi \in \mathcal{S}(\mathbb{R}^n).$$

2.2 Examples

Example 2.3 (Dirac delta).

$$\langle \delta_0, \phi \rangle = \phi(0), \quad |\phi(0)| \leq \|\phi\|_0 \implies \delta_0 \in \mathcal{S}'(\mathbb{R}^n),$$

with $k = 0, C = 1$ in the continuity criterion.

Example 2.4 (Schwartz functions as distributions). For $\psi \in \mathcal{S}(\mathbb{R}^n)$ define

$$\langle T_\psi, \phi \rangle := \int_{\mathbb{R}^n} \psi \phi \, dx.$$

Writing $\langle x \rangle := (1 + |x|^2)^{1/2}$ and inserting $1 = \langle x \rangle^{-n-1} \langle x \rangle^{n+1}$:

$$|\langle T_\psi, \phi \rangle| \leq \int |\psi| |\phi| \, dx = \int |\psi| \langle x \rangle^{-n-1} \langle x \rangle^{n+1} |\phi| \, dx \leq \|\psi\|_0 \left(\int \langle x \rangle^{-n-1} \, dx \right) \|\phi\|_{n+1}.$$

The middle integral converges because $n+1 > n$, so $T_\psi \in \mathcal{S}'$ with $k = n+1$. The point of the trick: one must *buy* integrability, and the price is paid with a higher seminorm of ϕ .

2.3 $\mathcal{S}$ is dense in $\mathcal{S}'$

Lemma 2.5. *The map*

$$\mathcal{S}(\mathbb{R}^n) \longrightarrow \mathcal{S}'(\mathbb{R}^n), \quad \psi \longmapsto T_\psi$$

is injective and has dense range (for the weak- topology).*

Proof. Injectivity. If $\int \psi \phi = 0$ for all $\phi \in \mathcal{S}$, take $\phi = \bar{\psi}$ to get $\|\psi\|_{L^2} = 0$, hence $\psi \equiv 0$ by continuity.

Dense range (non-constructive argument). We must show

$$\mathcal{U} \cap \text{range}(T) \neq \emptyset \quad \text{for every open } \mathcal{U} \neq \emptyset.$$

Without loss of generality, $\mathcal{U}$ is a basic weak-* neighborhood of some $u \in \mathcal{S}'$:

$$\mathcal{U} = \left\{ v \in \mathcal{S}'(\mathbb{R}^n) : \sup_{k=1, \dots, N} |\langle u - v, \phi_k \rangle| < \varepsilon \right\}$$

for finitely many test functions $\phi_1, \dots, \phi_N \in \mathcal{S}$. After a Gram–Schmidt procedure in L^2 we may assume $\phi_1, \dots, \phi_N$ are orthonormal. Put

$$\psi := \sum_{k=1}^N \langle u, \phi_k \rangle \overline{\phi_k} \in \mathcal{S}(\mathbb{R}^n).$$

Then by orthonormality, for each k ,

$$\langle u - T_\psi, \phi_k \rangle = \langle u, \phi_k \rangle - \langle u, \phi_k \rangle = 0,$$

so $T_\psi \in \mathcal{U}$ — in fact it hits u exactly on the finitely many directions that $\mathcal{U}$ tests.

Notation. From now on we suppress T and write $\langle T_\psi, \phi \rangle = \langle \psi, \phi \rangle$, and regard

$$\mathcal{S}(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n).$$

□

2.4 Extension of operations by duality

The density lemma is what makes the following definitions the *only* possible continuous ones.

Definition 2.6 (Multiplication by a smooth function). Let $a \in C_b^\infty(\mathbb{R}^n)$ (smooth with **all derivatives bounded**; e.g. $a(x) = x^\alpha$ is *not* allowed — its derivatives are unbounded — whereas $\langle x \rangle^{-1}$ is). For $u \in \mathcal{S}'(\mathbb{R}^n)$ define

$$\langle au, \phi \rangle := \langle u, a\phi \rangle.$$

This makes sense because $\phi \mapsto a\phi$ maps $\mathcal{S} \rightarrow \mathcal{S}$ continuously, which is exactly where boundedness of the derivatives is used (Leibniz).

Remark 2.7. Polynomial multipliers such as $a(x) = x^\alpha$ belong to a larger class of functions with polynomial growth; the assumption $a \in C_b^\infty$ gives a convenient sufficient condition here.

Definition 2.8 (Distributional differentiation).

$$\langle D^\alpha u, \phi \rangle := (-1)^{|\alpha|} \langle u, D^\alpha \phi \rangle.$$

(The same sign works for ∂^α and for $D^\alpha = i^{-|\alpha|} \partial^\alpha$, since the factors $i^{-|\alpha|}$ appear on both sides.) The formula is forced by integration by parts on the dense subspace.

Consistency. For $\psi \in \mathcal{S}$ one checks directly

$$T_{a\psi} = a T_\psi, \quad T_{D^\alpha \psi} = D^\alpha T_\psi.$$

Hence the operations defined above **restrict** on $\mathcal{S}$ to the classical ones; since $\mathcal{S}$ is dense in $\mathcal{S}'$ and the operations are weak-* continuous, they are the **unique** continuous extensions.

2.5 Support and singular support

Definition 2.9 (Support and singular support). Let $u \in \mathcal{S}'(\mathbb{R}^n)$.

$$x_0 \in \text{supp}(u)^c \stackrel{\text{def}}{\iff} \exists \chi \in C_c^\infty(\mathbb{R}^n), \chi(x_0) \neq 0, \text{ s.t. } \chi \cdot u = 0 \text{ in } \mathcal{S}'.$$

$$x_0 \in \text{sing supp}(u)^c \stackrel{\text{def}}{\iff} \exists \chi \in C_c^\infty(\mathbb{R}^n), \chi(x_0) \neq 0, \text{ s.t. } \chi \cdot u \in \mathcal{S}(\mathbb{R}^n)$$

(the last membership meaning: χu is *represented* by a Schwartz function, i.e. $\chi u = T_\psi$ for some $\psi \in \mathcal{S}$).

As illustrated in figure 2.1, the condition $x_0 \notin \text{supp}(u)$ means that a cutoff near x_0 annihilates u , whereas $x_0 \notin \text{sing supp}(u)$ means that such a cutoff makes u smooth.

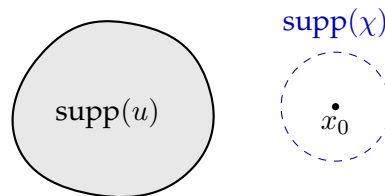


Figure 2.1: A cutoff supported near a point outside $\text{supp}(u)$.

Note that χu has compact support, so requiring $\chi u \in \mathcal{S}$ is the same as requiring $\chi u \in C_c^\infty$.

Example 2.10 (Support of the Dirac delta).

$$\text{supp}(\delta_0) = \{0\}, \quad \text{sing supp}(\delta_0) = \{0\}.$$

Example 2.11 (Support of a characteristic function). For a domain Ω , $\text{supp}(\mathbf{1}_\Omega) = \overline{\Omega}$, while

$$\text{sing supp}(\mathbf{1}_\Omega) = \partial\Omega.$$

To see this, let $x \in (\partial\Omega)^c$. One can choose a cutoff χ supported entirely inside Ω or entirely outside $\overline{\Omega}$, so that $\chi\mathbf{1}_\Omega$ equals χ or 0, both smooth. Hence $\text{sing supp}(\mathbf{1}_\Omega) \subset \partial\Omega$. Conversely, if a point $x \in \partial\Omega$ lay outside the singular support, then $\chi\mathbf{1}_\Omega$ would be continuous near x , contradicting its jump across the boundary when $\chi(x) \neq 0$.

The wave front set refines this to $\text{WF}(\mathbf{1}_\Omega) = N^*\partial\Omega \setminus 0$, recording not just *where* but *in which direction*.

General facts

- $\text{supp}(u)$ and $\text{sing supp}(u)$ are **closed** (their complements are open by construction: the defining conditions are stable under moving x_0 slightly, since $\chi(x_0) \neq 0$ is an open condition).
- $\text{sing supp}(u) \subseteq \text{supp}(u)$ (if $\chi u = 0$ then in particular $\chi u \in \mathcal{S}$).

A warning: no multiplication of distributions

Example 2.12 (Schwartz's impossibility theorem). There is **no** continuous bilinear map

$$M : \mathcal{S}' \times \mathcal{S}' \rightarrow \mathcal{S}'$$

whose restriction $M|_{\mathcal{S} \times \mathcal{S}}$ is the pointwise product (and which is associative/commutative in the usual way).

The standard obstruction is the pair of identities $x \cdot \text{p.v.} \frac{1}{x} = 1$ and $x \cdot \delta_0 = 0$, which together with associativity would give $\delta_0 = \delta_0 \cdot (x \cdot \text{p.v.} \frac{1}{x}) = (\delta_0 \cdot x) \cdot \text{p.v.} \frac{1}{x} = 0$.

This is one of the deep reasons microlocal analysis exists: one wants to know *when* products (and more generally pullbacks, restrictions, compositions) make sense. The wave front set answers this — products are definable whenever the wave front sets are not conormally opposed.

2.6 The Fourier transform

Definition 2.13 (Fourier transform on $\mathcal{S}$). For $\phi \in \mathcal{S}(\mathbb{R}^n)$ and $\xi \in \mathbb{R}^n$, define

$$\widehat{\phi}(\xi) = \mathcal{F}\phi(\xi) = \int_{\mathbb{R}^n} e^{-ix \cdot \xi} \phi(x) dx.$$

For $u \in \mathcal{S}'(\mathbb{R}^n)$, its Fourier transform is defined by duality:

$$\langle \mathcal{F}u, \psi \rangle := \langle u, \mathcal{F}\psi \rangle, \quad \psi \in \mathcal{S}(\mathbb{R}^n).$$

Facts.

- (1) $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$ is well defined and continuous.
 (2) It exchanges differentiation and multiplication:

$$\mathcal{F}(D^\alpha \phi) = \xi^\alpha \mathcal{F}(\phi), \quad \mathcal{F}(x^\alpha \phi) = (-1)^{|\alpha|} D_\xi^\alpha \mathcal{F}(\phi), \quad D_{x_j} = \frac{1}{i} \partial_{x_j}.$$

The first identity is integration by parts; the second is differentiation under the integral sign. Notice how the convention $D = \frac{1}{i} \partial$ makes the first formula constant-free.

- (3) $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \xrightarrow{\sim} \mathcal{S}(\mathbb{R}^n)$ is an isomorphism, with

$$\mathcal{F}^{-1}(\psi)(x) = \int_{\mathbb{R}^n} e^{ix \cdot \xi} \psi(\xi) \frac{d\xi}{(2\pi)^n}.$$

The normalizing factor is $(2\pi)^{-n}$, in agreement with the convention $\frac{d\xi}{(2\pi)^n} = d\xi / (2\pi)^n$.

Proof of the inversion formula. Set $A = \mathcal{F}^{-1} \circ \mathcal{F}$, with $\mathcal{F}^{-1}$ defined by the formula above. Using (2), one checks that A commutes with all D^α and all multiplications by x^α ; a functional-analytic argument then forces $A = c \cdot \text{id}$, and evaluation on a Gaussian gives $c = 1$.

- (4) **Duality (multiplication formula).**

$$\langle \mathcal{F}\phi, \psi \rangle = \langle \phi, \mathcal{F}\psi \rangle \quad \forall \phi, \psi \in \mathcal{S}(\mathbb{R}^n),$$

an immediate consequence of Fubini. This is the identity that lets us extend $\mathcal{F}$ to $\mathcal{S}'$.

- (5) **Plancherel.**

$$\|\mathcal{F}\phi\|_{L^2(d\xi)} = \|\phi\|_{L^2(dx)}$$

With the convention above,

$$\|\mathcal{F}\phi\|_{L^2(d\xi/(2\pi)^n)} = \|\phi\|_{L^2(dx)};$$

equivalently,

$$\|\widehat{\phi}\|_{L^2}^2 = (2\pi)^n \|\phi\|_{L^2}^2.$$

- (6) **Extension to tempered distributions.** The duality formula in the definition is well defined and continuous precisely by (1) and (4), and it is the unique continuous extension by density. Facts 1), 2), 3) then also hold on $\mathcal{S}'(\mathbb{R}^n)$.

This is exactly the Fourier-transform machinery used above, but now legitimate for distributional data.

2.7 The Schwartz kernel theorem

From kernels to operators Formally, an integral kernel K defines an operator

$$(O_K u)(x) = \int_{\mathbb{R}^m} K(x, y) u(y) dy.$$

When does this make sense? Take $K \in \mathcal{S}(\mathbb{R}_x^n \times \mathbb{R}_y^m)$, $u \in \mathcal{S}(\mathbb{R}_y^m)$, $v \in \mathcal{S}(\mathbb{R}_x^n)$. Then

$$\langle O_K(u), v \rangle = \iint K(x, y) u(y) v(x) dx dy = \underbrace{\langle K, v \otimes u \rangle}_{\text{makes sense for } K \text{ only a distribution}},$$

where $(v \otimes u)(x, y) := v(x)u(y)$. The right-hand side is the key observation: it is meaningful as soon as K is a **distribution** on the product, even though the integral formula is then purely formal.

The theorem

Theorem 2.14 (Schwartz kernel theorem). *The map $K \mapsto O_K$ is an isomorphism*

$$\mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^m) \xrightarrow{\sim} \mathcal{L}(\mathcal{S}(\mathbb{R}^m), \mathcal{S}'(\mathbb{R}^n)).$$

In words: **every** continuous linear operator from test functions to distributions is given by a distributional kernel, uniquely. This is a striking statement — no continuity, boundedness, or locality assumption is needed beyond continuity in the distributional topology.

Proof sketch. *Injectivity.* $O_K = 0 \iff \langle K, v \otimes u \rangle = 0$ for all $u, v \in \mathcal{S}$. Since the span of the pure tensors $v \otimes u$ is dense in $\mathcal{S}(\mathbb{R}^n \times \mathbb{R}^m)$, this forces $K = 0$.

Surjectivity. Given $O \in \mathcal{L}(\mathcal{S}, \mathcal{S}')$, regularize: with $\delta_x^\epsilon, \delta_y^\epsilon$ approximate identities (mollified point masses), set

$$\hat{K}(x, y) := \langle O \delta_y^\epsilon, \delta_x^\epsilon \rangle \in C_b(\mathbb{R}_x^n \times \mathbb{R}_y^m).$$

Continuity of O gives polynomial bounds on this smooth object; one then recovers the genuine kernel $K \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^m)$ by differentiating. The seminorm estimate for O says exactly that K is a finite number of derivatives of a polynomially bounded continuous function.

Example 2.15 (The identity operator). Consider $\text{Id} : \mathcal{S} \rightarrow \mathcal{S} \hookrightarrow \mathcal{S}'$. Its kernel is the **delta on the diagonal**

$$\Delta = \{(x, x) : x \in \mathbb{R}^n\} \subset \mathbb{R}^n \times \mathbb{R}^n, \quad \langle \delta_\Delta, \phi \rangle = \int_{\mathbb{R}^n} \phi(x, x) dx.$$

Indeed

$$\langle O_{\delta_\Delta} u, v \rangle = \langle \delta_\Delta, v \otimes u \rangle = \int (v \otimes u)(x, x) dx = \int u(x)v(x) dx = \langle u, v \rangle \implies O_{\delta_\Delta} = \text{id}.$$

The support of the kernel is the diagonal Δ , as shown in figure 2.2.

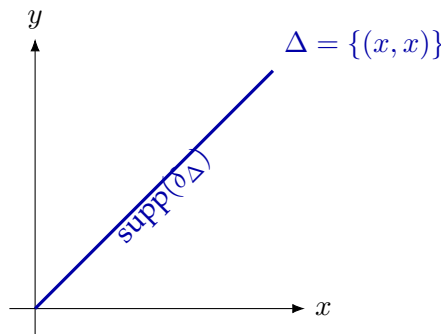


Figure 2.2: The Schwartz kernel of the identity is supported on the diagonal.

Example 2.16 (Differential operators).

$$D^\alpha \rightsquigarrow K = D_x^\alpha \delta_\Delta.$$

Again supported on the diagonal — this is no accident:

Locality and pseudolocality

Proposition 2.17 (Peetre’s theorem, kernel form). *If $K \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^n)$ has*

$$\text{supp}(K) \subset \Delta,$$

then $O_K \in \text{Diff}^(\mathbb{R}^n)$, i.e. O_K is a differential operator (of some finite order) with $C_b^\infty(\mathbb{R}^n)$ coefficients.*

The hypothesis $\text{supp}(K) \subset \Delta$ is exactly **locality**: $\text{supp}(O_K u) \subset \text{supp}(u)$, i.e. the operator does not move information around at all. Peetre’s theorem says locality *characterizes* differential operators — nothing else is strictly local.

Remark 2.18. Good news. If $A \in \Psi^m(\mathbb{R}^n)$, then its Schwartz kernel K_A satisfies

$$\text{sing supp}(K_A) \subset \Delta.$$

Pseudodifferential operators are therefore *not* local, but they are **pseudolocal**: off the diagonal their kernel is smooth, so they can move information around only in a smoothing way. Consequently

$$\text{sing supp}(Au) \subset \text{sing supp}(u) \quad \forall A \in \Psi^m.$$

This is the property that lets us use $\Psi^\bullet$ as a *microscope* for singularities: applying a pseudodifferential operator never creates new singularities, and by choosing the symbol to be supported in a small cone in ξ we can look at singularities one direction at a time. That is precisely the construction of $\text{WF}(u)$ promised in the overview.

Summary and outlook

Object	Where it lives	What it detects
$\text{supp}(u)$	$\mathbb{R}^n$	where u is nonzero
$\text{sing supp}(u)$	$\mathbb{R}^n$	where u fails to be smooth
$\text{WF}(u)$	$T^*\mathbb{R}^n \setminus 0$	where and in which direction u fails to be smooth
$\sigma^m(P)$	$T^*\mathbb{R}^n$	leading behavior of P ; generates the bicharacteristic flow
$\text{Char}(P)$	$T^*\mathbb{R}^n \setminus 0$	directions in which P fails to be elliptic — where singularities can live

The two structural theorems to be proved:

- **Elliptic regularity:** $Pu = 0 \Rightarrow \text{WF}(u) \subset \text{Char}(P)$.
- **Propagation of singularities:** $\text{WF}(u)$ is invariant under the Hamiltonian flow of $\sigma^m(P)$ inside $\text{Char}(P)$.

The tools are tempered distributions and duality, support and singular support, the Fourier transform on $\mathcal{S}'$, and the Schwartz kernel theorem with pseudolocality.

Chapter 3

Symbols and Pseudodifferential Operators on $\mathbb{R}^n$

Notation. Throughout, $\langle \xi \rangle = (1 + |\xi|^2)^{1/2}$, $D^\alpha = (-i\partial)^\alpha$, and $\frac{d\xi}{(2\pi)^n}$ denotes normalized Lebesgue measure, so that $u(x) = \int e^{ix \cdot \xi} \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n}$. The notation $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$ denotes the Schwartz space, $\mathcal{S}'$ its dual, and C_b^∞ the smooth functions whose derivatives of every order are bounded.

3.1 Motivation

Let P be a differential operator with bounded smooth coefficients:

$$P = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha, \quad a_\alpha \in C_b^\infty(\mathbb{R}^n).$$

Since $D^\alpha u(x) = \int e^{ix \cdot \xi} \xi^\alpha \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n}$, we get

$$Pu(x) = \int e^{ix \cdot \xi} \left(\underbrace{\sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha}_{=: a(x, \xi)} \right) \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n}.$$

Goal. Replace $a(x, \xi)$ by a smooth function with controlled growth in ξ .

3.2 The symbol classes S^m

Definition 3.1 (symbol of order $m \in \mathbb{R}$). $S^m(\mathbb{R}_x^n; \mathbb{R}_\xi^N) \subset C^\infty(\mathbb{R}_x^n \times \mathbb{R}_\xi^N)$ consists of those a satisfying

$$|D_x^\alpha D_\xi^\beta a(x, \xi)| \leq C_{\alpha, \beta} \langle \xi \rangle^{m - |\beta|} \quad \forall \alpha, \beta, \forall (x, \xi).$$

Two things to notice:

- The **gain in decay occurs only for** β (the ξ -derivatives). Differentiating in ξ improves decay by one power each time; differentiating in x does not. Slogan: ξ *controls the number of derivatives*.
- The estimates are **uniform in x** — this is the “uniform” or “global” symbol class.

Example 3.2 (Polynomial symbols).

$$a(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha \in S^m(\mathbb{R}^n, \mathbb{R}^N)$$

when $a_\alpha \in C_b^\infty$.

3.3 Topology

Definition 3.3 (Fréchet topology on S^m). The symbol space S^m is equipped with the Fréchet topology generated by the seminorms

$$\|a\|_{m,k} = \sup_{|\alpha|+|\beta| \leq k} \left\| \frac{D_x^\alpha D_\xi^\beta a(x, \xi)}{\langle \xi \rangle^{m-|\beta|}} \right\|_{L^\infty}$$

and convergence means

$$a_j \rightarrow a_* \iff \forall k, \|a_j - a_*\|_{m,k} \rightarrow 0.$$

Notation. Sometimes one writes S_∞^m and speaks of **uniform symbols**, to distinguish from local/compactly-supported variants.

3.4 Products of symbols

Proposition 3.4. *Multiplication of symbols gives a continuous bilinear map*

$$S^m(\mathbb{R}^n, \mathbb{R}^N) \times S^{m'}(\mathbb{R}^n, \mathbb{R}^N) \longrightarrow S^{m+m'}(\mathbb{R}^n, \mathbb{R}^N).$$

Proof. By Leibniz,

$$\begin{aligned} D_x^\alpha D_\xi^\beta (ab) &= \sum_{\substack{\alpha' + \alpha'' = \alpha \\ \beta' + \beta'' = \beta}} c_{\alpha' \alpha'' \beta' \beta''} \\ &\quad \times (D_x^{\alpha'} D_\xi^{\beta'} a) (D_x^{\alpha''} D_\xi^{\beta''} b). \end{aligned}$$

Each term satisfies

$$\langle \xi \rangle^{m-|\beta'|} \langle \xi \rangle^{m'-|\beta''|} = \langle \xi \rangle^{m+m'-|\beta|}.$$

The constant is controlled by $\|a\|_{m,k} \|b\|_{m',k}$, which also proves continuity. $\square$

3.5 Residual symbols

Definition 3.5 (Residual symbols). If $m' \leq m$ then $S^{m'} \subset S^m$ (less growth means a smaller class). The **residual class** is

$$S^{-\infty}(\mathbb{R}^n, \mathbb{R}^N) = \bigcap_{m \in \mathbb{R}} S^m(\mathbb{R}^n, \mathbb{R}^N),$$

i.e. symbols decaying faster than any power of $\langle \xi \rangle$, together with all their derivatives.

Density of residual symbols

Proposition 3.6. *Let $a \in S^m$ and let $\chi \in C_c^\infty(\mathbb{R}_\xi^N)$ with $\chi \equiv 1$ near $\xi = 0$. Put*

$$a_j(x, \xi) := \chi(\xi/j) a(x, \xi).$$

Then:

- (1) $a_j \in S^{-\infty}$ for every j ;
- (2) $\sup_{j \in \mathbb{N}} \|a_j\|_{m,k} < \infty$ for every k ;
- (3) $a_j \rightarrow a$ in $S^{m+\varepsilon}$ for every $\varepsilon > 0$.

Proof. (1) Clear: a_j is compactly supported in ξ , so all the estimates hold with any exponent.

(2), (3) By the multiplication proposition above it suffices to treat the case $a \equiv 1 \in S^0$, i.e. to study $\chi_j(\xi) := \chi(\xi/j)$.

Bound on the seminorms. We have

$$D_\xi^\beta \chi_j(\xi) = j^{-|\beta|} (D^\beta \chi)(\xi/j).$$

The factor $(D^\beta \chi)(\xi/j)$ is nonzero only when ξ/j lies in $\text{supp } \chi$, hence only when $|\xi| \lesssim j$. On that region

$$j^{-1} \lesssim |\xi|^{-1} \implies |D_\xi^\beta \chi_j(\xi)| \lesssim \langle \xi \rangle^{-|\beta|},$$

uniformly in j . Therefore $\|\chi_j\|_{0,k} \leq C_k$ with C_k independent of j — this is (2).

Convergence. For $\beta \neq 0$, $D^\beta(\chi_j - 1) = D^\beta \chi_j$ and

$$\left| \langle \xi \rangle^{-\varepsilon+|\beta|} D^\beta(\chi_j - 1) \right| = \left| \langle \xi \rangle^{-\varepsilon+|\beta|} j^{-|\beta|} (D^\beta \chi)(\xi/j) \right| \leq C \langle \xi \rangle^{-\varepsilon} \xrightarrow{j \rightarrow \infty} 0,$$

using again $|\xi| \lesssim j$ on the support (and, for $\beta = 0$, that $\chi_j - 1$ vanishes for $|\xi| \lesssim j$, so it is $O(\langle \xi \rangle^{-\varepsilon})$ there). Hence

$$\chi_j \rightarrow 1 \quad \text{in } S^\varepsilon \quad \text{for every } \varepsilon > 0.$$

Multiplying by $a \in S^m$ and using continuity of the product $S^\varepsilon \times S^m \rightarrow S^{m+\varepsilon}$ gives (3). $\square$

Conclusion. $S^{-\infty}$ is dense in S^m for the slightly weaker $S^{m+\varepsilon}$ topology. This permits one to prove statements first for rapidly decaying symbols and then pass to the limit.

3.6 Elliptic symbols

(The point: turn invertibility of the operator into an algebraic condition on the symbol.)

Definition 3.7 (elliptic). $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$ is **elliptic** iff

$$\exists b \in S^{-m} \quad \text{such that} \quad ab - 1 \in S^{-1}.$$

So b is an inverse modulo one order of smoothing. We want to connect invertibility of the operator with invertibility of the symbol.

Proposition 3.8 (TFAE). (1) a is elliptic;
 (2) $\exists C, c > 0$ such that $|\xi| \geq C \implies |a(x, \xi)| \geq c|\xi|^m$;
 (3) $\exists C, c > 0$ such that $|\xi| \geq 1 \implies |a(x, \xi)| \geq c|\xi|^m - C|\xi|^{m-1}$.

Proof. (1) $\Rightarrow$ (3). By hypothesis $1 - ab \in S^{-1}$, hence

$$|1 - ab| \leq C\langle \xi \rangle^{-1}.$$

Since $1 - |ab| \leq |1 - ab|$, we obtain

$$1 - C\langle \xi \rangle^{-1} \leq |ab| \leq |a| \cdot c'\langle \xi \rangle^{-m},$$

the last step because $b \in S^{-m}$ gives $|b| \leq c'\langle \xi \rangle^{-m}$. Rearranging,

$$|a| \geq \frac{1}{c'}\langle \xi \rangle^m - \frac{C}{c'}\langle \xi \rangle^{m-1}.$$

(3) $\Rightarrow$ (2). Factor:

$$|a| \geq c|\xi|^m - C|\xi|^{m-1} = c|\xi|^m \left(1 - \frac{C}{c}|\xi|^{-1}\right) \geq \frac{c}{2}|\xi|^m$$

as soon as $|\xi| \geq 2C/c$, i.e. for $|\xi|$ large enough.

(2) $\Rightarrow$ (1). Assume $|a| \geq c|\xi|^m$ for $|\xi| > C'$. Choose $\chi \in C_c^\infty(\mathbb{R}^N)$ with $\chi \equiv 1$ on $\{|\xi| < C'\}$ (a bump equal to 1 on $[-C', C']$). Set

$$b(x, \xi) := \frac{1 - \chi(\xi)}{a(x, \xi)},$$

which is well defined, since the numerator vanishes exactly where a might vanish. The cutoff profile is shown in figure 3.1.

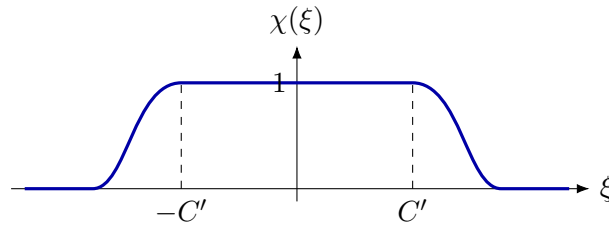


Figure 3.1: A compactly supported cutoff equal to 1 near $\xi = 0$.

Check that $b \in S^{-m}$:

- $|b| = 0$ for $|\xi| < C'$;
 - $|b| \leq \frac{1}{c}|\xi|^{-m}$ elsewhere, by the lower bound on a .
- (Exercise: carry out the corresponding computation for higher derivatives — one differentiates $1/a$ using the quotient rule and induction, each ξ -derivative producing an extra $\langle \xi \rangle^{-1}$.)

Hence $b \in S^{-m}(\mathbb{R}_x^n, \mathbb{R}_\xi^N)$ and

$$ab - 1 = -\chi(\xi) \in S^{-\infty}(\mathbb{R}^n, \mathbb{R}^N).$$

□

Remark 3.9. Remarks.

(1) The proof above actually gives the stronger characterization:

$$a \text{ elliptic} \iff \exists b \text{ with } ab - 1 \in S^{-\infty}(\mathbb{R}^n, \mathbb{R}^N).$$

(2) If $a, a' \in S^m$ with $a - a' \in S^{m-1}$, then a is elliptic $\iff a'$ is elliptic (immediate from criterion (3), where the S^{m-1} part is absorbed in the error term). That is:

$$\text{ellipticity is a property of the class } [a] \in S^m(\mathbb{R}^n, \mathbb{R}^N)/S^{m-1}(\mathbb{R}^n, \mathbb{R}^N).$$

3.7 Homogeneous and classical symbols

Definition 3.10. $S_{\text{hom}}^m(\mathbb{R}^n, \mathbb{R}^N \setminus \{0\})$ consists of smooth maps $a : \mathbb{R}_x^n \times (\mathbb{R}_\xi^N \setminus \{0\}) \rightarrow \mathbb{C}$ such that

- (1) $a(x, t\xi) = t^m a(x, \xi)$ for $t > 0$ (**positive homogeneity of degree m**);
- (2) $|D_x^\alpha D_\xi^\beta a(x, \xi)| \leq C_{\alpha, \beta} |\xi|^{m-|\beta|}$ for all $\xi \neq 0$.

Remark. A homogeneous symbol is in general singular at $\xi = 0$; to make it a genuine symbol one cuts the origin out. If $\chi \in C_c^\infty(\mathbb{R}^N)$ with $\chi \equiv 1$ near $\xi = 0$, then

$$a \in S_{\text{hom}}^m \implies (1 - \chi)a \in S^m(\mathbb{R}^n, \mathbb{R}^N).$$

Lemma 3.11. *The map*

$$S_{\text{hom}}^m(\mathbb{R}^n, \mathbb{R}^N \setminus \{0\}) \longrightarrow \frac{S^m(\mathbb{R}^n, \mathbb{R}^N \setminus \{0\})}{S^{m-1}(\mathbb{R}^n, \mathbb{R}^N \setminus \{0\})}, \quad a \longmapsto [(1 - \chi)a]$$

*is independent of the choice of χ and is **injective**.*

Exercise. $[(1 - \chi)a]$ is elliptic $\iff \inf_{\substack{x \in \mathbb{R}^n \\ |\xi|=1}} |a(x, \xi)| > 0$.

Proof of the lemma. Independence of χ : if χ_1, χ_2 are two such cutoffs, $(\chi_1 - \chi_2)a$ is compactly supported in ξ , hence lies in $S^{-\infty} \subset S^{m-1}$. $\checkmark$

Injectivity: suppose $[(1 - \chi)a] = 0$, i.e. $(1 - \chi)a \in S^{m-1}$. Then for all $|\xi| \gg 0$,

$$|a(x, \xi)| \leq C \langle \xi \rangle^{m-1}.$$

Fix ξ_0 with $|\xi_0| = 1$ and take $t > C$. Homogeneity gives

$$t^m |a(x, \xi_0)| = |a(x, t\xi_0)| \leq C t^{m-1}.$$

Dividing by t^m and letting $t \rightarrow \infty$ forces $|a(x, \xi_0)| = 0$. By homogeneity $a(x, \xi) \equiv 0$ — contradiction unless $a = 0$. $\square$

Classical (polyhomogeneous) symbols

Definition 3.12. $S_{\text{cl}}^m(\mathbb{R}^n, \mathbb{R}^N) \subset S^m(\mathbb{R}^n, \mathbb{R}^N)$ consists of all symbols a for which there exist

$$a_{m-j} \in S_{\text{hom}}^{m-j}(\mathbb{R}_x^n, \mathbb{R}_\xi^N), \quad j = 0, 1, 2, \dots$$

and a cutoff $\chi \in C_c^\infty(\mathbb{R}_\xi^N)$, $\chi \equiv 1$ near $\xi = 0$, such that

$$a - \sum_{j=0}^{J-1} (1 - \chi) a_{m-j} \in S^{m-J} \quad \forall J \in \mathbb{N}.$$

Remarks.

- (1) Each a_{m-j} is **uniquely determined** by a , by the injectivity Lemma above. (*Indeed, if a_{m_1} and a_{m_2} are two candidates for the leading term, subtracting the two expansions gives $(1 - \chi)(a_{m_1} - a_{m_2}) \in S^{m-1}$, hence $a_{m_1} = a_{m_2}$ by injectivity; then induct.*)
- (2) Differential operators are classical: regrouping by degree,

$$a(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha = \sum_{j=0}^m \underbrace{\sum_{|\alpha|=m-j} a_\alpha(x) \xi^\alpha}_{= a_{m-j}(x, \xi)},$$

and each a_{m-j} is homogeneous of degree $m - j$. It is smooth across $\xi = 0$, so inserting the cutoff changes only a residual symbol.

3.8 Asymptotic summation

Motivating question. Given complex numbers $a_k, k \in \mathbb{N}$, when does there exist f with

$$\frac{1}{k!} \frac{d^k}{dx^k} f(0) = a_k ?$$

Answer: always — this is **Borel's lemma**. The symbol version is the following.

Proposition 3.13 (asymptotic summation). *Let $(m_j) \subset \mathbb{R}$ be strictly decreasing with $m_j \rightarrow -\infty$ as $j \rightarrow \infty$. Given $a_j \in S^{m_j}(\mathbb{R}^n, \mathbb{R}^N)$, there exists*

$$a \in S^{m_0}(\mathbb{R}^n, \mathbb{R}^N) \quad \text{with} \quad a - \sum_{j=0}^{J-1} a_j \in S^{m_J} \quad \forall J,$$

and a is **unique modulo** $S^{-\infty}$. One writes $a \sim \sum_j a_j$.

Proof. Take a cutoff χ which is a smooth "reversed" bump:

$$\chi \equiv 0 \text{ near } 0, \quad \chi \equiv 1 \text{ for } |\xi| \geq 2,$$

(so χ rises from 0 to 1 between $|\xi| = 1$ and $|\xi| = 2$), and a sequence $\varepsilon_j \rightarrow 0$ as $j \rightarrow \infty$, to be chosen. Define

Figure 3.2 records the reversed cutoff used in the construction.

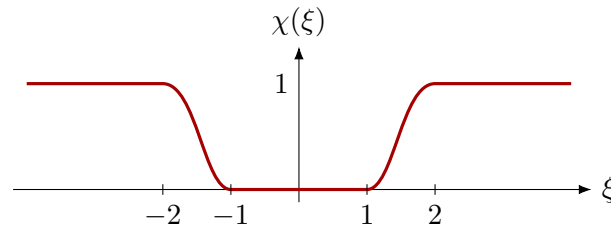


Figure 3.2: The reversed cutoff used for asymptotic summation.

$$a(x, \xi) := \sum_{j \geq 0} \chi(\varepsilon_j \xi) a_j(x, \xi).$$

For each fixed ξ this is a **finite sum** (the terms with $\varepsilon_j |\xi| \leq 1$ vanish, and $\varepsilon_j \rightarrow 0$), so $a \in C^\infty(\mathbb{R}^n \times \mathbb{R}^N)$ with no convergence issue. It remains to choose ε_j so that the partial sums are Cauchy in S^{m_0} .

After relabeling the orders, we may assume $m_j = m - j$. We claim that the numbers ε_j can be chosen so that

$$\|\chi(\varepsilon_j \cdot) a_j\|_{m-j', j'} \leq 2^{-j} \quad \forall 0 \leq j' < j.$$

Indeed, fix a pair (j', j) . Since $a_j \in S^{m-j}$ and $m - j < m - j'$, the cutoff lemma for residual symbols (in the reversed form: the cutoff *removes* a neighborhood of $\xi = 0$ that grows as $\varepsilon \rightarrow 0$) gives

$$\chi(\varepsilon \cdot) a_j \xrightarrow{\varepsilon \rightarrow 0} 0 \quad \text{in } S^{m-j'}.$$

For each fixed j there are only **finitely many** conditions ($j' = 0, \dots, j-1$), so a single small ε_j works for all of them. This proves the claim.

Consequently, for a fixed j' , the tails of $\sum_j \chi(\varepsilon_j \xi) a_j$ are bounded by $\sum_{j \geq J} 2^{-j}$ in $\|\cdot\|_{m, j'}$, so the partial sums

$$\sum_{j=0}^J \chi(\varepsilon_j \xi) a_j(x, \xi)$$

form a Cauchy sequence with respect to $\|\cdot\|_{m, j'}$. Since j' was arbitrary, the sequence is Cauchy in S^m , hence $a \in S^m$.

Finally, split the remainder:

$$a(x, \xi) - \sum_{j=0}^{J-1} a_j(x, \xi) = \underbrace{\sum_{j=0}^{J-1} [\chi(\varepsilon_j \xi) - 1] a_j(x, \xi)}_{\text{compactly supported in } \xi \Rightarrow \in S^{-\infty}} + \underbrace{\sum_{j \geq J} \chi(\varepsilon_j \xi) a_j(x, \xi)}_{\in S^{m-J}},$$

so the whole expression lies in S^{m-J} . □

(Uniqueness mod $S^{-\infty}$: if $a, \tilde{a}$ both work, then $a - \tilde{a} \in S^{m-J}$ for every J , i.e. $a - \tilde{a} \in S^{-\infty}$.)

3.9 Quantization and pseudodifferential operators

Definition 3.14 (Compound symbols). We now allow the amplitude to depend on **both** x and y , with possible polynomial growth in $x - y$:

$$\langle x - y \rangle^w S^m(\mathbb{R}_x^n \times \mathbb{R}_y^n; \mathbb{R}_\xi^n) = \left\{ a = \langle x - y \rangle^w \tilde{a} \mid \tilde{a} \in S^m(\mathbb{R}_x^n \times \mathbb{R}_y^n; \mathbb{R}_\xi^n) \right\}.$$

These are the **compound symbols**. The parameter $w \in \mathbb{R}$ measures growth off the diagonal.

Quantization

Definition 3.15 (quantization). For $u \in \mathcal{S}(\mathbb{R}^n)$,

$$\text{Op}(a)u(x) = \iint e^{i(x-y) \cdot \xi} a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n}.$$

Remark. As written, the integral converges absolutely only when $m < -n$ (so that $\langle \xi \rangle^m$ is integrable in $\xi \in \mathbb{R}^n$). Everything below is about removing this restriction.

Example 3.16 (A separated amplitude). The amplitude $a(x, y, \xi) = a_1(x) \xi^\sigma a_2(y)$ gives

$$\text{Op}(a)u = a_1 \cdot D^\sigma(a_2u).$$

Special cases

(1) a independent of y (**left quantization**). Then the y -integral produces $\widehat{u}$:

$$\text{Op}(a)u = \int_{\mathbb{R}^n} e^{ix\xi} a(x, \xi) \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n} =: \text{Op}_L(a)u,$$

which converges for **all** $m \in \mathbb{R}$ (since $\widehat{u} \in \mathcal{S}$).

(2) a independent of x (**right quantization**).

$$\text{Op}(a)u = \iint e^{i(x-y)\xi} a(y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n} =: \text{Op}_R(a)u.$$

(Mnemonic: "left" keeps the variable on the left of the operator, i.e. the coefficients act after differentiation.)

Goals

(1) Make sense of $\text{Op}(a) : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$ for general compound symbols.

(2) **Left/right reduction:** show

$$\text{Op}(a) = \text{Op}_L(a_L) = \text{Op}_R(a_R)$$

for some $a_L \in S^m(\mathbb{R}_x^n; \mathbb{R}_\xi^n)$ and $a_R \in S^m(\mathbb{R}_y^n; \mathbb{R}_\xi^n)$.

Exercise. Prove (2) for polynomial symbols (differential operators), by integration by parts / Taylor expansion.

First mapping property ($m < -n$)

Proposition 3.17. Let $a \in \langle x - y \rangle^w S^m$ with $m < -n$. Then

$$\text{Op}(a) : \mathcal{S}(\mathbb{R}^n) \longrightarrow \langle x \rangle^w C_b^0(\mathbb{R}_x^n).$$

In fact, for every $N > n + |w|$ there is $C > 0$ with

$$\|\text{Op}(a)u\|_{\langle x \rangle^w C_b^0} \stackrel{\text{def}}{=} \|\langle x \rangle^{-w} \text{Op}(a)u\|_{C_b^0} \leq C \|\tilde{a}\|_{m,0} \|u\|_N.$$

Here $\|u\|_N = \sup_{|\alpha|+|\beta| \leq N} \|x^\alpha D^\beta u\|_{L^\infty}$ is a Schwartz seminorm.

Proof. **Peetre's inequality:** for all $x, y \in \mathbb{R}^n$,

$$\langle x - y \rangle^w \leq 2^{|w|/2} \langle x \rangle^w \langle y \rangle^{|w|}.$$

Then, inserting $1 = \langle y \rangle^{-N} \langle y \rangle^N$,

$$|\text{Op}(a)u(x)| \leq \iint \left| e^{i(x-y)\xi} \tilde{a}(x, y, \xi) \langle x - y \rangle^w \langle y \rangle^{-N} \langle y \rangle^N u(y) \right| dy \frac{d\xi}{(2\pi)^n},$$

and estimate the three marked factors separately:

Hence

$$|\text{Op}(a)u(x)| \leq C \iint \langle \xi \rangle^m \langle x \rangle^w \langle y \rangle^{|w|-N} dy \frac{d\xi}{(2\pi)^n} \cdot \|\tilde{a}\|_{m,0} \|u\|_N.$$

- $m < -n \Rightarrow$ the ξ -integral converges;
- $N > n + |w| \Rightarrow$ the y -integral converges.

Dividing by $\langle x \rangle^w$ gives the claim. □

factor	bound
$\tilde{a}(x, y, \xi)$	$\leq \ \tilde{a}\ _{m,0} \langle \xi \rangle^m$
$\langle x - y \rangle^w$	$\lesssim \langle x \rangle^w \langle y \rangle^{ w }$
$\langle y \rangle^N u(y)$	$\leq \ u\ _N$

Removing the restriction: $\mathcal{S} \rightarrow \mathcal{S}$ for residual symbols We still need to make sense of

$$\text{Op}(a)u(x) = \iint e^{i(x-y)\cdot\xi} a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n}, \quad a = \langle x - y \rangle^w \tilde{a}, \quad \tilde{a} \in S^m.$$

Proposition 3.18. *Let $a \in \langle x - y \rangle^w S^{-\infty}(\mathbb{R}_x^n \times \mathbb{R}_y^n; \mathbb{R}_\xi^n)$, $w \in \mathbb{R}$. Then*

$$\text{Op}(a) : \mathcal{S}(\mathbb{R}^n) \longrightarrow \mathcal{S}(\mathbb{R}^n) \quad \text{continuously.}$$

Moreover, for all m, k there exist $N, C > 0$ with

$$\|\text{Op}(a)u\|_k \leq C \|a\|_{m,N} \|u\|_N.$$

Proof. Key insight (integration by parts in y). Since $D_y e^{i(x-y)\cdot\xi} = -\xi e^{i(x-y)\cdot\xi}$,

$$(1 - \xi \cdot D_y) e^{i(x-y)\cdot\xi} = (1 + |\xi|^2) e^{i(x-y)\cdot\xi} = \langle \xi \rangle^2 e^{i(x-y)\cdot\xi}.$$

Therefore, for any N ,

$$\begin{aligned} \text{Op}(a)u(x) &= \iint \frac{(1 - \xi D_y)^N}{\langle \xi \rangle^{2N}} e^{i(x-y)\cdot\xi} a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n} \\ &= \iint \frac{e^{i(x-y)\cdot\xi}}{\langle \xi \rangle^{2N}} (1 + \xi D_y)^N (a(x, y, \xi) u(y)) dy \frac{d\xi}{(2\pi)^n} \quad (\text{transpose } D_y) \\ &= \iint e^{i(x-y)\cdot\xi} \sum_{|r| \leq N} a_r(x, y, \xi) D_y^r u(y) dy \frac{d\xi}{(2\pi)^n} \\ &= \sum_{|r| \leq N} \text{Op}(a_r)(D_y^r u), \end{aligned}$$

where, after expanding by Leibniz,

$$a_r(x, y, \xi) = \sum_{|\delta|, |\epsilon| \leq N} c_{r,\delta,\epsilon} \frac{\xi^\delta}{\langle \xi \rangle^{2N}} D_y^\epsilon a(x, y, \xi), \quad \delta \geq \epsilon.$$

Check the symbol bound: $\|\tilde{a}_r\|_{m-N,0} \lesssim \|\tilde{a}\|_{m,N}$. Indeed, since $2N - |\delta| \geq |\delta|$ and $|\delta| \geq |\epsilon|$,

$$\frac{D_y^\epsilon a}{\langle \xi \rangle^{2N-|\delta|} \langle \xi \rangle^{m-N}} = \frac{D_y^\epsilon a}{\langle \xi \rangle^{m+N-|\delta|}} \leq \frac{D_y^\epsilon a}{\langle \xi \rangle^{m-|\epsilon|}},$$

because $m + N - |\delta| \geq m - |\epsilon|$. So each a_r is a symbol of order $m - N$, with seminorms controlled by the N -th seminorm of a .

Now **take** $N \gg 0$ **so that** $m - N < -n$, and apply the first mapping property to each $\text{Op}(a_r)$. With $M > n + |w|$:

$$\|\text{Op}(a)u\|_{\langle x \rangle^w C_b^0} \lesssim \sum_r \|\text{Op}(a_r) D_y^r u\|_{\langle x \rangle^w C_b^0} \leq \sum_r \|\tilde{a}\|_{m,N} \|D_y^r u\|_M \leq \|\tilde{a}\|_{m,N} \|u\|_{M+N}.$$

Now the derivatives and weights (to get Schwartz, not just bounded). Apply the two identities below to

$$x^\alpha D_x^\beta \text{Op}(a)u : \\ D_x e^{i(x-y)\xi} = -D_y e^{i(x-y)\xi}, \quad x^\alpha e^{i(x-y)\xi} = (y + D_\xi)^\alpha e^{i(x-y)\xi}.$$

(Goal: convert powers of x and x -derivatives into ξ -derivatives and y -derivatives, which fall on the amplitude.) Then

$$\begin{aligned} x^\alpha D_x^\beta \text{Op}(a)u &= \iint \left[(y + D_\xi)^\alpha (-D_y)^\beta e^{i(x-y)\xi} \right] a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n} \\ &= \iint e^{i(x-y)\xi} D_y^\beta \left[(y - D_\xi)^\alpha a(x, y, \xi) u(y) \right] dy \frac{d\xi}{(2\pi)^n} \\ &= \text{finite sum of } \text{Op}(a')u', \end{aligned}$$

with new amplitudes a' of the same type and $u' \in \mathcal{S}$ — so these can be estimated exactly as before. This gives control of every Schwartz seminorm $\|\text{Op}(a)u\|_k$, i.e. $\text{Op}(a)u \in \mathcal{S}$ with continuous dependence. $\square$

Extension to arbitrary order, and the space Ψ^m

Corollary 3.19. For arbitrary m, w , the map

$$\langle x - y \rangle^w S^{-\infty} \times \mathcal{S} \longrightarrow \mathcal{S}$$

extends to a **unique** map

$$\langle x - y \rangle^w S^m \times \mathcal{S} \longrightarrow \mathcal{S}$$

continuous with respect to the $\langle x - y \rangle^w S^{m'}$ -topology (for $m' > m$).

Proof. Take a cutoff χ (a bump equal to 1 near 0) and set $a\chi(\xi/j) \in \langle x - y \rangle^w S^{-\infty}$. By the residual-symbol approximation, $a\chi(\xi/j) \rightarrow a$ in $S^{m+\varepsilon}$. Define

$$\text{Op}(a)u := \lim_{j \rightarrow \infty} \text{Op}(a\chi(\xi/j))u$$

Applying theorem 3.18 with order $m + \varepsilon$, the limit exists in $\mathcal{S}(\mathbb{R}^n)$, and the uniform bounds give boundedness of

$$\langle x - y \rangle^w S^m \times \mathcal{S} \rightarrow \mathcal{S}.$$

$\square$

Definition 3.20 (Ψ DO). For $m \in \mathbb{R}$, the space of (**uniform**) pseudodifferential operators of order $\leq m$ is

$$\Psi^m(\mathbb{R}^n) = \{ A : \mathcal{S} \rightarrow \mathcal{S} \mid \exists w \in \mathbb{R}, a \in \langle x - y \rangle^w S^m \text{ s.t. } \text{Op}(a) = A \}.$$

Remarks.

- Also written Ψ_∞^m to emphasize *uniform*.
- $\Psi^{-\infty} = \bigcap_m \Psi^m$, and $\text{Op}(S^{-\infty}) \subset \Psi^{-\infty}$; the reverse inclusion $\supset$ will be proved later.
- Equivalently $\Psi^m = \bigcup_w \text{Op}(\langle x - y \rangle^w S^m) = \text{Op}(S^m)$, the last equality being the left-reduction theorem (later).

Action on distributions Pair $\text{Op}(a)u$ against $v \in \mathcal{S}$ and change the roles of the variables:

$$\begin{aligned} \langle \text{Op}(a)u, v \rangle &= \iiint e^{i(x-y)\cdot\xi} a(x, y, \xi) u(y) v(x) dy dx \frac{d\xi}{(2\pi)^n} \\ &= \iiint e^{i(x-y)\cdot\xi} u(x) v(y) \underbrace{a(y, x, -\xi)}_{= a^T(x, y, \xi)} dy dx \frac{d\xi}{(2\pi)^n} \\ &= \langle u, \text{Op}(a^T)v \rangle, \end{aligned}$$

where the **transposed symbol** is

$$a^T(x, y, \xi) := a(y, x, -\xi).$$

The map $a \mapsto a^T$ is an isomorphism of $\langle x-y \rangle^w S^m$ onto itself. Consequently, by duality, $\text{Op}(a)$ extends to

$$\langle x-y \rangle^w S^m \times \mathcal{S}' \longrightarrow \mathcal{S}',$$

i.e. every uniform pseudodifferential operator acts continuously on tempered distributions.

Part II

The Pseudodifferential Calculus

Chapter 4

Symbolic Calculus

This chapter develops left and right reduction, smoothing kernels, pseudolocality, composition, principal symbols, commutators, and classical operators. It concludes with a worked singular-integral example.

4.1 Conventions and Notation

Throughout, $n \geq 1$, $x, y \in \mathbb{R}^n$ are space variables and $\xi \in \mathbb{R}^n$ is the dual (frequency) variable.

Normalized measure. We use $\frac{d\xi}{(2\pi)^n}$ for normalized Lebesgue measure, so the Fourier transform and its inverse read

$$\widehat{u}(\xi) = \int e^{-iy \cdot \xi} u(y) dy, \quad u(x) = \int e^{ix \cdot \xi} \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n}.$$

Derivatives. $D = -i\partial$, so $D_x^\alpha e^{i(x-y) \cdot \xi} = \xi^\alpha e^{i(x-y) \cdot \xi}$, $D_y^\beta e^{i(x-y) \cdot \xi} = (-\xi)^\beta e^{i(x-y) \cdot \xi}$, and

$$D_\xi^\alpha e^{i(x-y) \cdot \xi} = (x-y)^\alpha e^{i(x-y) \cdot \xi}. \quad (4.1)$$

Japanese bracket. $\langle x \rangle = (1 + |x|^2)^{1/2}$. We use *Peetre's inequality* repeatedly:

$$\langle y \rangle^N \lesssim \langle x \rangle^N \langle x - y \rangle^{|N|}.$$

Symbol classes. For $m \in \mathbb{R}$, $a \in S^m(\mathbb{R}_z^N; \mathbb{R}_\xi^n)$ iff $a \in C^\infty$ and

$$|\partial_z^\beta \partial_\xi^\alpha a(z, \xi)| \leq C_{\alpha\beta} \langle \xi \rangle^{m-|\alpha|} \quad \forall \alpha, \beta.$$

These seminorms make S^m a Fréchet space. We write $S^{-\infty} = \bigcap_m S^m$ and, for $w \in \mathbb{R}$,

$$\langle x - y \rangle^w S^m(\mathbb{R}_x^n \times \mathbb{R}_y^n; \mathbb{R}_\xi^n) = \{ \langle x - y \rangle^w a : a \in S^m \}.$$

Quantizations. For an *amplitude* $a = a(x, y, \xi)$,

$$\text{Op}(a)u(x) = \iint e^{i(x-y) \cdot \xi} a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n},$$

and the two one-sided (*left* and *right*) quantizations of a symbol $a = a(\cdot, \xi)$:

$$\begin{aligned}\mathrm{Op}_L(a)u(x) &= \int e^{ix \cdot \xi} a(x, \xi) \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n}, \\ \mathrm{Op}_R(a)u(x) &= \iint e^{i(x-y) \cdot \xi} a(y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n}.\end{aligned}$$

Thus Op_L differentiates "first, then multiplies", Op_R the other way round.

Definition 4.1 (The operator classes). For $m \in \mathbb{R}$,

$$\Psi^m(\mathbb{R}^n) = \mathrm{Op} \left(\bigcup_{w \in \mathbb{R}} \langle x - y \rangle^w S^m(\mathbb{R}_x^n \times \mathbb{R}_y^n; \mathbb{R}_\xi^n) \right).$$

The point of allowing an arbitrary polynomial weight $\langle x - y \rangle^w$ in the amplitude is that it costs nothing: the key lemma below shows that such factors can be traded for ξ -derivatives, which only *improve* the order.

We set $\Psi^{-\infty} = \bigcap_m \Psi^m$.

4.2 The theorem

Theorem 4.2 (left/right reduction). *Let $a \in \langle x - y \rangle^w S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$. Then there exist unique symbols $a_L, a_R \in S^m(\mathbb{R}^n; \mathbb{R}^n)$ with*

$$\mathrm{Op}(a) = \mathrm{Op}_L(a_L) = \mathrm{Op}_R(a_R),$$

and they admit the asymptotic expansions

$$\begin{aligned}a_L(x, \xi) &\sim \sum_{\alpha \in \mathbb{N}_0^n} \frac{1}{\alpha!} D_\xi^\alpha \partial_y^\alpha a(x, y, \xi) \Big|_{y=x}, \\ a_R(y, \xi) &\sim \sum_{\alpha \in \mathbb{N}_0^n} \frac{1}{\alpha!} \widetilde{D}_\xi^\alpha \partial_x^\alpha a(x, y, \xi) \Big|_{x=y}, \quad \widetilde{D}_\xi := -D_\xi.\end{aligned}$$

Equivalently: the two maps

$$S^m(\mathbb{R}^n; \mathbb{R}^n) \begin{matrix} \xrightarrow{\mathrm{Op}_L} \\ \xrightarrow{\mathrm{Op}_R} \end{matrix} \Psi^m(\mathbb{R}^n)$$

are bijective. Their inverses are denoted σ_L, σ_R (the left, resp. right, full symbol).

Remarks.

- All the terms of the expansions are evaluated **on the diagonal** $\{x = y\}$. So the reduced symbol only sees the amplitude in an infinitesimal neighborhood of the diagonal — which is exactly the region where the operator is not smoothing. This is the structural reason why the whole theory is well behaved.
- Each term $\frac{1}{\alpha!} D_\xi^\alpha \partial_y^\alpha a|_{y=x}$ lies in $S^{m-|\alpha|}$: every ξ -derivative gains one order. So the sum is a genuine asymptotic series.
- $\alpha = 0$ gives $a_L(x, \xi) \equiv a(x, x, \xi)$ modulo S^{m-1} : to leading order, restriction to the diagonal.
- The sign $\widetilde{D}_\xi = -D_\xi$ in the expansion of a_R comes from $(x - y)^\alpha e^{i(x-y)\xi} = D_\xi^\alpha e^{i(x-y)\xi}$ (whereas for a_L one Taylor-expands in y and uses $(y - x)^\alpha = (-1)^{|\alpha|} (x - y)^\alpha$); see the key lemma below. Some sources absorb the sign into the convention and simply write $\partial_x^\alpha D_\xi^\alpha a|_{x=y}$.

4.3 Notation: asymptotic sums

Given $a_j \in S^{m_j}$ with $m_j \searrow -\infty$, we write

$$a \sim \sum_{j \geq 0} a_j \quad :\Longleftrightarrow \quad \forall J \in \mathbb{N} : a - \sum_{j=0}^{J-1} a_j \in S^{m_J}.$$

Borel's lemma guarantees that for any such sequence there *exists* $a \in S^{m_0}$ with $a \sim \sum_j a_j$, and a is unique modulo $S^{-\infty}$.

4.4 A baby case

Take the (differential-operator) amplitude

$$a(x, y, \xi) = a_1(x) a_2(y) \xi^r, \quad r \in \mathbb{N}_0^n.$$

Then $\text{Op}(a)u = a_1 D^r(a_2 u)$, and Leibniz gives

$$a_1 D^r(a_2 u) = \sum_{\delta \leq r} \frac{r!}{\delta! (r-\delta)!} a_1 (D^\delta a_2) D^{r-\delta} u = \sum_{\delta \leq r} \frac{1}{\delta!} a_1 (D^\delta a_2) \cdot \frac{r!}{(r-\delta)!} D^{r-\delta} u.$$

Hence

$$\text{Op}(a) = \text{Op}_L \left(\sum_{\delta \leq r} \frac{1}{\delta!} a_1(x) (D^\delta a_2)(x) \frac{r!}{(r-\delta)!} \xi^{r-\delta} \right).$$

Since $\partial_\xi^\delta \xi^r = \frac{r!}{(r-\delta)!} \xi^{r-\delta}$, this is precisely the asymptotic formula of the reduction theorem—and here the series **terminates**, so it is an exact identity.

4.5 The key lemma: $(y-x)^\alpha \rightsquigarrow D_\xi^\alpha$

Lemma 4.3. *Let $a \in \langle x-y \rangle^w S^m$ and $\alpha \in \mathbb{N}_0^n$. Then*

$$\text{Op}((y-x)^\alpha a) = \text{Op}(D_\xi^\alpha a).$$

Proof. By continuity (density of $S^{-\infty}$ in S^m in the topology of $S^{m'}$, $m' > m$), it suffices to treat $m = -\infty$, where all integrals converge absolutely. Using (4.1),

$$(y-x)^\alpha e^{i(x-y) \cdot \xi} = (-D_\xi)^\alpha e^{i(x-y) \cdot \xi},$$

so

$$\begin{aligned} \text{Op}((y-x)^\alpha a)u &= \iint [(-D_\xi)^\alpha e^{i(x-y) \cdot \xi}] a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n} \\ &= \iint e^{i(x-y) \cdot \xi} D_\xi^\alpha a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n}. \end{aligned}$$

the two sign changes (integration by parts in ξ , and $(-D_\xi)^\alpha$) cancelling. $\square$

This is why w does not matter: a weight $\langle x-y \rangle^w$ can be traded, order by order, for ξ -derivatives which lower the order of the symbol.

4.6 Smoothing operators and their kernels

Proposition 4.4. *Let $K \in \mathcal{S}'(\mathbb{R}_x^n \times \mathbb{R}_y^n)$. The following are equivalent.*

- (1) *K is the Schwartz kernel of an operator in $\Psi^{-\infty}(\mathbb{R}^n)$.*
- (2) *$K \in C^\infty(\mathbb{R}_x^n \times \mathbb{R}_y^n)$ and for all α, β, N ,*

$$|\partial_x^\alpha \partial_y^\beta K(x, y)| \lesssim \langle x - y \rangle^{-N}.$$

This proposition is what yields the **uniqueness** of a_L and a_R , because it identifies $\Psi^{-\infty}$ intrinsically.

Proof of (1) $\Rightarrow$ (2). Let $A \in \Psi^{-\infty}(\mathbb{R}^n)$ with kernel K . By definition, for every N there are $w_N \in \mathbb{R}$ and $a_N \in \langle x - y \rangle^{w_N} S^{-N}$ with $A = \text{Op}(a_N)$. Choosing $N \geq n + 1$, the ξ -integral converges absolutely and

$$K(x, y) = \int e^{i(x-y) \cdot \xi} a_N(x, y, \xi) \frac{d\xi}{(2\pi)^n}. \quad (4.2)$$

Decay off the diagonal. For a multi-index r , use (4.1) and integrate by parts in ξ :

$$(y - x)^r K(x, y) = \int (y - x)^r e^{i(x-y) \cdot \xi} a_N \frac{d\xi}{(2\pi)^n} = \int e^{i(x-y) \cdot \xi} D_\xi^r a_N(x, y, \xi) \frac{d\xi}{(2\pi)^n}.$$

Since $D_\xi^r a_N \in \langle x - y \rangle^{w_N} S^{-N-|r|}$,

$$|D_\xi^r a_N(x, y, \xi)| \lesssim \langle x - y \rangle^{w_N} \langle \xi \rangle^{-N-|r|},$$

and for $N + |r| \geq n + 1$ the integral converges, giving

$$|(y - x)^r K(x, y)| \lesssim \langle x - y \rangle^{w_N} \implies |K(x, y)| \lesssim \langle x - y \rangle^{w_N - |r|}.$$

Given any M , choose first N (hence w_N), then $|r| \geq M + w_N$; this gives

$$|K(x, y)| \lesssim \langle x - y \rangle^{-M} \quad \forall M.$$

Derivatives. Differentiate (4.2) under the integral sign and use that D_x (respectively, D_y) acting on the phase produces a factor ξ (respectively, $-\xi$):

$$D_x^\alpha D_y^\beta K(x, y) = \int e^{i(x-y) \cdot \xi} \underbrace{(D_x + \xi)^\alpha [(D_y - \xi)^\beta a_N]}_{=: \tilde{a}_N \in \langle x-y \rangle^{w_N} S^{-N+|\alpha|+|\beta|}} \frac{d\xi}{(2\pi)^n}.$$

Whenever $-N + |\alpha| + |\beta| \leq -(n + 1)$ — which we may always arrange by taking N large — the same estimate as before applies to $\tilde{a}_N$, and we obtain

$$|D_x^\alpha D_y^\beta K(x, y)| \lesssim \langle x - y \rangle^{-M} \quad \forall M.$$

In particular K is smooth. □

Proof of (2) $\Rightarrow$ (1). We construct a_L . We look for $a_L = a_L(x, \xi)$ with

$$K(x, y) = \int e^{i(x-y) \cdot \xi} a_L(x, \xi) \frac{d\xi}{(2\pi)^n}.$$

Substituting $z = x - y$ this reads

$$K(x, x - z) = \int e^{iz \cdot \xi} a_L(x, \xi) \frac{d\xi}{(2\pi)^n} \iff \boxed{a_L(x, \xi) = \int K(x, x - z) e^{-iz \cdot \xi} dz}$$

i.e. $a_L(x, \cdot)$ is the Fourier transform of $z \mapsto K(x, x - z)$. By hypothesis (2), $z \mapsto K(x, x - z)$ is smooth and rapidly decaying *uniformly in x* , together with all its derivatives; hence its Fourier transform is rapidly decaying in ξ with all x -derivatives, i.e.

$$a_L \in S^{-\infty}.$$

Therefore K is the kernel of $\text{Op}_L(a_L) \in \Psi^{-\infty}$. □

4.7 Proof of the reduction theorem (existence)

Proof of existence. Let $a \in \langle x - y \rangle^w S^m$. **Taylor-expand in y around $y = x$ to order N :**

$$\begin{aligned}
 a(x, y, \xi) &= \underbrace{\sum_{k=0}^{N-1} \sum_{|\alpha|=k} \frac{(y-x)^\alpha}{\alpha!} \partial_y^\alpha a(x, y, \xi) \Big|_{y=x}}_{\text{Taylor polynomial}} \\
 &+ \underbrace{N \sum_{|\alpha|=N} \frac{(y-x)^\alpha}{\alpha!} \int_0^1 (1-t)^{N-1} \partial_y^\alpha a(x, x+t(y-x), \xi) dt}_{\text{integral remainder}}.
 \end{aligned} \tag{4.3}$$

Now apply the key lemma to each occurrence of $(y-x)^\alpha$, replacing it by D_ξ^α acting on what follows. Define

$$\begin{aligned}
 a_k(x, \xi) &= \sum_{|\alpha|=k} \frac{1}{\alpha!} D_\xi^\alpha \partial_y^\alpha a(x, y, \xi) \Big|_{y=x} \\
 r_N &= N \sum_{|\alpha|=N} \frac{1}{\alpha!} D_\xi^\alpha \int_0^1 (1-t)^{N-1} \\
 &\quad \times \partial_y^\alpha a(x, x+t(y-x), \xi) dt.
 \end{aligned}$$

Check the orders.

- $a_k \in S^{m-k}$ (each of the k ξ -derivatives gains one order; the y -derivatives cost nothing, and restriction to $y = x$ removes the $\langle x - y \rangle^w$ weight).
- $r_N \in \langle x - y \rangle^w S^{m-N}$ (the N ξ -derivatives gain N orders; the remaining x, y -dependence keeps a polynomial weight, uniformly in $t \in [0, 1]$).

By the key lemma, (4.3) becomes the operator identity

$$\text{Op}(a) = \sum_{k=0}^{N-1} \text{Op}_L(a_k) + \text{Op}(r_N). \tag{4.4}$$

Borel. Since $a_k \in S^{m-k}$ with orders decreasing to $-\infty$, Borel's lemma provides

$$b \in S^m(\mathbb{R}_x^n; \mathbb{R}_\xi^n), \quad b \sim \sum_{k \geq 0} a_k, \quad \text{i.e.} \quad S_N := b - \sum_{k=0}^{N-1} a_k \in S^{m-N} \quad \forall N.$$

Substituting into (4.4) gives

$$\text{Op}(a) = \text{Op}_L(b) + \underbrace{\text{Op}_L(-S_N) + \text{Op}(r_N)}_{\in \Psi^{m-N}}.$$

Since N was arbitrary,

$$\text{Op}(a) - \text{Op}_L(b) \in \bigcap_N \Psi^{m-N} = \Psi^{-\infty} = \text{Op}_L(S^{-\infty}),$$

where the last equality follows from the smoothing-kernel proposition. Absorbing that $S^{-\infty}$ correction into b , we obtain $a_L := b$ with $\text{Op}(a) = \text{Op}_L(a_L)$ and $a_L \sim \sum_k a_k$. This proves **existence of the left reduction**. $\square$

4.8 Injectivity of Op_L

Proof of injectivity. Fix $\xi_0 \in \mathbb{R}^n \setminus \{0\}$ and let

$$e_{\xi_0}(x) = e^{ix \cdot \xi_0} \in \mathcal{S}'(\mathbb{R}_x^n).$$

Then

$$\boxed{\text{Op}_L(a_L) e_{\xi_0}(x) = a_L(x, \xi_0) e_{\xi_0}(x).}$$

(Indeed $\widehat{e_{\xi_0}} = (2\pi)^n \delta_{\xi_0}$, and the defining formula for Op_L collapses to evaluation at ξ_0 .)

Proof: first for $a_L \in S^{-\infty}$, where everything converges absolutely; then for general $a_L \in S^m$ by continuity/density. (**Exercise.**)

Consequently $\text{Op}_L(a_L) = 0$ forces $a_L(x, \xi_0) = 0$ for all x and all $\xi_0 \neq 0$, hence $a_L \equiv 0$ by continuity. So Op_L is injective, and a_L is unique. The same argument (or transposition, below) gives uniqueness of a_R . $\square$

4.9 Right reduction by transposition

Proof of right reduction. Define, for an amplitude a ,

$$a^T(x, y, \xi) := a(y, x, -\xi) \quad \implies \quad \text{Op}(a)^T = \text{Op}(a^T),$$

where T is the transpose with respect to the bilinear pairing $\langle \cdot, \cdot \rangle_{S \times S'}$ (no complex conjugation).

Now take $\tilde{a}_L$ with $\text{Op}(a)^T = \text{Op}_L(\tilde{a}_L)$ (possible by 1.6). Transposing back,

$$\text{Op}(a) = \text{Op}_L(\tilde{a}_L)^T = \text{Op}_R(\tilde{a}_L^T), \quad \tilde{a}_L^T(y, \xi) = \tilde{a}_L(y, -\xi).$$

So the right reduction follows from the left one, with the expected sign flip $D_\xi \mapsto -D_\xi$. $\square$

4.10 Topology on Ψ^m

Definition 4.5 (Operator topology).

The topology on $\Psi^m(\mathbb{R}^n)$ is transported from $S^m(\mathbb{R}^n; \mathbb{R}^n)$ by requiring

$$\sigma_L: \Psi^m \longrightarrow S^m$$

to be a homeomorphism. Equivalently, one may use Op_L . By theorem 4.2, using σ_R instead gives the same topology.

4.11 Schwartz kernels and pseudolocality

Proposition 4.6 (pseudolocality). *Let $A \in \Psi^m(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$. Then*

$$\text{sing supp}(Au) \subseteq \text{sing supp}(u).$$

The proof rests on the following.

Lemma 4.7. $A \in \Psi^{-\infty}(\mathbb{R}^n) \implies A : \mathcal{S}'(\mathbb{R}^n) \longrightarrow \mathcal{S}'(\mathbb{R}^n) \cap C^\infty(\mathbb{R}^n)$.

Proof of the lemma. Step 1: $u \in \mathcal{S}$. With K the kernel of A (smooth, rapidly decaying off the diagonal by the smoothing-kernel proposition),

$$Au(x) = \int K(x, y)u(y) dy = \langle K(x, \cdot), u \rangle \leq \|K(x, \cdot)\|_N \cdot \sup_{\|\varphi\|_N \leq 1} |\langle u, \varphi \rangle|,$$

where $\|\cdot\|_N$ denotes the N -th Schwartz seminorm. Now

$$\sup \|y^\alpha D_y^\beta K(x, y)\|_{L_y^\infty} \lesssim \langle y \rangle^N \langle x - y \rangle^{-N} \lesssim \langle x \rangle^N$$

by Peetre. Hence

$$|\langle x \rangle^{-N} Au(x)| \leq C \sup_{\|\varphi\|_N \leq 1} |\langle u, \varphi \rangle|, \quad \text{i.e.} \quad \|Au\|_{\langle x \rangle^N C_b^0} < \infty.$$

Step 2: $u \in \mathcal{S}'$. Choose $(u_j) \subset \mathcal{S}$ with $\langle u - u_j, \varphi \rangle \rightarrow 0$. Because u is a continuous functional on the Fréchet space $\mathcal{S}$, there is a single N with

$$\sup_{\|\varphi\|_N \leq 1} |\langle u_j - u, \varphi \rangle| \longrightarrow 0.$$

This uses the standard characterization of continuity for a linear map out of a Fréchet space F : if the topology is generated by an increasing family of seminorms p_α , then $\|f(v)\| \leq Cp_\alpha(v)$ for some α and C .

Passing to the limit in Step 1 gives $Au \in \langle x \rangle^N C_b^0$.

Step 3: smoothness. For any α , $D^\alpha A \in \Psi^{-\infty}$ as well, and $D^\alpha(Au) = (D^\alpha A)u$, so the same bound applies to every derivative. Hence $Au \in C^\infty$. $\square$

Proof of the proposition. Let $x_0 \notin \text{sing supp}(u)$. Choose

$$\chi \in C_c^\infty, \quad \chi \equiv 1 \text{ near } x_0, \quad \text{supp } \chi \cap \text{sing supp } u = \emptyset,$$

and a second cutoff $\tilde{\chi} \in C_c^\infty(\mathbb{R}^n)$ with

$$\text{supp}(\tilde{\chi}) \cap \text{sing supp } u = \emptyset, \quad \text{supp } \chi \subset \{\tilde{\chi} = 1\}.$$

Figure 4.1 illustrates this nesting. **Goal:** χAu is smooth. Split

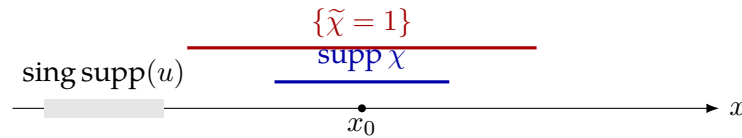


Figure 4.1: Nested cutoffs used in the pseudolocality argument.

$$\chi Au = \underbrace{\chi A \tilde{\chi} u}_{\text{smooth}} + \underbrace{\chi A (1 - \tilde{\chi}) u}_{= A^\chi u}.$$

The first term is smooth because $\tilde{\chi}u \in C_c^\infty$ and A maps $\mathcal{S} \rightarrow \mathcal{S}$.

Claim 4.8. $A^\chi := \chi A (1 - \tilde{\chi}) \in \Psi^{-\infty}(\mathbb{R}^n)$.

Proof of the Claim. Write $A = \text{Op}_R(a)$. Then

$$A^\chi = \text{Op}_R(\chi(x) a(x, \xi) (1 - \tilde{\chi}(y))) = \text{Op}_L(a_L^\chi),$$

and by the reduction formula,

$$a_L^\chi \sim \sum_{\alpha} \frac{1}{\alpha!} \chi(x) D_\xi^\alpha a(x, \xi) \partial_x^\alpha (1 - \tilde{\chi}(x)) = 0 \quad \forall \alpha,$$

because on $\text{supp } \chi$ we have $\tilde{\chi} \equiv 1$, so $1 - \tilde{\chi}$ vanishes identically together with all its derivatives there. Hence every term of the expansion vanishes and $a_L^\chi \in S^{-\infty}$.

By the Lemma, $A^\chi u$ is smooth, so $\chi A u$ is smooth, i.e. $x_0 \notin \text{sing supp}(A u)$. $\square$

Corollary 4.9. For $A \in \Psi^m(\mathbb{R}^n)$ with Schwartz kernel K_A ,

$$\text{sing supp}(K_A) \subseteq \Delta = \{(x, x)\}.$$

Figure 4.2 depicts the possible singular support of the kernel.

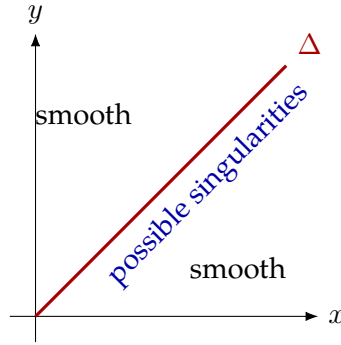


Figure 4.2: A pseudodifferential kernel can be singular only on the diagonal.

4.12 Composition

Theorem 4.10. Let $A \in \Psi^m(\mathbb{R}^n)$, $B \in \Psi^{m'}(\mathbb{R}^n)$. Then $A \circ B : \mathcal{S} \rightarrow \mathcal{S}$ belongs to $\Psi^{m+m'}(\mathbb{R}^n)$, and

$$\sigma_L^{m+m'}(A \circ B) \sim \sum_{\alpha} \frac{1}{\alpha!} D_\xi^\alpha [\sigma_L^m(A)] \partial_x^\alpha [\sigma_L^{m'}(B)].$$

Each term of the expansion carries one ξ -derivative and one x -derivative per unit of order gained. This is precisely the pattern of the Poisson bracket and leads to the commutator formula below.

Proof. Write $A = \text{Op}_L(a)$, $B = \text{Op}_R(b)$ (left reduction for A , right reduction for B — this is the combination that composes cleanly). Assume first both are of order $-\infty$, so all integrals converge absolutely; the general case follows by continuity.

Step 1: compute $\widehat{Bu}$.

$$Bu(x) = \iint e^{i(x-y)\cdot\xi} b(y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n} = \int e^{ix\cdot\xi} \left[\int e^{-iy\cdot\xi} b(y, \xi) u(y) dy \right] \frac{d\xi}{(2\pi)^n},$$

so

$$\widehat{Bu}(\xi) = \int e^{-iy \cdot \xi} b(y, \xi) u(y) dy.$$

Step 2: apply A .

$$A(Bu)(x) = \int e^{ix \cdot \xi} a(x, \xi) \widehat{Bu}(\xi) \frac{d\xi}{(2\pi)^n} = \iint e^{i(x-y) \cdot \xi} a(x, \xi) b(y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n}.$$

Hence the exact identity

$$\boxed{\text{Op}_L(a) \circ \text{Op}_R(b) = \text{Op}(a(x, \xi) b(y, \xi)).}$$

By continuity this persists for $a \in S^m, b \in S^{m'}$, and obviously

$$c(x, y, \xi) := a(x, \xi) b(y, \xi) \in S^{m+m'}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n).$$

So $A \circ B = \text{Op}(c) \in \Psi^{m+m'}$.

Step 3: reduce c to the left. By the reduction theorem,

$$\sigma_L^{m+m'}(AB)(x, \xi) \sim \sum_{\alpha} \frac{1}{\alpha!} D_{\xi}^{\alpha} \partial_y^{\alpha} c(x, y, \xi) \Big|_{y=x}.$$

Since ∂_y only hits b , Leibniz in ξ gives

$$D_{\xi}^{\alpha} \partial_y^{\alpha} c \Big|_{y=x} = \sum_{\beta+\gamma=\alpha} \binom{\alpha}{\beta} D_{\xi}^{\beta} a(x, \xi) D_{\xi}^{\gamma} \partial_y^{\alpha} b(y, \xi) \Big|_{y=x},$$

so, regrouping the sum over (β, γ) instead of α ,

$$\sigma_L(AB) \sim \sum_{\beta, \gamma} \frac{1}{\beta! \gamma!} D_{\xi}^{\beta} a(x, \xi) D_{\xi}^{\gamma} \partial_x^{\beta+\gamma} b(x, \xi) = \sum_{\beta} \frac{1}{\beta!} D_{\xi}^{\beta} a(x, \xi) \underbrace{\partial_x^{\beta} \left[\sum_{\gamma} \frac{1}{\gamma!} D_{\xi}^{\gamma} \partial_x^{\gamma} b \right]}_{= \sigma_L(B)}.$$

The inner sum is exactly the left symbol of $B = \text{Op}_R(b)$ (apply the reduction theorem to the amplitude $b(y, \xi)$). Therefore

$$\sigma_L(A \circ B) \sim \sum_{\beta} \frac{1}{\beta!} D_{\xi}^{\beta} \sigma_L(A) \partial_x^{\beta} \sigma_L(B).$$

□

4.13 The principal symbol

Definition 4.11. The **principal symbol** of $A \in \Psi^m(\mathbb{R}^n)$ is the class

$$\sigma^m(A) = [\sigma_L(A)] \in S^m(\mathbb{R}^n; \mathbb{R}^n) / S^{m-1}(\mathbb{R}^n; \mathbb{R}^n).$$

Note that $[\sigma_L(A)] = [\sigma_R(A)]$: the two reductions differ by $\frac{1}{i} \sum_j \partial_{\xi_j} \partial_{x_j} a + \dots \in S^{m-1}$. So the principal symbol is independent of the choice of quantization.

The symbol sequence

Remark 1. There is a short exact sequence

$$0 \longrightarrow \Psi^{m-1}(\mathbb{R}^n) \hookrightarrow \Psi^m(\mathbb{R}^n) \xrightleftharpoons[\sigma_L]{\sigma^m} \frac{S^m(\mathbb{R}^n; \mathbb{R}^n)}{S^{m-1}(\mathbb{R}^n; \mathbb{R}^n)} \longrightarrow 0,$$

with σ_L providing a (non-canonical) splitting.

Multiplicativity and adjoints

Remark 2.

$$\sigma^{m+m'}(AB) = \sigma^m(A) \sigma^{m'}(B), \quad \sigma^m(A^*)(x, y, \xi) = \overline{\sigma^m(A)(y, x, \xi)}.$$

Two different adjoints are in play:

operation	defined by
$A^T = \text{Op}(a^T)$	adjoint w.r.t. the bilinear pairing $\langle \cdot, \cdot \rangle_{S \times S'}$
A^*	adjoint w.r.t. the L^2 inner product — involves complex conjugation

Commutators and the Poisson bracket Let $A \in \Psi^m$, $B \in \Psi^{m'}$. A priori $[A, B] = AB - BA \in \Psi^{m+m'}$, but

$$\sigma^{m+m'}([A, B]) = [\sigma^m(A), \sigma^{m'}(B)] = 0$$

since the principal symbols are functions and hence commute. Therefore

$$[A, B] \in \Psi^{m+m'-1}.$$

The next-order term is the Poisson bracket:

Proposition 4.12.

$$\sigma^{m+m'-1}(i[A, B]) = \{\sigma^m(A), \sigma^{m'}(B)\}$$

where for $a = a(x, \xi)$, $b = b(x, \xi)$,

$$\{a, b\} = \sum_j (\partial_{\xi_j} a \partial_{x_j} b - \partial_{x_j} a \partial_{\xi_j} b).$$

In particular $\{\cdot, \cdot\}$ **descends to the quotients:**

$$\frac{S^m}{S^{m-1}} \times \frac{S^{m'}}{S^{m'-1}} \longrightarrow \frac{S^{m+m'-1}}{S^{m+m'-2}}.$$

Proof. Put $a = \sigma_L(A)$, $b = \sigma_L(B)$. By the composition formula, keeping the terms $|\alpha| \leq 1$,

$$\sigma_L(AB) = ab + \sum_{j=1}^n D_{\xi_j} a \partial_{x_j} b + \text{h.o.t.} = ab + \frac{1}{i} \sum_j \partial_{\xi_j} a \partial_{x_j} b \mod S^{m+m'-2}.$$

Interchanging the roles of A and B and subtracting,

$$\sigma_L(AB - BA) = \frac{1}{i} \sum_j (\partial_{\xi_j} a \partial_{x_j} b - \partial_{\xi_j} b \partial_{x_j} a) \mod S^{m+m'-2},$$

i.e. $\sigma^{m+m'-1}(i[A, B]) = \{a, b\}$. □

4.14 Classical (polyhomogeneous) operators

Definition 4.13. $S_{cl}^m(\mathbb{R}^n; \mathbb{R}^n)$ denotes the classical symbols, i.e. those admitting an asymptotic expansion into terms positively homogeneous of degrees $m, m-1, m-2, \dots$ in ξ for $|\xi| \geq 1$. Set

$$\text{Op}(S_{cl}^m(\mathbb{R}^n; \mathbb{R}^n)) = \Psi_{cl}^m(\mathbb{R}^n) \subset \Psi^m(\mathbb{R}^n), \quad \Psi_{cl}^{-\infty} = \Psi^{-\infty}.$$

Remarks.

- (1) $\Psi_{cl}^\bullet$ is closed under multiplication (composition) and adjoints.
- (2) There is a short exact sequence

$$0 \longrightarrow \Psi_{cl}^{m-1} \longrightarrow \Psi_{cl}^m \xrightarrow{\sigma^m} S_{\text{hom}}^m(\mathbb{R}^n; \mathbb{R}^n \setminus \{0\}) \longrightarrow 0,$$

i.e. for classical operators the principal symbol can be taken to be an honest **homogeneous function** on $\mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$, rather than a class in a quotient.

Summary so far.

$$S^m(\mathbb{R}_x^n; \mathbb{R}_\xi^n) \xrightleftharpoons[\sigma_L]{\text{Op}_L} \Psi^m(\mathbb{R}^n).$$

Example 4.14 (Homogeneous singular integral operators). Let $A : \mathcal{S}(\mathbb{R}) \rightarrow \mathcal{S}(\mathbb{R})$ be given by the principal-value convolution-type formula

$$Au(x) = \lim_{\varepsilon \rightarrow 0} \int_{|x-y| > \varepsilon} \underbrace{K(x, x-y) u(y)}_{\text{convolution in the 2nd slot}} \chi(x-y) dy, \quad \chi \in C_c^\infty(\mathbb{R}), \quad \chi \equiv 1 \text{ near } 0.$$

The cutoff χ ensures uniform estimates at large distances without changing the singularity structure.

Assumptions on the kernel.

- $K \in C^\infty(\mathbb{R}_x \times (\mathbb{R} \setminus \{0\})_z)$;
- **homogeneity of degree -1** : $K(x, \lambda z) = \lambda^{-1} K(x, z)$ for $\lambda > 0$;
- $K(x, 1) \in C_b^\infty(\mathbb{R}_x)$;
- (for the computation below) K is **odd** in z : $K(x, -z) = -K(x, z)$.

Claim 4.15. $A \in \Psi_{cl}^0(\mathbb{R})$, with principal symbol

$$\sigma^0(A)(x, \xi) = \frac{\text{sgn}(\xi)}{i} \pi K(x, 1).$$

(The model case $K(x, z) = \frac{1}{\pi z}$ is the Hilbert transform, whose symbol is $-i \text{sgn } \xi$.)

Proof. Write the truncated operator, substituting $z = x - y$:

$$A_\varepsilon u(x) = \int_{|y-x| > \varepsilon} \chi(x-y) K(x, x-y) u(y) dy = \int e^{ix \cdot \xi} a_\varepsilon(x, \xi) \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n},$$

where

$$a_\varepsilon(x, \xi) = \int_{|z| > \varepsilon} e^{-iz \cdot \xi} \chi(z) K(x, z) dz.$$

Note a_ε is **not** a symbol, but $a_\varepsilon \in C_b^\infty(\mathbb{R} \times \mathbb{R})$.

Exploit oddness. Splitting $\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{+\infty}$ and using $K(x, -z) = -K(x, z)$, χ even,

$$a_{\varepsilon}(x, \xi) = \int_{\varepsilon}^{+\infty} [e^{-iz\xi} - e^{iz\xi}] \chi(z) K(x, z) dz, \quad e^{-iz\xi} - e^{iz\xi} = -2i \sin(z\xi).$$

Rescale. Set $t = \pm z|\xi|$ (sign according to $\text{sgn } \xi$), so $dt = \pm|\xi| dz$ and $z = t/|\xi|$. Homogeneity gives $K(x, t/|\xi|) = \frac{|\xi|}{t} K(x, 1)$, hence

$$a_{\varepsilon}(x, \xi) = \mp \int_{\varepsilon|\xi|}^{+\infty} 2i \sin(t) \chi\left(\frac{t}{|\xi|}\right) K\left(x, \frac{t}{|\xi|}\right) \frac{dt}{|\xi|} = \mp 2i K(x, 1) \int_{\varepsilon|\xi|}^{+\infty} \frac{\sin t}{t} \chi\left(\frac{t}{|\xi|}\right) dt.$$

Let $\varepsilon \rightarrow 0$.

$$a_{\varepsilon}(x, \xi) \xrightarrow{\varepsilon \rightarrow 0} a(x, \xi) = K(x, 1) \cdot b(\xi), \quad b(\xi) = \mp 2i \int_0^{+\infty} \frac{\sin t}{t} \chi\left(\frac{t}{|\xi|}\right) dt.$$

Claim (check). $b \in S^0(\mathbb{R}_{\xi})$. Consequently $a \in S^0$ and $A = \text{Op}_L(a) \in \Psi^0(\mathbb{R})$.

Claim. $b \in S_{cl}^0(\mathbb{R})$; more precisely $b(\xi) - b_0(\xi) \in S^{-\infty}$, where

$$b_0(\xi) = \mp 2i \int_0^{\infty} \frac{\sin t}{t} dt = \mp 2i \cdot \frac{\pi}{2} = \mp i\pi = \frac{\text{sgn}(\xi)}{i} \pi.$$

(The difference $b - b_0$ involves $\chi(t/|\xi|) - 1$, which vanishes for $t \lesssim |\xi|$, giving rapid decay in ξ .)

Therefore $A \in \Psi_{cl}^0(\mathbb{R})$ with $\sigma^0(A)(x, \xi) = K(x, 1) b_0(\xi) = \frac{\text{sgn} \xi}{i} \pi K(x, 1)$. $\square$

4.15 Elliptic operators

Definition 4.16. $A \in \Psi^m(\mathbb{R}^n)$ is called **elliptic** iff its principal symbol $\sigma^m(A) \in S^m/S^{m-1}$ is elliptic.

The next chapter constructs parametrices for elliptic operators.

Chapter 5

Elliptic Theory and Mapping Properties

5.1 Notation and Conventions

For $a \in S^m(\mathbb{R}_x^n, \mathbb{R}_\xi^n)$ the **left quantization** is

$$\text{Op}_L(a)u(x) = \int_{\mathbb{R}^n} e^{ix \cdot \xi} a(x, \xi) \widehat{u}(\xi) \frac{d\xi}{(2\pi)^n} = \iint e^{i(x-y) \cdot \xi} a(x, \xi) u(y) dy \frac{d\xi}{(2\pi)^n}.$$

More generally, for an **amplitude** $a \in S^m(\mathbb{R}_x^n \times \mathbb{R}_y^n; \mathbb{R}_\xi^n)$,

$$\text{Op}(a)u(x) = \iint e^{i(x-y) \cdot \xi} a(x, y, \xi) u(y) dy \frac{d\xi}{(2\pi)^n}.$$

We write $\Psi^m(\mathbb{R}^n) = \text{Op}_L(S^m)$, $\Psi^{-\infty} = \bigcap_m \Psi^m$ for operators with smooth Schwartz kernels, and K_A for the Schwartz kernel of A .

Facts used repeatedly:

- **(Left reduction.)** $\text{Op}(a) = \text{Op}_L(a_L)$ with $a_L(x, \xi) \sim \sum_\alpha \frac{1}{\alpha!} \partial_\xi^\alpha D_y^\alpha a(x, y, \xi)|_{y=x}$; in particular $a_L(x, \xi) \equiv a(x, x, \xi) \bmod S^{m-1}$.
- **(Symbol map.)** $\sigma^m : \Psi^m \rightarrow S^m/S^{m-1}$ is the principal symbol, and $\sigma^{m+m'}(AB) = \sigma^m(A) \sigma^{m'}(B)$, $\sigma^m(A^*) = \overline{\sigma^m(A)}$.
- **(Asymptotic summation.)** Given $A_j \in \Psi^{m-j}$ there is $A \in \Psi^m$, unique mod $\Psi^{-\infty}$, with $A - \sum_{j < J} A_j \in \Psi^{m-J}$ for every J ; we write $A \sim \sum_j A_j$.
- **(Pseudolocality.)** $\text{sing supp}(Au) \subset \text{sing supp}(u)$ for $A \in \Psi^m$.

5.2 Elliptic operators

Definition 5.1. $A \in \Psi^m(\mathbb{R}^n)$ is called **elliptic** iff $\sigma(A) \in S^m/S^{m-1}$ is elliptic, i.e. for one (equivalently, any) representative a of $\sigma^m(A)$ there are constants $C, R > 0$ with

$$|a(x, \xi)| \geq \frac{1}{C} \langle \xi \rangle^m, \quad |\xi| \geq R, \quad x \in \mathbb{R}^n.$$

(The condition is well posed on the quotient: changing a by an element of S^{m-1} changes it by $O(\langle \xi \rangle^{m-1}) = o(\langle \xi \rangle^m)$, which is absorbed for $|\xi|$ large.)

5.3 Symbolic inverses

Lemma 5.2. *If $a \in S^m$ is elliptic then there is $b \in S^{-m}$ with $ab - 1 \in S^{-\infty} \subset S^{-1}$.*

Proof. Pick $\chi \in C^\infty(\mathbb{R}_\xi^n)$ with $\chi \equiv 0$ on $\{|\xi| \leq R\}$ and $\chi \equiv 1$ on $\{|\xi| \geq 2R\}$, and set

$$b(x, \xi) = \frac{\chi(\xi)}{a(x, \xi)}.$$

This is smooth (the denominator does not vanish on $\text{supp } \chi$). To see $b \in S^{-m}$: by induction on $|\alpha| + |\beta|$, $\partial_\xi^\alpha \partial_x^\beta (1/a)$ is a finite linear combination of terms

$$a^{-(k+1)} \prod_{i=1}^k \partial_\xi^{\alpha_i} \partial_x^{\beta_i} a, \quad \sum_i \alpha_i = \alpha, \sum_i \beta_i = \beta, \quad k \geq 1,$$

each of which is $O(\langle \xi \rangle^{-(k+1)m} \cdot \langle \xi \rangle^{km-|\alpha|}) = O(\langle \xi \rangle^{-m-|\alpha|})$ on $|\xi| \geq R$. Finally $ab - 1 = \chi - 1$ is supported in $\{|\xi| \leq 2R\}$, hence lies in $S^{-\infty}$. $\square$

5.4 Existence of a parametrix

Theorem 5.3. *Let $A \in \Psi^m(\mathbb{R}^n)$ be elliptic. Then there exists $B \in \Psi^{-m}(\mathbb{R}^n)$ such that*

$$AB - I, \quad BA - I \in \Psi^{-\infty}(\mathbb{R}^n),$$

and B is unique modulo $\Psi^{-\infty}$.

Proof. Write $A = \text{Op}_L(a)$ with a elliptic. By the Lemma choose $b \in S^{-m}$ with $ab - 1 \in S^{-1}$, and put $B_0 = \text{Op}_L(b) \in \Psi^{-m}$. Then

$$\sigma^0(AB_0 - I) = ab - 1 \in S^{-1} \implies AB_0 = I - R, \quad R \in \Psi^{-1}.$$

Since R has **negative order**, $I - R$ is invertible modulo smoothing, by a Neumann series made rigorous through asymptotic summation. Indeed $R^j \in \Psi^{-j}$, so there is $Q \in \Psi^0$ with

$$Q - \sum_{j=0}^{J-1} R^j \in \Psi^{-J}(\mathbb{R}^n) \quad \forall J, \quad \text{i.e.} \quad Q \sim \sum_{j \geq 0} R^j.$$

Then, modulo Ψ^{-J} ,

$$AB_0Q = (I - R)Q \equiv (I - R) \left(\sum_{j=0}^{J-1} R^j \right) = \underbrace{I - R^J}_{\text{telescoping}} \equiv I \pmod{\Psi^{-J}}.$$

As J was arbitrary, $A(B_0Q) = I \pmod{\Psi^{-\infty}}$, so $B_r := B_0Q \in \Psi^{-m}$ is a **right** parametrix.

The same procedure applied to $\text{Op}_L(b)A$ produces $B_\ell \in \Psi^{-m}$ with $B_\ell A \equiv I \pmod{\Psi^{-\infty}}$, a **left** parametrix.

Uniqueness: work in the ring $\Psi^\infty/\Psi^{-\infty}$, where identities hold modulo smoothing. A left and a right inverse of the same element coincide:

$$B_\ell \equiv B_\ell(AB_r) = (B_\ell A)B_r \equiv B_r \pmod{\Psi^{-\infty}}.$$

In particular the inverse is unique, and $B := B_r$ satisfies both identities. $\square$

Definition 5.4 (Elliptic parametrix). An operator $B \in \Psi^{-m}$ satisfying

$$AB - I, BA - I \in \Psi^{-\infty}$$

is called an **elliptic parametrix** of A .

5.5 Elliptic regularity

Proposition 5.5. Let $A \in \Psi^m(\mathbb{R}^n)$ be elliptic and $u \in \mathcal{S}'(\mathbb{R}^n)$. Then

(1) $Au \in C^\infty(\mathbb{R}^n) \implies u \in C^\infty(\mathbb{R}^n)$;

(2) $\text{sing supp}(u) \subset \text{sing supp}(Au)$.

Combined with pseudolocality, which gives $\text{sing supp}(Au) \subset \text{sing supp}(u)$, one gets the *equality*

$$\text{sing supp}(Au) = \text{sing supp}(u).$$

Proof. Let B be a parametrix, $BA - I = -R$ with $R \in \Psi^{-\infty}$. Then

$$u = BAu + Ru.$$

Since R has Schwartz kernel in $\mathcal{S}(\mathbb{R}^{2n})$, it has the mapping property

$$R : \mathcal{S}'(\mathbb{R}^n) \longrightarrow \mathcal{S}(\mathbb{R}^n).$$

Thus Ru is smooth and $\text{sing supp}(Ru) = \emptyset$. By pseudolocality of B ,

$$\text{sing supp}(u) \subset \text{sing supp}(BAu) \cup \underbrace{\text{sing supp}(Ru)}_{=\emptyset} \subset \text{sing supp}(Au).$$

This is (2), and (1) is the special case $\text{sing supp}(Au) = \emptyset$. □

5.6 Examples

Example 5.6 (The Laplacian). The operator $\Delta = \sum_j D_{x_j}^2$ on $\mathbb{R}^n$ has symbol

$$a(x, \xi) = |\xi|^2 \in S_{\text{hom}}^2(\mathbb{R}^n; \mathbb{R}^n \setminus \{0\}),$$

which vanishes only at $\xi = 0$; hence Δ is elliptic.

Example 5.7 (The Cauchy–Riemann operator). Identify $\mathbb{R}^2 = \mathbb{C}$ and let

$$\partial_{\bar{z}} = \frac{1}{2}(\partial_{x_1} + i\partial_{x_2}), \quad a(x, \xi) = i\left(\frac{1}{2}\xi_1 + \frac{i}{2}\xi_2\right) \in S_{\text{hom}}^1(\mathbb{R}^2; \mathbb{R}^2 \setminus \{0\}).$$

Again $a(x, \xi) = \frac{i}{2}(\xi_1 + i\xi_2) \neq 0$ for $\xi \neq 0$, so $\partial_{\bar{z}}$ is elliptic. (Its parametrix is the Cauchy transform, and elliptic regularity recovers the smoothness of holomorphic functions.)

5.7 Boundedness on L^2 and on Sobolev spaces

Schur test

Lemma 5.8. *Let $(X, \mu), (Y, \nu)$ be measure spaces and $K : X \times Y \rightarrow \mathbb{C}$ measurable with*

$$(1) \sup_x \int_Y |K(x, y)| d\nu(y) = C_1 < +\infty,$$

$$(2) \sup_y \int_X |K(x, y)| d\mu(x) = C_2 < +\infty.$$

Then $Au(x) = \int_Y K(x, y)u(y) d\nu(y)$ is bounded $L^2(\nu) \rightarrow L^2(\mu)$, with $\|A\|_{\text{op}} \leq \sqrt{C_1 C_2}$.

Proof. For $u \in L^2(\nu), v \in L^2(\mu)$, split $|K| = |K|^{1/2} \cdot |K|^{1/2}$ and use Cauchy–Schwarz on the product measure:

$$\begin{aligned} |(Au, v)_{L^2(\mu)}| &\leq \iint |K(x, y)| |u(y)| |v(x)| d\nu d\mu = \iint (|K|^{1/2} |u(y)|) (|K|^{1/2} |v(x)|) \\ &\leq \left(\iint |K| |u(y)|^2 \right)^{1/2} \left(\iint |K| |v(x)|^2 \right)^{1/2} \leq \sqrt{C_2} \|u\|_{L^2} \cdot \sqrt{C_1} \|v\|_{L^2}, \end{aligned}$$

where in the first factor we integrated in x first (hypothesis 2) and in the second factor in y first (hypothesis 1). Taking the supremum over $\|v\| \leq 1$ gives $\|A\|_{\text{op}} \leq \sqrt{C_1 C_2}$. $\square$

Boundedness of smoothing operators

Proposition 5.9. $A \in \Psi^{-\infty}(\mathbb{R}^n) \implies A : L^2 \rightarrow L^2$ is bounded.

Proof. The Schwartz kernel K of A satisfies, for all α, β, N ,

$$|\partial_x^\alpha \partial_y^\beta K(x, y)| \leq C_N \langle x - y \rangle^{-N}.$$

Take $N = n + 1$. Then

$$\sup_x \int \langle x - y \rangle^{-(n+1)} dy < +\infty, \quad \sup_y \int \langle x - y \rangle^{-(n+1)} dx < +\infty,$$

so both Schur hypotheses hold and the Schur lemma applies. $\square$

Calderón–Vaillancourt, order 0

Proposition 5.10.

$$A \in \Psi^0(\mathbb{R}^n) \text{ is bounded } L^2(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n).$$

Hörmander’s square-root trick. **Idea.** Find $C > 0$ and $B \in \Psi^0$ with

$$C - A^* A = B^* B \pmod{\Psi^{-\infty}(\mathbb{R}^n)}, \quad \text{i.e.} \quad R := C - A^* A - B^* B \in \Psi^{-\infty}(\mathbb{R}^n).$$

Granting this, for $u \in \mathcal{S}$ (dense in L^2),

$$\begin{aligned} \|Au\|_{L^2}^2 &= (A^* Au, u)_{L^2} \\ &= C\|u\|_{L^2}^2 - \underbrace{(B^* Bu, u)}_{\|Bu\|_{L^2}^2 \geq 0} - (Ru, u) \\ &\leq C\|u\|_{L^2}^2 + \|Ru\|_{L^2} \|u\|_{L^2} \leq C'\|u\|_{L^2}^2. \end{aligned}$$

using that $R \in \Psi^{-\infty}$ is L^2 -bounded by theorem 5.9. Hence A extends boundedly to L^2 .

How to construct B . Write $A = \text{Op}_L(a)$, $a \in S^0(\mathbb{R}^n; \mathbb{R}^n)$, and set

$$C = 2 \sup_{x, \xi} |a(x, \xi)|^2 < +\infty, \quad b_0(x, \xi) = \sqrt{C - |a(x, \xi)|^2} \in S^0.$$

(Exercise: the expression under the root is $\geq C/2 > 0$, so $\sqrt{\cdot}$ is smooth there; the symbol estimates follow from the chain rule exactly as in the elliptic-symbol lemma above.) Put $B_0 = \text{Op}_L(b_0) \in \Psi^0(\mathbb{R}^n)$. Then

$$R_1 := C - A^*A - B_0^*B_0 \in \Psi^{-1},$$

by checking the principal symbol: $\sigma^0(A^*A) = |a|^2$ and $\sigma^0(B_0^*B_0) = b_0^2 = C - |a|^2$, so $\sigma^0(R_1) = C - |a|^2 - (C - |a|^2) = 0$.

Improve the error inductively. Suppose $B_0, \dots, B_{J-1}$ have been found with $B_j \in \Psi^{-j}$ and

$$R_J = C - A^*A - \left(\sum_{j=0}^{J-1} B_j \right)^* \left(\sum_{j=0}^{J-1} B_j \right) \in \Psi^{-J}.$$

We look for $B_J \in \Psi^{-J}$ such that

$$R_J - B_J^* \left(\sum_{j=0}^{J-1} B_j \right) - \left(\sum_{j=0}^{J-1} B_j^* \right) B_J \in \Psi^{-(J+1)}.$$

At the level of principal symbols only B_0 contributes to the leading order, so this is equivalent to

$$0 = \sigma^{-J}(R_J) - \sigma^{-J}(B_J) \sigma_0(B_0) - \sigma_0(B_0) \sigma^{-J}(B_J).$$

Since $\sigma_0(B_0) = b_0 \geq \sqrt{C/2} > 0$, we may define

$$b_J = \frac{\sigma^{-J}(R_J)}{2 \sigma_0(B_0)} \in S^{-J}, \quad B_J = \text{Op}_L(b_J).$$

Because $B_J^*B_J \in \Psi^{-2J} \subset \Psi^{-(J+1)}$ for $J \geq 1$, this gives $R_{J+1} \in \Psi^{-(J+1)}$ and the induction continues. Finally take the asymptotic sum

$$B \sim \sum_{J \geq 0} B_J \in \Psi^0,$$

for which $C - A^*A - B^*B \in \Psi^{-\infty}$, as required. $\square$

Continuity of the quantization into $\mathcal{L}(L^2)$

$$\boxed{\Psi^0(\mathbb{R}^n) \longrightarrow \mathcal{L}(L^2(\mathbb{R}^n)), \quad A \mapsto (u \mapsto Au)}$$

is **continuous** (for the Fréchet topology on Ψ^0 induced by the symbol seminorms).

Example 5.11 (Closed graph theorem argument). Consider

$$\begin{array}{ccc} \Psi^0 & \longrightarrow & \mathcal{L}(L^2) \\ \text{cts } \downarrow & & \downarrow \text{ injection} \\ \mathcal{L}(\mathcal{S}, L^2) & = & \mathcal{L}(\mathcal{S}, L^2) \end{array}$$

The two maps

$$\Psi^0 \longrightarrow \mathcal{L}(\mathcal{S}, L^2), \quad \mathcal{L}(L^2) \hookrightarrow \mathcal{L}(\mathcal{S}, L^2)$$

are respectively continuous and continuous injective. Hence the graph of $\Psi^0 \rightarrow \mathcal{L}(L^2)$ is closed. The closed graph theorem (Fréchet source, Banach target) now yields continuity.

Compactness

Proposition 5.12. *Let $A \in \Psi^m(\mathbb{R}^n)$ with $\boxed{m < 0}$ and $\text{supp}(K_A) \Subset \mathbb{R}^n \times \mathbb{R}^n$. Then $A : L^2 \hookrightarrow$ is **compact**.*

Proof. Choose $K \subset \mathbb{R}^n$ compact with $\text{supp}(K_A) \subset K \times K$. Recall

$$C_K^k(\mathbb{R}^n) = \{u \in C^k(\mathbb{R}^n) : \text{supp } u \subset K\}$$

is a Banach space and $C_K^{k+1}(\mathbb{R}^n) \hookrightarrow C_K^k(\mathbb{R}^n)$ is **compact** by the Arzelà–Ascoli theorem.

Case 1: $A \in \Psi^{-\infty}(\mathbb{R}^n)$. The kernel is smooth and compactly supported, so A maps L^2 into C_K^1 boundedly (differentiate under the integral, Cauchy–Schwarz). Then

$$A : L^2 \longrightarrow C_K^1(\mathbb{R}^n) \xrightarrow{\text{cpt}} C_K^0(\mathbb{R}^n) \hookrightarrow L^2,$$

a bounded map followed by a compact one; hence $A : L^2 \hookrightarrow$ is compact.

Case 2: $A \in \Psi^m$, $m < 0$. Fix $\varepsilon > 0$ with $m + \varepsilon < 0$. Truncating the symbol, $a_j(x, \xi) = a(x, \xi)\chi(\xi/j)$, produces $A_j \in \Psi^{-\infty}(\mathbb{R}^n)$ with $\text{supp}(K_{A_j}) \subset K \times K$ and

$$A_j \longrightarrow A \quad \text{in } \Psi^{m+\varepsilon}(\mathbb{R}^n) \subset \Psi^0(\mathbb{R}^n).$$

By the continuity statement above, $A_j \rightarrow A$ in $\mathcal{L}(L^2)$. A norm-limit of compact operators is compact, so A is compact. $\square$

Action on Sobolev spaces

$$H^s(\mathbb{R}^n) = \left\{ u \in \mathcal{S}'(\mathbb{R}^n) : \|\langle \xi \rangle^s \hat{u}\|_{L^2(\mathbb{R}_\xi^n)} < +\infty \right\}, \quad \langle D \rangle^s u = \mathcal{F}^{-1}(\langle \xi \rangle^s \hat{u}),$$

$$\|u\|_{H^s} = \|\langle D \rangle^s u\|_{L^2}.$$

E.g. $s = 2$ gives $\langle D \rangle^2 = 1 + \Delta$; in general $s \in \mathbb{R}$. (Exercise: this is the same as the old definition of H^s .)

Moreover

$$\langle D \rangle^\sigma : H^s \xrightarrow{\sim} H^{s-\sigma} \quad \text{is an isomorphism,} \quad \langle D \rangle^\sigma \in \Psi^\sigma(\mathbb{R}^n).$$

The Sobolev mapping property

Proposition 5.13. $A \in \Psi^m(\mathbb{R}^n) \implies A : H^s \rightarrow H^{s-m}$ is bounded, for every $s \in \mathbb{R}$.

Proof. Consider the commutative diagram

$$\begin{array}{ccc} H^s & \xrightarrow{A} & H^{s-m} \\ \langle D \rangle^s \downarrow \wr & & \wr \downarrow \langle D \rangle^{s-m} \\ L^2 & \xrightarrow{B} & L^2(\mathbb{R}^n) \end{array} \quad B \triangleq \langle D \rangle^{s-m} A \langle D \rangle^{-s}.$$

The vertical maps are isomorphisms, so A is bounded $H^s \rightarrow H^{s-m}$ iff B is bounded on L^2 . By the composition calculus,

$$B \in \Psi^{(s-m)+m+(-s)}(\mathbb{R}^n) = \Psi^0(\mathbb{R}^n),$$

which is L^2 -bounded by the Calderón–Vaillancourt proposition. $\square$

Example 5.14 (Characterization of H^s by elliptic regularity). If $A \in \Psi^s(\mathbb{R}^n)$ is elliptic, then

$$H^s(\mathbb{R}^n) = \{ u \in \mathcal{S}'(\mathbb{R}^n) : Au \in L^2(\mathbb{R}^n) \}.$$

Proof. For the forward inclusion, theorem 5.13 gives $A : H^s \rightarrow H^0 = L^2$. Conversely, let $B \in \Psi^{-s}$ be a parametrix, so that $u = BAu + Ru$ with $R \in \Psi^{-\infty}$. If $Au \in L^2 = H^0$, then $BAu \in H^s$, while $Ru \in \mathcal{S} \subset H^s$. Hence $u \in H^s$. $\square$

5.8 Coordinate invariance

For $\Omega \subset \mathbb{R}^n$ open, set

$$\Psi_c^m(\Omega) = \{ A \in \Psi^m(\mathbb{R}^n) : \text{supp}(K_A) \Subset \Omega \times \Omega \}.$$

Let $\kappa : \Omega \xrightarrow{\sim} \Omega'$ be a diffeomorphism.

Invariance under changes of coordinates

Theorem 5.15. Let $\Omega, \Omega' \subset \mathbb{R}^n$, let $\kappa : \Omega \rightarrow \Omega'$ be a diffeomorphism and $A \in \Psi_c^m(\Omega')$. For $u \in \mathcal{S}(\mathbb{R}^n)$ define

$$A_\kappa u := \kappa^* A \kappa_* (u|_\Omega).$$

Then

- (1) $A_\kappa \in \Psi_c^m(\Omega)$;
- (2) $\sigma^m(A_\kappa)(x, \xi) = \sigma^m(A)(\kappa(x), D\kappa(x)^{-T}\xi)$
- (3) $A \mapsto A_\kappa$ is a bijection $\Psi_c^m(\Omega') \rightarrow \Psi_c^m(\Omega)$.

Proof. Write $A = \text{Op}_L(a_L)$ with $a_L \in S^m(\mathbb{R}_x^n, \mathbb{R}_\xi^n)$; write a for a_L .

(i) **Localize.** Take $\gamma \in C_c^\infty(\Omega')$ such that

$$(x, y) \in \text{supp}(K_A) \implies \gamma(x)\gamma(y) = 1.$$

Such a function exists because $\text{supp } K_A \Subset \Omega' \times \Omega'$. Then, as an amplitude quantization,

$$\text{Op}(\gamma(x)\gamma(y)a(x, \xi)) = \gamma A(\gamma \cdot) \quad \text{since} \quad K_A(x, y) \rightsquigarrow \gamma(x)K_A(x, y)\gamma(y) = K_A(x, y),$$

so this operator $= A$.

(ii) **Cut off near the diagonal.** Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi \equiv 1$ near 0, and set

$$a_\varepsilon(x, y, \xi) = \chi\left(\frac{|x - y|}{\varepsilon}\right) \gamma(x) \gamma(y) a(x, \xi).$$

Figure 5.1 shows the region retained by this cutoff.

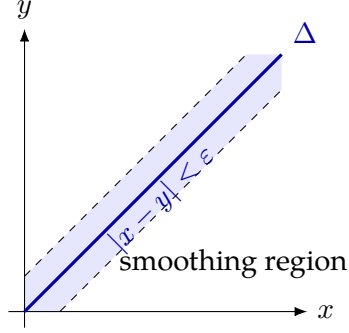


Figure 5.1: The amplitude is retained near the diagonal; the off-diagonal remainder is smoothing.

Then

$$A = \text{Op}(a_\varepsilon) + R_\varepsilon,$$

where the amplitude of R_ε vanishes on a neighborhood of the diagonal. Repeated integration by parts in ξ (allowed since $|x - y| \geq \varepsilon$ there) shows $K_{R_\varepsilon} \in C_c^\infty$, i.e. R_ε is a **residual (smoothing) operator, by construction**. So it suffices to treat $\text{Op}(a_\varepsilon)$ and R_ε separately.

Step 1: the residual part R_ε .

K_{R_ε} is a compactly supported smooth function. Then

$$\begin{aligned} R_{\varepsilon, \kappa} u(x) &= \int_{\mathbb{R}^n} K_{R_\varepsilon}(\kappa(x), z) u(\kappa^{-1}(z)) dz \\ &= \int_{\mathbb{R}^n} K_{R_\varepsilon}(\kappa(x), \kappa(y)) u(y) |\det D\kappa(y)| dy. \end{aligned}$$

Hence

$$\begin{aligned} K_{R_{\varepsilon, \kappa}}(x, y) &= K_{R_\varepsilon}(\kappa(x), \kappa(y)) |\det D\kappa(y)| \in C_c^\infty(\Omega \times \Omega), \\ R_{\varepsilon, \kappa} &\in \Psi_c^{-\infty}(\Omega). \end{aligned}$$

Step 2: the main part a_ε .

Assume first $m = -\infty$, so that all the integrals below converge absolutely; the general case follows by continuity at the end. Let $u \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\begin{aligned} A_{\varepsilon, \kappa} u(x) &= \iint e^{i(\kappa(x) - z) \cdot \eta} a_\varepsilon(\kappa(x), z, \eta) u(\kappa^{-1}(z)) dz d\eta \\ &\stackrel{z=\kappa(y)}{=} \iint e^{i(\kappa(x) - \kappa(y)) \cdot \eta} a_\varepsilon(\kappa(x), \kappa(y), \eta) u(y) |\det D\kappa(y)| dy d\eta. \end{aligned}$$

Linearize the phase. By Taylor with integral remainder,

$$\kappa(x) - \kappa(y) = \underbrace{\left(\int_0^1 D\kappa(y + t(x-y)) dt \right)}_{=: \Xi(x,y) \in \mathbb{R}^{n \times n}} (x-y), \quad \Xi(x,x) = D\kappa(x) \in GL_n(\mathbb{R}).$$

Choose ε small enough that on the (compact) set

$$\{(x,y) : \gamma(\kappa(x))\gamma(\kappa(y)) \neq 0, |x-y| < \varepsilon\}$$

we still have $\Xi(x,y) \in GL_n(\mathbb{R})$ — possible by continuity of $\det \Xi$ and compactness. Then $(\kappa(x) - \kappa(y)) \cdot \eta = (x-y) \cdot \Xi^T(x,y)\eta$, so

$$A_{\varepsilon,\kappa}u(x) = \iint \exp(i(x-y) \cdot \Xi^T(x,y)\eta) \cdots dy d\eta.$$

Substitute $\xi = \Xi^T(x,y)\eta$, a **linear** (hence measure-friendly) change of variables in η for fixed (x,y) , with $d\xi = |\det \Xi^T(x,y)| d\eta$:

$$A_{\varepsilon,\kappa}u(x) = \iint e^{i(x-y) \cdot \xi} \underbrace{a_\varepsilon(\kappa(x), \kappa(y), \Xi^{-T}\xi) \frac{|\det D\kappa(y)|}{|\det \Xi(x,y)|}}_{=: b_\varepsilon(x,y,\xi)} u(y) dy d\xi.$$

Therefore

$$\text{Op}(a_\varepsilon)_\kappa = \text{Op}(b_\varepsilon), \quad a \in S^{-\infty}(\mathbb{R}^n; \mathbb{R}^n). \quad (5.1)$$

Check: $b_\varepsilon \in S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$ is a legitimate amplitude — the map $\xi \mapsto \Xi^{-T}\xi$ is a smooth family of linear isomorphisms with uniformly bounded inverse on the relevant compact set, and the Jacobian factor is smooth and compactly supported in (x,y) .

Moreover $a \mapsto b_\varepsilon$ is **continuous** from $S^{m'} \rightarrow S^{m'}$ for every $m' \geq m$. Since $S^{-\infty}$ is dense in S^m with respect to the topology of $S^{m'}$ for $m' > m$, (5.1) extends to all $a \in S^m(\mathbb{R}^n; \mathbb{R}^n)$.

Consequently $A_{\varepsilon,\kappa} \in \Psi_c^m(\Omega)$, and **by left reduction** of the amplitude,

$$\begin{aligned} \sigma^m(A_{\varepsilon,\kappa})(x, \xi) &= b_\varepsilon(x, x, \xi) \pmod{S^{m-1}(\mathbb{R}^n; \mathbb{R}^n)} \\ &= a_\varepsilon(\kappa(x), \kappa(x), D\kappa(x)^{-T}\xi) = a(\kappa(x), D\kappa(x)^{-T}\xi), \end{aligned}$$

Here $\Xi(x,x) = D\kappa(x)$, so the two Jacobian factors cancel. We also used

$$\chi(0) = 1, \quad \gamma(\kappa(x))^2 = 1$$

on the relevant set.

Adding back Step 1 (a smoothing operator, which does not affect the principal symbol) gives (1) and (2). Statement (3) follows because κ^{-1} induces the inverse map $\Psi_c^m(\Omega) \rightarrow \Psi_c^m(\Omega')$. $\square$

Compactly supported Sobolev spaces

$$H_c^s(\Omega) = \{ u \in H^s(\mathbb{R}^n) : \text{supp } u \Subset \Omega \}.$$

Diffeomorphism invariance of compactly supported Sobolev spaces

Proposition 5.16. $\kappa^* : H_c^s(\Omega') \rightarrow H_c^s(\Omega)$.

Proof. Let $v \in H_c^s(\Omega')$ and $u = \kappa^* v \in \mathcal{S}'(\mathbb{R}^n)$, with $\text{supp}(u) \Subset \Omega$.

Goal: find an elliptic $A \in \Psi^s$ such that $Au \in L^2$; then $u \in H^s$ by the Sobolev characterization above.

Take $\phi, \tilde{\phi} \in C_c^\infty(\Omega)$ with

- $\phi \equiv 1$ on $\text{supp } u$,
- $\tilde{\phi}(x) = 1$ whenever $\text{dist}(x, \text{supp } \phi) \leq \varepsilon$.

Then $\phi u = u$. Choose $A \in \Psi^s(\mathbb{R}^n)$ elliptic whose kernel is supported in the ε -neighborhood Δ_ε of the diagonal (e.g. cut off the kernel of $\langle D \rangle^s$ near the diagonal; this changes it by a smoothing operator and preserves ellipticity). Then

$$Au = \tilde{\phi} A \phi u,$$

and $\tilde{\phi} A \phi \in \Psi_c^s(\Omega)$ (its kernel is compactly supported inside $\Omega \times \Omega$). By part (3) of the coordinate-invariance theorem there is $A' \in \Psi_c^s(\Omega')$ with

$$\tilde{\phi} A \phi = A'_\kappa.$$

Hence

$$Au = A'_\kappa u = \kappa^*(A'v) \in L^2,$$

because $A'v \in L^2$ ($v \in H^s$, $A' \in \Psi^s$) and κ^* preserves L^2 for compactly supported functions (change of variables; the Jacobian is bounded above and below on the relevant compact set). Therefore $u \in H^s$ with compact support in Ω . $\square$

Compactly supported symbols and the twisted pullback We now have the three compactly supported objects

$$\Psi_c^m(\Omega), \quad H_c^s(\Omega), \quad S_c^m(\Omega; \mathbb{R}^N),$$

where

$$S_c^m(\Omega; \mathbb{R}^N) = \{ a \in S^m(\mathbb{R}^n; \mathbb{R}^N) : \text{supp } a \subset K \times \mathbb{R}^N, K \subset \Omega \text{ compact} \}.$$

The relevant geometry is a bundle map covering κ :

$$\begin{array}{ccc} \Omega \times \mathbb{R}^N & \xrightarrow{\hat{\kappa}} & \Omega' \times \mathbb{R}^N \\ \downarrow & & \downarrow \\ \Omega & \xrightarrow{\kappa} & \Omega' \end{array} \quad \hat{\kappa}(x, \xi) = (\kappa(x), \gamma_x(\xi)), \quad \gamma : \Omega \rightarrow GL_N(\mathbb{R}).$$

Figure 5.2 displays the resulting bundle map.

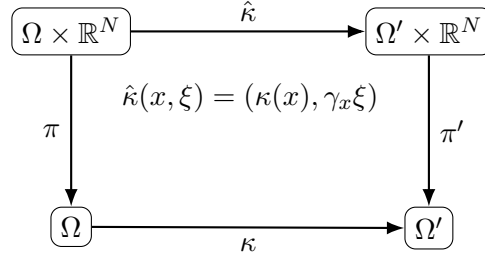


Figure 5.2: The fiberwise-linear bundle map covering the coordinate change.

Fiberwise-linear invariance of symbols

Proposition 5.17. *Let $\kappa : \Omega \xrightarrow{\sim} \Omega'$ be a diffeomorphism and $\gamma : \Omega \rightarrow GL_N(\mathbb{R})$ smooth. Then*

$$\hat{\kappa}^* : S_c^m(\Omega'; \mathbb{R}^N) \longrightarrow S_c^m(\Omega; \mathbb{R}^N), \quad \hat{\kappa}^* a(x, \xi) = a(\kappa(x), \gamma_x(\xi)).$$

(The point is that the symbol classes S_c^m are invariant not only under diffeomorphisms of the base but also under fiberwise linear changes of the covariable — exactly what is needed to transplant the calculus onto a manifold, with $\gamma_x = D\kappa(x)^{-T}$ realizing the cotangent-bundle transition maps.)

Part III

Global and Microlocal Analysis

Chapter 6

Analysis and Pseudodifferential Calculus on Manifolds

Throughout, M and N are smooth, second-countable Hausdorff manifolds of dimension n , while $E \rightarrow M$ and $F \rightarrow N$ are smooth vector bundles.

Before globalizing the calculus, we establish two invariance statements that make chart-wise definitions independent of the chosen chart.

6.1 Invariance of H_c^s under diffeomorphisms

Proposition 6.1. *Let $\kappa : \Omega \xrightarrow{\sim} \Omega'$ be a diffeomorphism of open sets in $\mathbb{R}^n$. Then*

$$\kappa^* : H_c^s(\Omega') \longrightarrow H_c^s(\Omega)$$

is well defined and continuous, for every $s \in \mathbb{R}$.

Proof. Let $v \in H_c^s(\Omega')$ and put $u = \kappa^*v \in \mathcal{S}'(\mathbb{R}^n)$; then $\text{supp } u \subset \Omega$ is compact (the pullback of a compactly supported distribution under a diffeomorphism is defined by duality and has compact support).

Goal: find an *elliptic* A of order s with $Au \in L^2$. This suffices: for an elliptic $A \in \Psi^s$ the elliptic estimate on $\mathbb{R}^n$ gives $\|u\|_s \leq C(\|Au\|_0 + \|u\|_{-N}) < \infty$ for compactly supported u .

Take $A = \langle D \rangle^s$ and choose cutoffs

$$\phi, \tilde{\phi} \in C_c^\infty(\Omega), \quad \phi \equiv 1 \text{ on } \text{supp } u, \quad \tilde{\phi}(x) = 1 \text{ if } \text{dist}(x, \text{supp } \phi) \leq \varepsilon.$$

Then $\phi u = u$, and

$$Au = \tilde{\phi}A\phi u + (1 - \tilde{\phi})A\phi u.$$

The second term is smoothing: $(1 - \tilde{\phi})$ and ϕ have disjoint supports, so $(1 - \tilde{\phi})A\phi \in \Psi^{-\infty}$ has a smooth, rapidly decaying kernel and maps $\mathcal{E}' \rightarrow \mathcal{S} \subset L^2$.

The first term $A_0 := \tilde{\phi}A\phi \in \Psi_c^s(\Omega)$ has $\text{supp } k_{A_0} \subset \Delta_\varepsilon$, an ε -neighborhood of the diagonal, hence is properly supported and compactly supported in $\Omega \times \Omega$. By coordinate invariance of compactly supported pseudodifferential operators, there is $A' \in \Psi_c^s(\Omega')$ with

$$\tilde{\phi}A\phi = \kappa^* \circ A' \circ (\kappa^{-1})^*, \quad \text{i.e.} \quad A_0 u = A_0(\kappa^*v) = \kappa^*(A'v).$$

Since $v \in H_c^s(\Omega')$ and A' has order s , $A'v \in L^2$ with compact support; the pullback by a diffeomorphism preserves L^2 on compact sets (the Jacobian is bounded above and below there). Hence $A_0 u \in L^2$, so $Au \in L^2$ and therefore $u \in H^s$. $\square$

6.2 Invariance of symbols (twisted pullback)

Let

$$S_c^m(\Omega; \mathbb{R}^N) = \{a \in S^m(\mathbb{R}^n; \mathbb{R}^N) \mid \text{supp } a \subset K \times \mathbb{R}^N, K \Subset \Omega\}.$$

Proposition 6.2. *Let $\kappa : \Omega \xrightarrow{\sim} \Omega'$ be a diffeomorphism and $\gamma : \Omega \rightarrow GL_N(\mathbb{R})$ smooth. Define the bundle map*

$$\widehat{\kappa} : \Omega \times \mathbb{R}^N \rightarrow \Omega' \times \mathbb{R}^N, \quad (x, \xi) \mapsto (\kappa(x), \gamma_x(\xi)).$$

Then

$$\widehat{\kappa}^* : S_c^m(\Omega'; \mathbb{R}^N) \rightarrow S_c^m(\Omega; \mathbb{R}^N), \quad \widehat{\kappa}^* a(x, \xi) = a(\kappa(x), \gamma_x(\xi)).$$

*This is exactly the statement needed to define symbols **on the total space of a vector bundle**: the fiber coordinate transforms linearly, and symbol estimates are invariant under such twists. (For Ψ DOs the relevant γ is $\gamma_x = (D\kappa(x)^T)^{-1}$, so symbols live on T^*M .)*

Two generalizations are now made simultaneously:

$$\text{uniform} \rightsquigarrow \text{non-uniform}, \quad \mathbb{R}^n \rightsquigarrow \text{manifold}.$$

Figure 6.1 summarizes the two compatible projections involved in the twisted pullback.

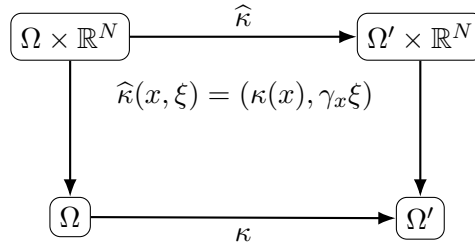


Figure 6.1: A fiberwise-linear bundle map covering a change of coordinates.

6.3 Test functions

Let M be an n -dimensional manifold. The two test function spaces are

$$C^\infty(M), \quad C_c^\infty(M).$$

Definition 6.3 (Test-function topologies on a manifold). For $(f_j) \subset C^\infty(M)$, convergence in $C^\infty(M)$ means

$$\begin{aligned} f_j \rightarrow 0 \text{ in } C^\infty(M) &\iff \text{for every chart } F : U \xrightarrow{\sim} \Omega \subset \mathbb{R}^n \\ &\text{and every } \chi \in C_c^\infty(U), \\ &(F^{-1})^*(\chi f_j) \rightarrow 0 \text{ in } \mathcal{S}(\mathbb{R}^n). \end{aligned}$$

(Since χf_j has compact support in U , convergence in $\mathcal{S}(\mathbb{R}^n)$ is the same as uniform convergence of all derivatives on that fixed compact set.)

Convergence in $C_c^\infty(M)$ means

$$f_j \rightarrow 0 \text{ in } C_c^\infty(M) \iff f_j \rightarrow 0 \text{ in } C^\infty(M) \text{ and } \exists K \Subset M \text{ with } \text{supp}(f_j) \subset K \forall j.$$

Remark 6.4. • $C^\infty(M)$ is a **Fréchet space**: a countable partition of unity subordinate to a countable atlas, each chart contributing countably many seminorms, gives a countable separating family of seminorms; completeness is local.

- $C_c^\infty(M)$ is **generally not Fréchet** (it is an LF-space: the strict inductive limit of the Fréchet spaces $C_K^\infty(M)$, $K \Subset M$).
- If $E \rightarrow M$ is a vector bundle, the same recipe (charts + local trivializations) defines the topology on

$$C^\infty(M; E) = \{\text{smooth sections}\}, \quad C_c^\infty(M; E).$$

6.4 Densities

Recall $\Lambda_x^k M = \{\omega : \underbrace{T_x M \times \cdots \times T_x M}_k \rightarrow \mathbb{R} \text{ alternating}\}$, and $\omega \in C^\infty(M; \Lambda^k M)$ is a k -form.

Example 6.5 (From forms to densities). For an n -form ω and $\alpha \geq 0$,

$$\mu_x(v_1, \dots, v_n) = |\omega(v_1, \dots, v_n)|^\alpha$$

is not a form, but transforms by $|\det|^\alpha$. This is the model for:

Definition 6.6 (α -density bundle). For $0 \leq \alpha \leq 1$ the bundle $\Omega^\alpha M \rightarrow M$ has fibers

$$\Omega_x^\alpha M = \{\mu_x : \Lambda^n T_x M \rightarrow \mathbb{R} : \mu_x(tv) = |t|^\alpha \mu_x(v)\}, \quad v = v_1 \wedge \cdots \wedge v_n.$$

$\Omega^\alpha M \rightarrow M$ is a **trivializable line bundle** ($\text{rk} = 1$).

Why trivializable: the transition functions are $|\det D\kappa|^\alpha > 0$; a partition of unity produces a global **positive** section $\mu > 0$, and any other section is $u\mu$ with $u \in C^\infty(M)$. (Contrast $\Lambda^n M$, which is trivial only if M is orientable.)

Remarks.

(1) **Local form.** In a chart,

$$\mu = u(x) |dx^1 \wedge \cdots \wedge dx^n|^\alpha.$$

(2) **Pullback.** If $f : M \rightarrow N$ is smooth (a local diffeomorphism, e.g. a chart change) there is a pullback

$$f^* : C^\infty(N, \Omega^\alpha N) \rightarrow C^\infty(M, \Omega^\alpha M),$$

locally

$$f^*(v(y) |dy^1 \wedge \cdots \wedge dy^n|^\alpha) = [v \circ f(x)] |\det Df(x)|^\alpha |dx^1 \wedge \cdots \wedge dx^n|^\alpha.$$

(3) **Integration.** Analogously to n -forms, 1-densities can be integrated:

$$\int_M : C_c^\infty(M, \Omega^1 M) \longrightarrow \mathbb{R}.$$

This needs no orientation: the factor $|\det Df|$ in Remark 2 with $\alpha = 1$ is exactly the change-of-variables Jacobian, so the chart-by-chart definition (via a partition of unity) is consistent. This is precisely why analysis on manifolds is done with densities rather than forms — on $\mathbb{R}^n$ one simply uses $\int dx$.

6.5 Distributions

Definition 6.7. Let $E \rightarrow M$ be a real vector bundle. Set

$$\mathcal{D}'(M; E) = \mathcal{L}(C_c^\infty(M, \Omega^1 M \otimes E^*), \mathbb{C})$$

$$\mathcal{E}'(M; E) = \mathcal{L}(C^\infty(M, \Omega^1 M \otimes E^*), \mathbb{C}).$$

The point of the twist by $\Omega^1 M \otimes E^*$ is that *smooth sections* can then be used as test objects: the pairing of a section with a test object is a 1-density, which can be integrated.

Embedding. There is a canonical embedding

$$C^\infty(M; E) \hookrightarrow \mathcal{D}'(M; E), \quad u \mapsto \left(\phi \mapsto \int_M \langle u, \phi \rangle_{E \otimes E^*} \right).$$

Remarks.

- For $M = \mathbb{R}^n$: $\mathcal{E}' \subsetneq \mathcal{S}' \subsetneq \mathcal{D}'$.
- $\mathcal{E}'(M) = \{u \in \mathcal{D}'(M) : \text{supp } u \text{ compact}\}$.
- **Alternative.**

Fix a nonvanishing density μ once and for all. Then $\Omega^1 M \cong M \times \mathbb{R}$, and one may simply define $\mathcal{D}'(M) = [C_c^\infty(M)]'$. All identifications below, including adjoints and L^2 spaces, are then relative to μ .

6.6 Schwartz kernels and properly supported operators

The Schwartz kernel theorem With π_M, π_N the two projections off $N \times M$, put

$$\Omega_R = \pi_M^* \Omega M, \quad \Omega_L = \pi_N^* \Omega N.$$

Figure 6.2 records these projections.

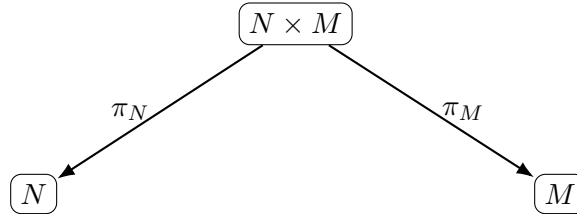


Figure 6.2: The two projections used to place the left and right density factors.

Theorem 6.8 (Schwartz kernels).

$$\mathcal{L}[C_c^\infty(M; E), \mathcal{D}'(N; F)] \cong \mathcal{D}'(N_x \times M_y; F \boxtimes E^* \otimes \Omega_R)$$

the isomorphism being $A \leftrightarrow k = k_A$ with

$$A_k u(x) = \int_M \langle k(x, y), u(y) \rangle_{E^* \otimes E} \in F_x$$

The density factor Ω_R sits on the **integration variable** $y \in M$: that is exactly what makes the integral invariantly defined; the output is still a section (or distribution) in $x \in N$.

Properly supported operators Let $A : C_c^\infty(M; E) \rightarrow C^\infty(N; F)$ be continuous linear.

Definition 6.9. A is **properly supported** iff both restricted projections

$$\pi_M|_{\text{supp } k_A}, \quad \pi_N|_{\text{supp } k_A} : \underbrace{\text{supp } k_A}_{\text{closed}} \longrightarrow M \text{ resp. } N$$

are **proper** maps (preimages of compacta are compact).

Figure 6.3 contrasts proper and nonproper support.

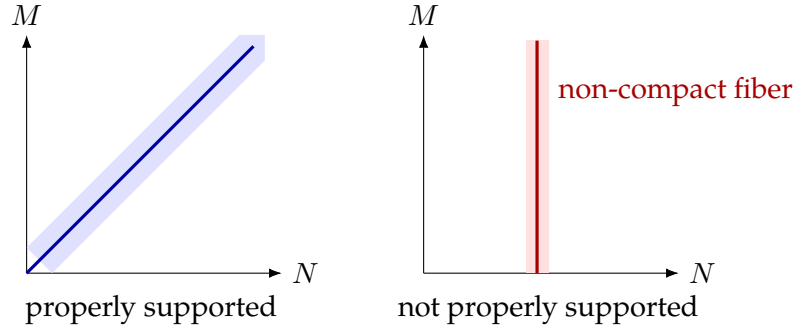


Figure 6.3: Proper support requires both restricted projections of $\text{supp } k_A$ to be proper.

Example 6.10 (Support propagation). If A has Schwartz kernel k , then

$$\text{supp}(Au) \subset \{x \in N : \exists y \in M \text{ with } (x, y) \in \text{supp}(k)\}.$$

Remark 6.11. why proper support matters. If A is properly supported then

$$A : C_c^\infty(M; E) \longrightarrow \mathcal{E}'(N; E),$$

and A extends by locality to

$$A : C^\infty(M; E) \longrightarrow \mathcal{D}'(N; E), \quad u = \sum_i \phi_i u \longmapsto \hat{A}u(w) := \sum_{i \geq 1} A(\phi_i u)(w),$$

where $\{\phi_i\}$ is a partition of unity. **The sum is locally finite**, hence convergent: for a test object w with $\text{supp } w =: L \Subset N$ compact, only those i with

$$\text{supp } \phi_i \cap \pi_M(\text{supp } k_A \cap \pi_N^{-1}(L)) \neq \emptyset$$

contribute, and the set in brackets is compact by properness.

This is the mechanism by which composition of Ψ DOs on a manifold makes sense.

6.7 Sobolev spaces

Definition 6.12. $H_{loc}^s(M) \subset \mathcal{D}'(M)$ consists of all u such that **locally**

$$F_*(\chi u) \in H^s(\mathbb{R}^n)$$

for every chart $F : M \supset U \xrightarrow{\sim} \Omega \subset \mathbb{R}^n$ and every $\chi \in C_c^\infty(U)$. By diffeomorphism invariance this is **independent of the choice of chart**. Further

$$H_c^s(M) = \{u \in H_{loc}^s(M) : \text{supp}(u) \text{ compact}\}.$$

Remark 6.13. If M is compact then $H_{loc}^s = H_c^s =: H^s(M)$, and this can be equipped with a Hilbert space norm

$$\|u\|_{H^s(M)}^2 = \sum_{i \text{ (finite)}} \|(F_i)_*(\chi_i u)\|_{H^s(\mathbb{R}^n)}^2,$$

for a finite atlas $\{(U_i, F_i)\}$ and subordinate partition of unity $\{\chi_i\}$. Different choices give **equivalent** norms — *but there is no canonical choice!*

6.8 The pseudodifferential calculus

Definition 6.14 (Symbol spaces on vector bundles). Recall $S^m(\mathbb{R}^n; \mathbb{R}^N)$ and $S_c^m(\Omega; \mathbb{R}^N)$, where the subscript c means compact support in the **first** variable. Given a vector bundle $E \rightarrow M$ of rank N , define

$$S^m(E) \subset C^\infty(E) \quad (\text{functions on the total space}),$$

declared **locally**, via a chart F and a cutoff χ : $a \in S^m(E)$ iff $F_*(\chi a) \in S_c^m(\Omega; \mathbb{R}^N)$ for all charts and cutoffs. The twisted-pullback proposition guarantees that this is independent of the trivialization and chart.

The case of interest is $E = T^*M$, giving $S^m(T^*M)$.

Pseudodifferential operators on a manifold Let $A : C_c^\infty(M) \rightarrow C^\infty(M)$ be continuous linear.

Definition 6.15 (smoothing operators).

$$A \in \Psi^{-\infty}(M) \iff k_A \in C^\infty(M \times M, \Omega_R) \iff A \text{ extends continuously } \mathcal{E}' \rightarrow C^\infty(M).$$

Definition 6.16 (Ψ DOs). $A \in \Psi^m(M)$ iff — informally, “smooth off the diagonal” + “locally a Ψ_c^m ”:

- (1) k_A is **smooth on** $M \times M \setminus \Delta$; equivalently: for all $\phi, \psi \in C_c^\infty(M)$ with **disjoint supports**, $\phi A \psi \in \Psi^{-\infty}(M)$.
- (2) For every $x_0 \in M$ there exist a chart $F : M \supset U \xrightarrow{\sim} \Omega \subset \mathbb{R}^n$ and $\psi \in C_c^\infty(U)$ with $\psi(x_0) \neq 0$, and an operator $B \in \Psi_c^m(\Omega)$, such that the diagram

$$\begin{array}{ccc} C_c^\infty(U) & \xrightarrow{\psi A \psi} & C^\infty(U) \\ F_* \downarrow & & \downarrow F_* \\ C_c^\infty(\Omega) & \xrightarrow{B} & C^\infty(\Omega) \end{array}$$

Bookkeeping remark. For scalar operators $k_A \in \mathcal{D}'(M^2, \Omega_R \otimes \mathbb{R} \otimes \mathbb{R}^*) = \mathcal{D}'(M^2, \Omega_R)$, and the pairing $\mathcal{D}'(M^2, \Omega_R) \times C_c^\infty \rightarrow \mathcal{D}'(M)$ is the one from the Schwartz-kernel discussion above.

Remarks.

- (1) For $M = \mathbb{R}^n$: $\Psi_{\text{unif}}^m(\mathbb{R}^n) \subsetneq \Psi^m(\mathbb{R}^n)$, the right-hand side being the class just defined. The *uniform* operators are the ones considered so far; the new class is genuinely larger (no uniformity in x required).
- (2) By coordinate invariance, condition 2. then automatically holds **for all charts and all cutoffs**, not just for one.
- (3) Globalization: with $M = \bigcup_i U_i$ and a subordinate partition of unity one may always write

$$A = \sum_i A_i + R, \quad A_i \in \Psi_c^m(U_i), \quad R \in \Psi^{-\infty}(M).$$

This decomposition reduces most proofs on manifolds to the local $\mathbb{R}^n$ case.

Structural properties of Ψ DOs (*Proofs: exercise — they all follow from Remark 3 above plus the corresponding statements on $\mathbb{R}^n$.*)

Composition and adjoints Let $A \in \Psi^m(M)$, $B \in \Psi^{m'}(M)$, **one of them properly supported**. Then

$$A \circ B \in \Psi^{m+m'}(M).$$

The proper-support hypothesis is what makes $A \circ B$ defined at all, as explained above.

Fix a nonvanishing density μ . Then the formal $L^2(\mu)$ -adjoint satisfies

$$A^* \in \Psi^m(M), \quad \langle Au, v \rangle_{L^2(\mu)} = \langle u, A^*v \rangle_{L^2(\mu)}.$$

Definition 6.17 (Principal symbol on a manifold). The local principal symbols can be **glued together** to a morphism

$$\sigma^m : \Psi^m(M) \longrightarrow S^m(T^*M) / S^{m-1}(T^*M).$$

Why it glues: under a chart change $\kappa : \Omega \rightarrow \Omega'$ one has, modulo S^{m-1} ,

$$\sigma_\kappa(x, \xi) = \sigma(\kappa(x), D\kappa^{-T}(\xi)),$$

This is precisely the transformation law of a function on T^*M ; it is the twisted-pullback formula with $\gamma_x = (D\kappa(x)^T)^{-1}$. The subprincipal information is *not* invariant, which is why one quotients by S^{m-1} .

Properties.

$$\sigma^{m+m'}(A \circ B) = \sigma^m(A) \sigma^{m'}(B), \quad \sigma^m(A^*) = \overline{\sigma^m(A)}.$$

The symbol sequence

$$0 \longrightarrow \Psi^{m-1}(M) \longrightarrow \Psi^m(M) \xrightarrow{\sigma^m} S^m / S^{m-1}(T^*M) \longrightarrow 0.$$

Exactness on the right is *symbol quantization*: every symbol is realized by an operator.

Classical operators For Ψ_{cl}^m (symbols with asymptotic expansions into homogeneous terms) one gets versions of (I), (II), (III); see the subsection on classical operators below.

Commutators A, B , one properly supported. Then

$$\sigma^{m+m'-1}(i[A, B]) = \{\sigma^m(A), \sigma^{m'}(B)\},$$

the Poisson bracket on T^*M . (Note the order drop: the leading terms cancel.)

Asymptotic summation If $m_0 > m_1 > \dots \rightarrow -\infty$ and $A_k \in \Psi^{m_k}(M)$, then there exists $A \in \Psi^{m_0}(M)$ with

$$A \sim \sum_{k \geq 0} A_k, \quad \text{i.e.} \quad A - \sum_{k < K} A_k \in \Psi^{m_K}(M) \quad \forall K.$$

A is unique modulo $\Psi^{-\infty}(M)$. (Borel-type construction.)

Action on distributions $A \in \Psi^m(M)$ acts as $A : C_c^\infty(M) \rightarrow C^\infty(M)$, and extends to continuous linear maps

$$\mathcal{E}'(M) \longrightarrow \mathcal{D}'(M), \quad H_c^s(M) \longrightarrow H_{loc}^{s-m}(M).$$

If A is **properly supported**, one gets in addition

$$\mathcal{E}' \rightarrow \mathcal{E}', \quad \mathcal{D}' \rightarrow \mathcal{D}', \quad H_c^s \rightarrow H_c^{s-m}, \quad H_{loc}^s \rightarrow H_{loc}^{s-m}.$$

Classical operators and homogeneous principal symbols

Theorem 6.18. If $A \in \Psi_{cl}^m(M)$, then $\sigma^m(A)$ has a **unique homogeneous representative** $a_m \in S_{\text{hom}}^m(T^*M \setminus 0)$ (the "dominant part"), characterized by

$$e^{-\frac{i}{h}S} A(e^{\frac{i}{h}S} f)(x) = f(x) a_m(x, dS_x) h^{-m} + o(h^{-m}), \quad h \rightarrow 0,$$

for all $f \in C_c^\infty(M)$ and all $S \in C^\infty(M)$ with $dS_x \neq 0$ for $x \in \text{supp } f$.

Proof sketch (in a chart). Write $A = \text{Op}(a)$ and compute

$$e^{-iS/h} A(e^{iS/h} f)(x) = (2\pi)^{-n} \iint e^{i[(x-y)\xi + S(y) - S(x)]/h} a(x, \xi) f(y) dy d\xi.$$

Substituting $\xi = \eta/h$ gives

$$h^{-n} \iint e^{i[(x-y)\eta + S(y) - S(x)]/h} a(x, \eta/h) f(y) dy d\eta.$$

For stationary phase in (y, η) , the phase is critical exactly at $y = x$ and $\eta = dS_x$, with nondegenerate Hessian and signature 0. By homogeneity, the leading term is

$$a(x, dS_x/h) f(x) = h^{-m} a_m(x, dS_x) f(x) + o(h^{-m}).$$

□

Because a_m is *unique*, the sequence sharpens to

$$0 \longrightarrow \Psi_{cl}^{m-1} \longrightarrow \Psi_{cl}^m \longrightarrow S_{\text{hom}}^m(T^*M \setminus 0) \longrightarrow 0.$$

Example 6.19 (Principal symbol of a vector field). Let X be a vector field on M , acting on $C^\infty(M)$:

$$X(e^{iS/h}f)(x) = \frac{i}{h} XS \cdot e^{iS/h}f + e^{iS/h}Xf = \frac{i}{h}dS_x(X) e^{iS/h}f(x) + e^{iS/h}Xf,$$

hence

$$\sigma^1(X)(x, \xi) = i \xi_x(X_x), \quad (x, \xi) \in T^*M.$$

Example 6.20 (Principal symbol of the Laplace–Beltrami operator).

$$\sigma^2(\Delta_g)(x, \xi) = -|\xi|_g^2.$$

Example 6.21 (Vector fields are not elliptic). X a vector field on M , $X \in \text{Diff}^1(M) \subset \Psi_{cl}^1(M)$:

$$\sigma^1(X)(x, \xi) = i XS(x) = i dS_x(X) = i \langle \xi, X(x) \rangle, \quad \xi = dS_x.$$

Not elliptic: for $n \geq 2$ the symbol vanishes on the hyperplane $\xi \perp X(x)$ (and identically where $X(x) = 0$).

Example 6.22 (The Laplace–Beltrami operator). On (M, g) , locally

$$\Delta_g f = \frac{1}{\sqrt{|g|}} \partial_{x_i} \left(\sqrt{|g|} g^{ij}(x) \partial_{x_j} f \right),$$

$$\sigma^2(\Delta_g)(x, \xi) = -\xi_i g^{ij} \xi_j = -|\xi|_{g(x)}^2.$$

This never vanishes for $\xi \neq 0$: Δ_g **is elliptic**.

Example 6.23 (The Dirichlet-to-Neumann map). Let $\Omega \subset \mathbb{R}^n$ be a smooth domain, $M = \partial\Omega$, and g a Riemannian metric on $\bar{\Omega}$. For $h \in C^\infty(M)$ solve

$$\begin{cases} \Delta_g u = 0 & \text{in } \Omega, \\ u = h & \text{on } \partial\Omega. \end{cases}$$

Define the **Dirichlet-to-Neumann map**

$$\Lambda_g h = (\partial_\nu u)|_M \in C^\infty(M).$$

Then (Taylor, *PDE II*)

$$\Lambda_g \in \Psi^1(M), \quad \sigma^1(\Lambda_g)(x, \xi) = |\xi|_g.$$

In particular Λ_g is elliptic of order 1 — the starting point of Calderón’s inverse problem.

Chapter 7

Global Elliptic Theory

Let M be **closed** (compact without boundary), $A \in \Psi^m(M)$, $m \in \mathbb{R}$.

Definition 7.1.

A elliptic $:\iff \sigma^m(A) \in S^m/S^{m-1}(T^*M)$ is elliptic

i.e. some (hence any) representative a satisfies $|a(x, \xi)| \geq c|\xi|^m$ for $|\xi| \geq R$. For **classical** A this is equivalent to

$\sigma^m(A) \in S_{\text{hom}}^m(T^*M \setminus 0)$ nowhere vanishing.

(For operators between bundles: $\sigma^m(A)(x, \xi) : E_x \rightarrow F_x$ invertible for all $\xi \neq 0$.)

Example 7.2 (Lower-order perturbations of the Laplacian). On (M, g) , the operator $\Delta_g + X + q \in \text{Diff}^2(M)$ is elliptic: its principal symbol is that of Δ_g , so the lower-order terms do not matter.

Example 7.3 (Elliptic complexes). Let $E_k \rightarrow M$ be vector bundles with metrics h_k , g a Riemannian metric, and a complex of differential operators

$$\cdots \longrightarrow C^\infty(M; E_k) \xrightarrow{d_k} C^\infty(M; E_{k+1}) \longrightarrow \cdots, \quad d_k \circ d_{k-1} = 0.$$

The complex is **elliptic** iff, by definition, the Hodge Laplacians

$$\Delta_k = d_k^* d_k + d_{k-1} d_{k-1}^* \in \text{Diff}^2(M; E_k)$$

are elliptic. Since

$$\sigma^1(d_k)(x, \xi) : E_{k,x} \rightarrow E_{k+1,x},$$

this is equivalent to **exactness of the principal symbol sequence** for $\xi \neq 0$:

$$\cdots \longrightarrow E_{k-1,x} \xrightarrow{\sigma(d_{k-1})} E_{k,x} \xrightarrow{\sigma(d_k)} E_{k+1,x} \longrightarrow \cdots.$$

Indeed,

$$\begin{aligned} \sigma^2(\Delta_k) &= \sigma(d_k)^* \sigma(d_k) + \sigma(d_{k-1}) \sigma(d_{k-1})^*, \\ \sigma^2(\Delta_k) \text{ injective} &\iff \ker \sigma(d_k) = \text{ran } \sigma(d_{k-1}). \end{aligned}$$

The de Rham complex is the model case.

7.1 Parametrices and elliptic regularity

Basic observations for elliptic $A \in \Psi^m(M)$, M closed.

1) Parametrix. There exists $B \in \Psi^{-m}(M)$ with

$$I - A \circ B, \quad B \circ A - I \in \Psi^{-\infty}(M).$$

Construction. Locally, ellipticity gives $|a| \geq c|\xi|^m$ for $|\xi| \geq R$; set $b_0 = \chi(\xi)/a$ with $\chi = 0$ for $|\xi| \leq R$, $\chi = 1$ for $|\xi| \geq 2R$. Then $b_0 \in S^{-m}$ and, by the composition formula,

$$A \operatorname{Op}(b_0) = I - R_1, \quad R_1 \in \Psi^{-1}.$$

By asymptotic summation (VI), choose $R \sim \sum_{k \geq 0} R_1^k \in \Psi^0$; then $B := \operatorname{Op}(b_0)R$ satisfies $AB = I$ modulo $\Psi^{-\infty}$. A left parametrix is built symmetrically, and the two agree modulo $\Psi^{-\infty}$:

$$B_L = B_L(AB_R) = (B_L A)B_R = B_R \pmod{\Psi^{-\infty}}.$$

The partition-of-unity construction globalizes the argument.

2) Elliptic regularity.

$$\boxed{\text{if } u \in \mathcal{D}'(M) \text{ and } Au = f \in H^{s-m}(M) \implies u \in H^s(M).} \quad (7.1)$$

Proof. M compact $\implies$ every $u \in \mathcal{D}'(M)$ lies in some $H^{-N}(M)$ (distributions on a compact manifold have finite order). With $BA = I - R$, $R \in \Psi^{-\infty}$,

$$u = BAu + Ru = Bf + Ru.$$

Now $B: H^{s-m} \rightarrow H^s$ by the Sobolev mapping property, so $Bf \in H^s$; and R is smoothing, so $Ru \in C^\infty(M)$. Hence $u \in H^s(M)$. $\square$

Remark 7.4. The same argument applies on $\mathbb{R}^n$ and uses only the basic pseudodifferential-calculus properties established above.

Goal. Solve equations

$$Au = f,$$

using the regularity statement (7.1).

7.2 The Fredholm theorem

Theorem 7.5. Let M be closed and $A \in \Psi^m(M)$, $m \in \mathbb{R}$, *elliptic*.

(I) *Elliptic estimates.* For all $s \in \mathbb{R}$ and $N \in \mathbb{N}$ there is $C > 0$ with

$$\|u\|_s \leq C(\|Au\|_{s-m} + \underbrace{\|u\|_{-N}}_{\text{"easy" term}}).$$

(II) *Kernel.* $\ker(A) \subset C^\infty(M)$ and is finite dimensional. Moreover, if $V \subset C^\infty(M)$ satisfies

$$\overline{V}^{\|\cdot\|_s} \cap \ker(A) = \{0\},$$

then the "easy" term can be dropped on V :

$$\|u\|_s \leq C\|Au\|_{s-m} \quad \forall u \in V.$$

(III) *Cokernel.* Fix a nonvanishing density μ . Then for $f \in H^{s-m}(M)$:

$$f \in \operatorname{ran}(A|_{H^s}) \iff f \perp \ker(A^*) \text{ w.r.t. } L^2(\mu).$$

Corollary 7.6. *A is a Fredholm operator in all these settings:*

$$A : C^\infty(M) \rightarrow C^\infty(M), \quad A : H^s(M) \rightarrow H^{s-m}(M),$$

with $\ker$, coker finite dimensional and $\text{ran } A$ closed.

Corollary 7.7 (index).

$$\text{Ind}(A) = \dim \ker A - \dim \text{coker } A \quad \text{is independent of } s.$$

Proof of the index statement. By elliptic regularity (7.1),

$$\ker(A : H^s \rightarrow H^{s-m}) = \ker(A : C^\infty \rightarrow C^\infty)$$

for every s ; this is a fixed finite-dimensional space of smooth functions. By part (III) of theorem 7.5, $\text{ran}(A|_{H^s}) = \{f \in H^{s-m} : f \perp \ker A^*\}$, so

$$\text{coker}(A|_{H^s}) \cong \ker(A^*) \subset C^\infty(M),$$

again independent of s (note A^* is elliptic of order m , so its kernel is smooth and finite dimensional). $\square$

7.3 Proof of the Fredholm theorem

Sobolev facts used

- $(H^s(M), \|\cdot\|_s)$ is a **Hilbert space**.
- The seminorms $\{\|\cdot\|_s\}_{s \in \mathbb{R}}$ induce the usual Fréchet topology on $C^\infty(M)$, and

$$C^\infty(M) = \bigcap_{s \in \mathbb{R}} H^s(M).$$

- $H^{s+\varepsilon}(M) \hookrightarrow H^s(M)$ is **compact** ($\varepsilon > 0$; Rellich).
- $|\langle u, v \rangle_{L^2_\mu}| \leq C \|u\|_{-s} \|v\|_s$, i.e. $(H^s)' \cong H^{-s}$ via the $L^2(\mu)$ -pairing.

Elliptic estimates

Proof. Let B be a parametrix, $BA = I - R$ with $R \in \Psi^{-\infty}$. For $u \in C^\infty(M)$,

$$u = BAu + Ru.$$

Hence

$$\|u\|_s \leq \|BAu\|_s + \|Ru\|_s.$$

Now $B : H^{s-m} \rightarrow H^s$ and $R : H^{-N} \rightarrow H^s$ are bounded, so

$$\|u\|_s \lesssim \|Au\|_{s-m} + \|u\|_{-N}.$$

$\square$

(The estimate extends to all $u \in H^s$ by density.)

Finite-dimensional kernel

Proof. First, $\ker A \subset C^\infty(M)$: if $Au = 0$, then $Au \in H^t$ for every t , so (7.1) gives $u \in H^t$ for every t , and therefore $u \in C^\infty(M)$.

Now let $(u_n) \subset \ker(A)$ with $\|u_n\|_s \leq C$.

- By Banach–Alaoglu and reflexivity of the Hilbert space H^s , a subsequence converges weakly: $u_n \rightharpoonup u_* \in H^s$.
- Since $H^s \hookrightarrow H^{-N}$ is **compact**, weak convergence in H^s implies

$$\|u_n - u_*\|_{-N} \rightarrow 0.$$

- $A(u_n - u_*) = 0$ (weak limits of solutions solve the equation: A is continuous $H^s \rightarrow H^{s-m}$, hence weakly continuous).
- By (I),

$$\|u_n - u_*\|_s \leq C \underbrace{(\|A(u_n - u_*)\|_{s-m})}_{=0} + \|u_n - u_*\|_{-N} \rightarrow 0.$$

So every bounded sequence in $\ker A$ has a norm-convergent subsequence: the closed unit ball of $\ker A$ (a closed subspace of H^s) is compact, hence by **Riesz' lemma**

$$\dim \ker A < \infty.$$

Second part of (II). Let $V \subset C^\infty(M)$ with $\overline{V}^{\|\cdot\|_s} \cap \ker A = \{0\}$. Suppose the estimate $\|u\|_s \leq C\|Au\|_{s-m}$ fails. Then there is a sequence $(u_n) \subset V$ with

$$\|u_n\|_s = 1, \quad \|Au_n\|_{s-m} \rightarrow 0.$$

As above, after a subsequence

$$u_n \rightharpoonup u_* \text{ in } H^s, \quad u_n \rightarrow u_* \text{ in } H^{-N}, \quad Au_* = 0.$$

Apply (I) to u_n :

$$1 = \|u_n\|_s \leq C(\|Au_n\|_{s-m} + \|u_n\|_{-N}) \xrightarrow{n \rightarrow \infty} C\|u_*\|_{-N},$$

so $1 \leq C\|u_*\|_{-N}$, in particular $u_* \neq 0$. But $u_* \in \overline{V}^{\|\cdot\|_s}$ and $u_* \in \ker A$ — **contradiction** with $\overline{V} \cap \ker A = \{0\}$. $\square$

The cokernel

Proof. ($\Rightarrow$) If $f = Au$ and $\phi \in \ker A^*$ then $\langle f, \phi \rangle = \langle Au, \phi \rangle = \langle u, A^*\phi \rangle = 0$.

($\Leftarrow$) Assume $f \in H^{s-m}(M)$ with $f \perp \ker A^*$. Note A^* is **elliptic** of order m , so part (II) applies to it. Put

$$V = \ker(A^*)^\perp \cap C^\infty(M) \quad (\perp \text{ w.r.t. } L^2(\mu)).$$

Since $\ker A^* \subset C^\infty(M)$ is finite dimensional, the orthogonality conditions $\langle \cdot, \psi_j \rangle = 0$ (for a basis ψ_j of $\ker A^*$) are continuous on H^{m-s} ; hence

$$\overline{V}^{\|\cdot\|_{m-s}} \cap \ker A^* = \{0\},$$

and (II) applied to A^* (which maps $H^{m-s} \rightarrow H^{-s}$) gives

$$\|\phi\|_{-s+m} \leq C\|A^*\phi\|_{-s}, \quad \phi \in V. \quad (7.2)$$

Define

$$D = \{A^*\phi : \phi \in V\} \subset C^\infty(M), \quad T : D \rightarrow \mathbb{C}, \quad T(A^*\phi) = \langle \phi, f \rangle.$$

- **T is well defined:** if $A^*\phi_1 = A^*\phi_2$ with $\phi_i \in V$ then $\phi_1 - \phi_2 \in \ker A^* \cap V = \{0\}$; alternatively, $\langle \phi_1 - \phi_2, f \rangle = 0$ **because** $f \perp \ker A^*$.
- **T is bounded for $\|\cdot\|_{-s}$:** for $\psi = A^*\phi \in D$, (7.2) gives

$$|T(\psi)| \leq |\langle \phi, f \rangle| \lesssim \|\phi\|_{-s+m} \|f\|_{s-m} \lesssim \|A^*\phi\|_{-s} \|f\|_{s-m} = \|\psi\|_{-s} \|f\|_{s-m}.$$

By **Hahn–Banach**, T extends to a continuous functional $\tilde{T}$ on $H^{-s}(M) \supset D$ (equivalently, on $C^\infty(M)$ with the $\|\cdot\|_{-s}$ norm). Since $(H^{-s})' \cong H^s$ via the $L^2(\mu)$ -pairing, there is

$$u \in H^s(M) \quad \text{with} \quad \tilde{T}(\psi) = \langle \psi, u \rangle \quad \forall \psi \in H^{-s}.$$

Then for $\phi \in V$:

$$\langle A^*\phi, u \rangle = \langle \phi, f \rangle.$$

For $\phi \in \ker A^*$ both sides vanish (left: $A^*\phi = 0$; right: $f \perp \ker A^*$). Since $C^\infty(M) = V \oplus \ker A^*$, the identity

$$\langle A^*\phi, u \rangle = \langle \phi, Au \rangle = \langle \phi, f \rangle$$

holds for all $\phi \in C^\infty(M)$. Therefore

$$Au = f \quad \text{in } \mathcal{D}'(M),$$

and (7.1) gives $u \in H^s(M)$. □

Fredholmness

Proof. • $\dim \ker A < \infty$ by (II).

- $\text{ran}(A|_{H^s}) = \{f \in H^{s-m} : \langle f, \psi_j \rangle = 0, j = 1, \dots, k\}$ by (III), which is an intersection of finitely many kernels of continuous functionals, hence **closed**, of finite codimension $k = \dim \ker A^* < \infty$.
- The C^∞ statement follows since $\ker$ and coker consist of smooth objects. □

Example 7.8 (The Laplacian on a closed manifold). Let (M, g) be closed and consider $\Delta_g u = f$. Then

$$\ker \Delta_g \equiv \{\text{constant functions}\}, \quad \Delta_g^* = \Delta_g \text{ w.r.t. the Riemannian volume.}$$

Hence by (III):

$$\Delta_g u = f \text{ is solvable} \iff \int_M f dV_g = 0.$$

Moreover the index is $\text{Ind}(\Delta_g) = 1 - 1 = 0$.

Chapter 8

Wave Front Sets and Microlocalization

8.1 The composition formula is local in (x, ξ)

Let $A, B \in \Psi^*(\mathbb{R}^n)$, $A = \text{Op}_L(a)$, $B = \text{Op}_L(b)$ (left quantization). Then

$$\sigma_L(A \circ B) \sim \sum_{\alpha} \frac{1}{\alpha!} D_{\xi}^{\alpha} a \partial_x^{\alpha} b = \sum_{\alpha} \frac{1}{\alpha!} \partial_{\xi}^{\alpha} a D_x^{\alpha} b, \quad D = -i\partial.$$

Key observation: the right-hand side at a point (x, ξ) only involves a and b (and their derivatives) **at that same point** — the expansion is *local in (x, ξ)* , not merely in x . Everything not seen by both symbols disappears. This is the entire content of microlocalization.

Definition 8.1 (informal).

$$\text{WF}'(A) \text{ ''} = \text{''} \{ (x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n \setminus \{0\} : A \text{ is non-trivial at } (x, \xi) \}.$$

Upshot.

$$\text{WF}'(AB) \subset \text{WF}'(A) \cap \text{WF}'(B).$$

We now make "non-trivial at (x, ξ) " precise.

8.2 Essential support of a symbol

Definition 8.2 (essential support). Let $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$. Its essential support is a subset of

$$\mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})$$

defined by

$$(x_0, \xi_0) \notin \text{ess supp}(a) \iff \exists \varepsilon > 0 \text{ s.t. } \forall \alpha \in \mathbb{N}^n, \beta \in \mathbb{N}^N, \forall M :$$

$$|\partial_x^{\alpha} \partial_{\xi}^{\beta} a(x, \xi)| \leq C_{\alpha\beta M} \langle \xi \rangle^{-M} \quad \text{whenever} \quad |x - x_0| + \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \varepsilon.$$

In words: a is *rapidly decaying* (of order $-\infty$) on a **conic neighborhood** of (x_0, ξ_0) — a neighborhood in x times a cone of directions in ξ .

Figure 8.1 illustrates the conic geometry of the definition.

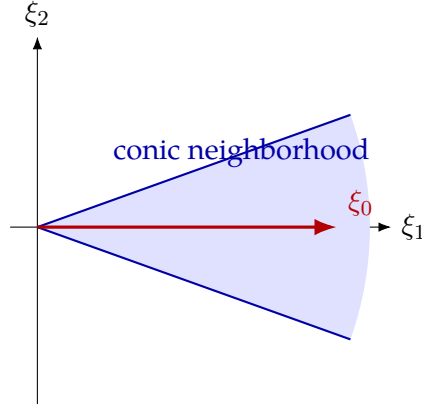


Figure 8.1: A conic neighborhood depends on the direction of ξ , not its length.

Remark 8.3. $\text{ess supp}(a)$ is **closed and conical**. Recall

$$A \subset \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\}) \text{ is } \mathbf{conical} \iff \begin{cases} \forall (x, \xi) \in A, \forall t > 0 : \\ (x, t\xi) \in A. \end{cases}$$

Closedness is immediate from the definition (the complement is manifestly open and conic).

8.3 Microsupport / operator wave front set

Definition 8.4 (microsupport, operator wave front set). For $A \in \Psi^m(\mathbb{R}^n)$,

$$\boxed{\text{WF}'(A) = \text{ess supp}(\sigma_L(A)).}$$

This is well defined: two symbols giving the same operator modulo $\Psi^{-\infty}$ differ by a symbol of order $-\infty$, whose essential support is empty; and the definition is independent of the quantization (left/right/Weyl), since the transition formulas are again local in phase space.

Basic properties (all consequences of the locality of the composition formula):

- $\text{WF}'(A) = \emptyset \iff A \in \Psi^{-\infty}$.
- $\text{WF}'(A + B) \subset \text{WF}'(A) \cup \text{WF}'(B)$.
- $\text{WF}'(A \circ B) \subset \text{WF}'(A) \cap \text{WF}'(B)$ — the composition kills everything outside the *intersection*, which is the microlocal analogue of $\text{supp}(fg) \subset \text{supp } f \cap \text{supp } g$.
- $\text{WF}'(A^*) = \text{WF}'(A)$.

On a manifold, the same definition works chart by chart and yields a closed conic subset

$$\text{WF}'(A) \subset T^*M \setminus 0,$$

by exactly the coordinate-invariance of the principal symbol established above.

Notation index

Symbol	Meaning
$\Omega^\alpha M$	α -density bundle; $\Omega_R = \pi_M^* \Omega M$, $\Omega_L = \pi_N^* \Omega N$
$\mathcal{D}'(M; E)$	$\mathcal{L}(C_c^\infty(M, \Omega^1 M \otimes E^*), \mathbb{C})$
$\mathcal{E}'(M; E)$	$\mathcal{L}(C^\infty(M, \Omega^1 M \otimes E^*), \mathbb{C})$
H_{loc}^s, H_c^s	Sobolev spaces defined chartwise; equal on compact M
$S^m(E)$	symbols on the total space of E , defined locally via $S_c^m(\Omega; \mathbb{R}^N)$
$\Psi^{-\infty}(M)$	smoothing operators ($k_A \in C^\infty(M^2, \Omega_R)$)
$\Psi^m(M)$	k_A smooth off Δ + locally a Ψ_c^m in charts
σ^m	principal symbol $\Psi^m \rightarrow S^m/S^{m-1}(T^*M)$
a_m	homogeneous principal symbol, $A \in \Psi_{cl}^m$
ess supp	conic set where a symbol fails to be $O(\langle \xi \rangle^{-\infty})$
$\text{WF}'(A)$	$\text{ess supp}(\sigma_L(A)) \subset T^*M \setminus 0$

Notation. Throughout, M is a smooth manifold and

$$\Psi^{-\infty}(M) = \bigcap_m \Psi^m(M).$$

We write σ_L and σ_R for the full left and right symbols, σ^m for the principal symbol, and K_A for the Schwartz kernel of A . Moreover,

$$S^*M = (T^*M \setminus 0)/\mathbb{R}_{>0} \xrightarrow{\pi} M$$

the cosphere bundle. Closed conic subsets of $T^*M \setminus 0$ are identified with closed subsets of S^*M . We write $\langle \xi \rangle = (1 + |\xi|^2)^{1/2}$.

8.4 Operator microlocalization

The motivating computation Let $A, B \in \Psi^*(\mathbb{R}^n)$ with $A = \text{Op}_L(a)$, $B = \text{Op}_L(b)$. The composition formula gives the asymptotic expansion

$$\sigma_L(A \circ B) \sim \sum_{\alpha} \frac{1}{\alpha!} D_{\xi}^{\alpha} a \partial_x^{\alpha} b = \sum_{\alpha} \frac{1}{\alpha!} \partial_{\xi}^{\alpha} a D_x^{\alpha} b.$$

The key observation: every term on the right-hand side is *local in* (x, ξ) — it is a product of derivatives of a and b evaluated at the *same* point of phase space. So if a is negligible near (x_0, ξ_0) , the whole expansion is negligible near (x_0, ξ_0) , and likewise for b .

This suggests a notion of “the phase-space region where A genuinely lives”:

Definition 8.5 (Provisional definition).

$$\text{WF}'(A) = \{(x, \xi) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus 0) : A \text{ is non-trivial at } (x, \xi)\}.$$

Upshot (to be proved):

$$\boxed{\text{WF}'(AB) \subseteq \text{WF}'(A) \cap \text{WF}'(B).}$$

To make this precise we need the essential support of a symbol.

Essential support

Definition 8.6 (Essential support).

Let $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$. Its essential support $\text{ess supp}(a) \subset \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})$ is defined by:

$$\begin{aligned} & \exists \varepsilon > 0 \text{ such that for all } \alpha \in \mathbb{N}^n, \beta \in \mathbb{N}^N, M > 0, \\ (x_0, \xi_0) \notin \text{ess supp}(a) & \iff \exists C > 0 : \quad |\partial_x^\alpha \partial_\xi^\beta a(x, \xi)| \leq C \langle \xi \rangle^{-M}, \\ & \text{whenever } |x - x_0| + \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \varepsilon. \end{aligned}$$

In words: a is a symbol of order $-\infty$ in a conic neighborhood of the direction ξ_0 , over a neighborhood of x_0 .

Remark 8.7. $\text{ess supp}(a)$ is **closed** and **conic**. Recall:

$$A \subset \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\}) \text{ is conic} \iff \forall (x, \xi) \in A, \forall t > 0 : (x, t\xi) \in A.$$

Because the condition only constrains ξ through the *direction* $\xi/|\xi|$ (and requires decay as $|\xi| \rightarrow \infty$), the excluded region is a full open cone:

Consequently $\text{ess supp}(a)$ descends to a closed subset of $\mathbb{R}^n \times S^{N-1}$.

Figure 8.2 illustrates the conic neighborhoods used to test essential support.

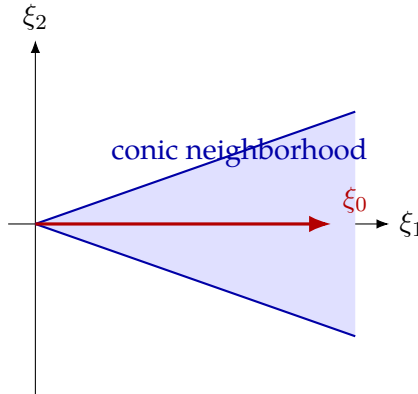


Figure 8.2: Essential support is tested in conic neighborhoods of frequency directions.

The operator wave front set

Definition 8.8 (microsupport / operator wave front set). For $A \in \Psi^m(\mathbb{R}^n)$,

$$\text{WF}'(A) := \text{ess supp}(\sigma_L(A)).$$

Properties of the operator wave front set

Proposition 8.9 (Operator wave front set calculus). Let $A, B \in \Psi^*(\mathbb{R}^n)$.

(i) If $\text{supp}(K_A) \subset \mathbb{R}^n \times \mathbb{R}^n$ is compact, then

$$\boxed{\text{WF}'(A) = \emptyset \iff A \in \Psi^{-\infty}(\mathbb{R}^n).}$$

(ii) $\text{WF}'(A) = \text{ess supp}(\sigma_R(A))$ — i.e. the definition does not depend on the quantization.

(iii) $\text{WF}'(AB) \subseteq \text{WF}'(A) \cap \text{WF}'(B)$.

- (iv) $\text{WF}'(A + B) \subseteq \text{WF}'(A) \cup \text{WF}'(B)$.
(v) $\text{WF}'(A^*) = \text{WF}'(A)$.
(vi) (Invariance) If $\kappa : \Omega \xrightarrow{\sim} \Omega'$ is a diffeomorphism, with the induced canonical transformation

$$\kappa_{\#}(x, \xi) = (\kappa(x), D\kappa(x)^{-T}\xi),$$

then $\kappa_{\#}^{-1}(\text{WF}'(A)) = \text{WF}'(A_{\kappa})$, where A_{κ} is the pullback of A .

Proof sketch. All of these follow by inspecting the asymptotic expansions.

- For (iii): by the motivating computation above, each term $\frac{1}{\alpha!} D_{\xi}^{\alpha} a \partial_x^{\alpha} b$ vanishes to infinite order in a conic neighborhood of any point outside $\text{ess supp}(a)$, and likewise outside $\text{ess supp}(b)$; hence so does the (Borel-)sum, up to $S^{-\infty}$.
- For (ii) and (vi): the change-of-quantization and change-of-coordinates formulas are again asymptotic sums of terms local in (x, ξ) ; the leading term is a (resp. $a \circ \kappa_{\#}$) and all corrections are supported in the same conic set.
- For (v): $\sigma_L(A^*) \sim \sum \frac{1}{\alpha!} \partial_{\xi}^{\alpha} D_x^{\alpha} \bar{a}$, again local.
- For (i) ($\Leftarrow$ is trivial): if $\text{ess supp}(a) = \emptyset$ then every point of $\mathbb{R}^n \times S^{n-1}$ has a conic neighborhood on which a is $O(\langle \xi \rangle^{-\infty})$; compactness of $\text{supp } K_A$ in x lets us pass from local uniformity to global uniformity by a finite subcover, giving $a \in S^{-\infty}$.

□

Definition 8.10. For M a manifold and $A \in \Psi^*(M)$, the set $\text{WF}'(A) \subset T^*M \setminus 0$ is defined via local charts; this is well defined by (vi).

Figure 8.3 shows the induced transformation on the cotangent bundle.

$$\begin{array}{ccc} \boxed{\Omega} & \xrightarrow{\kappa} & \boxed{\Omega'} \\ & \xrightarrow{\kappa_{\#}} & \\ \kappa_{\#}(x, \xi) = (\kappa(x), D\kappa(x)^{-T}\xi) & & \end{array}$$

Figure 8.3: The canonical transformation governing coordinate invariance of microsupport.

Remark 8.11. Coordinate invariance.

- (iii)–(vi) hold verbatim on M .
- (i) holds on M **without** the support restriction (properness is built into the manifold conventions).
- $\text{WF}'(A) \subset T^*M \setminus 0$ is closed and conic, hence may be viewed as a **closed subset of S^*M** .

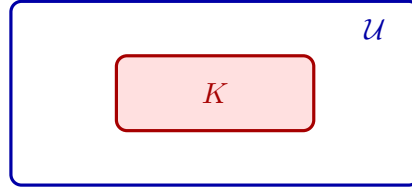
8.5 Microlocal cutoffs and ellipticity

Microlocal partition of unity

Lemma 8.12 (Microlocal partition of unity). *Suppose $S^*M = \bigcup_i \mathcal{U}_i$ with each $\mathcal{U}_i$ open. Then there exist $A_i \in \Psi^0(M)$ such that*

- (1) $\{\text{supp } K_{A_i}\}$ is locally finite;
- (2) $\sum_i A_i = \text{Id}$;
- (3) $\text{WF}'(A_i) \subset \mathcal{U}_i$.

Figure 8.4 depicts the support and elliptic region of a microlocalizer.



$A \equiv I$ on K , while $\text{WF}'(A) \subset \mathcal{U}$

Figure 8.4: A microlocalizer supported in a prescribed open neighborhood.

Proof. After replacing the cover by a locally finite refinement, we may assume

$$\mathcal{U}_i \subset S_{V_i}^* M = \pi^{-1}(V_i),$$

where $V_i \subset M$ is a chart domain.

Step 1: a first, approximate partition. Choose $\chi_i \in C^\infty(\mathcal{U}_i)$ with

$$\sum_i \chi_i \equiv 1, \quad \text{supp}(\chi_i) \subset \mathcal{U}_i,$$

i.e. an ordinary partition of unity on the manifold S^*M . Choose also $\gamma_i \in C_c^\infty(V_i)$ with $\gamma_i \equiv 1$ near $\pi(\text{supp } \chi_i)$.

Working in the chart, extend χ_i to a homogeneous degree-0 symbol

$$\tilde{\chi}_i(x, \xi) = \chi_i\left(x, \frac{\xi}{|\xi|}\right) \quad (|\xi| \geq 1),$$

smoothly cut off near $\xi = 0$; then $\tilde{\chi}_i \in S^0$ and $\text{ess sup}(\tilde{\chi}_i) = \text{supp } \chi_i$. Set

$$A'_i = \gamma_i \text{Op}(\tilde{\chi}_i) \gamma_i \in \Psi^0(M),$$

which is defined via the chart but makes global sense because of the cutoffs γ_i .

Then property (1) holds (the kernels are supported in $\text{supp } \gamma_i \times \text{supp } \gamma_i$, and $\{V_i\}$ is locally finite), and

$$(3) \text{ } \text{WF}'(A'_i) \subset \mathcal{U}_i.$$

Why: multiplying by γ_i on the left does not enlarge the essential support. Multiplication on the right produces the expansion

$$\sum_{\alpha} \frac{1}{\alpha!} \partial_{\xi}^{\alpha} (\gamma_i \tilde{\chi}_i) D_x^{\alpha} \gamma_i.$$

Every term is supported in $\text{ess sup}(\tilde{\chi}_i)$. Hence

$$\text{WF}'(A'_i) \subseteq \text{ess sup}(\tilde{\chi}_i) = \text{supp } \chi_i \subset \mathcal{U}_i.$$

This construction works because χ_i depends on both x and the direction $\xi/|\xi|$; its support may therefore be an arbitrary closed subset of $\mathcal{U}_i$, rather than only a product set.

Step 2: correcting to an exact partition. By construction the principal symbols add up correctly:

$$\sigma^0\left(\sum_i A'_i\right) = \sum_i \sigma^0(A'_i) = \sum_i \chi_i \equiv 1.$$

Hence

$$\sum_i A'_i = I - Q, \quad Q \in \Psi^{-1}(M).$$

Since Q has negative order, the Neumann series can be summed asymptotically: there exists $B \in \Psi^0(M)$ with $B \sim \sum_{k \geq 0} Q^k$, and then

$$\left(\sum_i A'_i\right)B = (I - Q)B = I - R, \quad R \in \Psi^{-\infty}(M).$$

Step 3: absorb the smoothing error. Fix an index 0 and set

$$A_0 := A'_0 B + R, \quad A_i := A'_i B \quad (i \neq 0).$$

Then $\sum_i A_i = I$, which is (2). For (3):

$$\text{WF}'(A'_0 B + R) \subseteq \text{WF}'(A'_0 B) \cup \underbrace{\text{WF}'(R)}_{=\emptyset} \subseteq \text{WF}'(A'_0) \cap \text{WF}'(B) \subseteq \text{WF}'(A'_0) \subset \mathcal{U}_0,$$

using parts (iii) and (iv) of the operator wave front set calculus and $R \in \Psi^{-\infty}$. Similarly $\text{WF}'(A_i) \subset \mathcal{U}_i$ for all i . Property (1) is preserved after arranging B to be properly supported. $\square$

Microlocalizers

Lemma 8.13 (Microlocalizers). *Let $K \subset \mathcal{U} \subset S^*M$ with K closed and $\mathcal{U}$ open. Then there exists $A \in \Psi^0(M)$ such that*

$$\text{WF}'(A - I) \cap K = \emptyset \quad \text{and} \quad \text{WF}'(A) \subset \mathcal{U}.$$

The first condition is read as: " $A \equiv 1$ microlocally on K ". It is the exact analogue of the scalar statement

$$\chi \equiv 1 \text{ on } K \iff \text{supp}(\chi - 1) \cap K = \emptyset.$$

Proof. The two sets $\mathcal{U}$ and $V := K^c$ form an open cover of S^*M (indeed $K \subset \mathcal{U}$ means $\mathcal{U} \cup K^c = S^*M$). Apply the microlocal partition-of-unity lemma to this cover: we get

$$A + B = \text{Id}, \quad \text{WF}'(A) \subset \mathcal{U}, \quad \text{WF}'(B) \subset V.$$

Then $A - I = -B$, so by the operator wave front set calculus

$$\text{WF}'(A - I) = \text{WF}'(B) \subset V = K^c,$$

i.e. $\text{WF}'(A - I) \cap K = \emptyset$. $\square$

Elliptic and characteristic sets Write $A = \text{Op}_L(a)$.

Definition 8.14 (Elliptic and characteristic sets). For $A \in \Psi^m(\mathbb{R}^n)$, the **elliptic set** $\text{Ell}(A) \subset \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$ is defined by

$$(x_0, \xi_0) \in \text{Ell}(A) \iff \exists C, c, \varepsilon > 0 \text{ s.t. } |a(x, \xi)| \geq c \langle \xi \rangle^m$$

whenever

$$|\xi| \geq C, \quad |x - x_0| + \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \varepsilon.$$

On a manifold this is transported through charts (it is invariantly defined). The **characteristic set** is the complement:

$$\text{Char}(A) = (T^*M \setminus \{0\}) \setminus \text{Ell}(A).$$

Both are closed conic, hence subsets of S^*M ; $\text{Ell}(A)$ is open and $\text{Char}(A)$ is closed.

Example 8.15 (Classical operators). If $A \in \Psi_{cl}^m(M)$ then $\sigma^m(A) \in S_{\text{hom}}^m(T^*M \setminus 0) [\simeq C^\infty(S^*M)]$ and

$$\text{Ell}(A) = \{(x, \xi) \in T^*M \setminus 0 : \sigma^m(A)(x, \xi) \neq 0\}.$$

Example 8.16 (The wave operator). On $\mathbb{R}_t \times \mathbb{R}_x^n$ let $\square = \partial_t^2 - \Delta$. Then

$$\sigma^2(\square)(t, x, \tau, \xi) = -\tau^2 + |\xi|^2 \implies \text{Char}(\square) = \{\tau^2 = |\xi|^2\},$$

the **light cone**:

Figure 8.5 shows the characteristic cone in a two-dimensional slice.

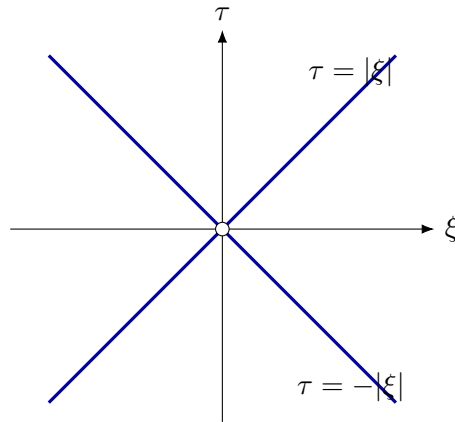


Figure 8.5: The characteristic set of the wave operator is the punctured light cone.

The microlocal elliptic parametrix

Proposition 8.17 (Microlocal elliptic parametrix). Let $A \in \Psi^m(M)$ and let $K \subset \text{Ell}(A)$ be closed, $K \subset S^*M$. Then there exists $B \in \Psi^{-m}(M)$ such that

$$\boxed{AB - I \text{ satisfies } \text{WF}'(AB - I) \cap K = \emptyset} \quad \text{and} \quad \text{WF}'(BA - I) \cap K = \emptyset.$$

Proof. **Step 1: nested microlocalizers.** By the microlocalizer lemma choose $Q, \tilde{Q} \in \Psi^0(M)$ with

$$\text{WF}'(Q - I) \cap K = \emptyset, \quad \text{WF}'(\tilde{Q} - I) \cap \text{WF}'(Q) = \emptyset, \quad \text{WF}'(Q), \text{WF}'(\tilde{Q}) \subset \text{Ell}(A).$$

That is: $Q \equiv 1$ microlocally on K , and $\tilde{Q} \equiv 1$ microlocally on $\text{WF}'(Q)$; both are microsupported inside the elliptic set.

Step 2: invert the symbol locally. On $\text{Ell}(A)$ the principal symbol $\sigma^m(A)$ is invertible with the right bounds, so we may define a symbol

$$\sigma^{-m}(B') = \frac{\sigma^0(\tilde{Q})}{\sigma^m(A)}$$

(the quotient is well defined because $\tilde{Q}$ is microsupported where A is elliptic), and choose $B' \in \Psi^{-m}(M)$ with this principal symbol. Changing the representative of $\sigma^m(A)$ by an element of S^{m-1} changes the construction only by a lower-order term.

Step 3: first-order error. By the symbol calculus,

$$AB' = I - R, \quad R \in \Psi^0(M), \quad \sigma^0(R) = 1 - \sigma^0(\tilde{Q}).$$

Decompose the error using Q :

$$AB' = I - QR - (I - Q)R.$$

The first error term is **one order better**:

$$\sigma^0(QR) = \sigma^0(Q) \underbrace{\sigma^0(I - AB')}_{=1-\sigma^0(\tilde{Q})} = 0,$$

because $\tilde{Q} \equiv 1$ microlocally on $\text{WF}'(Q)$. Hence $QR \in \Psi^{-1}(M)$.

Step 4: sum the Neumann series. Since QR has negative order, choose $T \in \Psi^0(M)$ with

$$T \sim \sum_{k \geq 0} (QR)^k, \quad \text{so} \quad (I - QR)T = I - \tilde{R}, \quad \tilde{R} \in \Psi^{-\infty}.$$

Then, writing $R = QR + (I - Q)R$,

$$AB'T = (I - R)T = \underbrace{(I - QR)T}_{=I-\tilde{R}} - (I - Q)RT = I - \tilde{R} - (I - Q)RT.$$

Step 5: conclude. Put $B := B'T \in \Psi^{-m}(M)$. Then

$$\text{WF}'(AB - I) = \text{WF}'(-\tilde{R} - (I - Q)RT) \subseteq \text{WF}'((I - Q)RT) \subseteq \text{WF}'(I - Q),$$

which is disjoint from K by the choice of Q . This gives the right parametrix.

A left parametrix is constructed symmetrically (or obtained from the right one by the usual uniqueness-of-parametrix argument: if $B_\ell A \equiv I$ and $AB_r \equiv I$ microlocally on K , then $B_\ell \equiv B_\ell AB_r \equiv B_r$ microlocally on K). $\square$

Summary of the toolkit For $A \in \Psi^m(M)$ we have the three phase-space sets $\text{WF}'(A)$, $\text{Ell}(A)$, $\text{Char}(A)$, and two construction principles:

(a) Microlocal cutoffs. Given $K \subset \mathcal{U} \subset S^*M$ (K closed, $\mathcal{U}$ open):

$$\exists A \in \Psi^0(M) : \quad \text{WF}'(\text{Id} - A) \cap K = \emptyset, \quad \text{WF}'(A) \subset \mathcal{U}.$$

(b) **Microlocal elliptic parametrices.** Given $A \in \Psi^m(M)$ and $K \subset \text{Ell}(A)$ closed:

$$\exists B \in \Psi^{-m}(M) : \quad \text{WF}'(I - BA) \cap K = \emptyset.$$

8.6 The wave front set of distributions

Let M be a manifold.

Definition 8.18 (Wave front set). Let $u \in \mathcal{D}'(M)$ and $\alpha \in S^*M$. Then

$$\alpha \notin \text{WF}(u) \iff \exists \mathcal{U} \ni \alpha \text{ open s.t. } \forall A \in \Psi^0(M) \text{ with } \text{WF}'(A) \subset \mathcal{U}, \quad Au \text{ is smooth.}$$

Lemma 8.19 (One elliptic operator suffices).

$$\alpha \notin \text{WF}(u) \iff \exists A \in \Psi^0(M) \text{ elliptic at } \alpha \text{ with } Au \in C^\infty.$$

Proof. " $\Rightarrow$ ": Let $\mathcal{U}$ be as in the definition above. By the microlocalizer lemma (with $K = \{\alpha\}$) choose A with $\text{WF}'(A) \subset \mathcal{U}$ and $A \equiv 1$ microlocally near α ; such an A has $\sigma^0(A) = 1$ near α , hence is elliptic at α , and $Au \in C^\infty$ by assumption.

" $\Leftarrow$ ": Assume A is elliptic at α and $Au \in C^\infty$. Put

$$\mathcal{U} := \text{Ell}(A) \ni \alpha \quad (\text{open}).$$

Let $B \in \Psi^0(M)$ be arbitrary with $\text{WF}'(B) \subset \mathcal{U}$.

Goal: Bu is also smooth.

Set $K := \text{WF}'(B)$, a closed subset of $\text{Ell}(A)$. By the microlocal elliptic-parametrix proposition there is a microlocal (left) parametrix $Q \in \Psi^{-m}$ for A over K :

$$\underbrace{\text{WF}'(I - QA)}_{=:R} \cap K = \emptyset.$$

Therefore

$$Bu = B(R + QA)u = \underbrace{BQAu}_{\text{smooth}} + \underbrace{BRu}_{\text{smooth}}.$$

- $BQAu$ is smooth because $Au \in C^\infty$ and BQ is a (properly supported) pseudodifferential operator, hence maps $C^\infty \rightarrow C^\infty$.
- $\text{WF}'(BR) \subseteq \text{WF}'(B) \cap \text{WF}'(R) = K \cap \text{WF}'(R) = \emptyset$, so $BR \in \Psi^{-\infty}$ is smoothing and $BRu \in C^\infty$.

Hence $Bu \in C^\infty$ for every such B , i.e. $\alpha \notin \text{WF}(u)$. □

Remark 8.20. The non-compact case. If M is compact the above is fine as written. If M is non-compact, choose B and Q **properly supported** so that all compositions and mapping properties are legitimate. This is always possible by:

Lemma 8.21. Every $A \in \Psi^m(M)$ decomposes as

$$A = A_0 + A_1, \quad A_0 \text{ properly supported, } A_1 \in \Psi^{-\infty}.$$

(Cut the kernel with a proper cutoff equal to 1 near the diagonal.)

Corollary 8.22.

$$\text{WF}(u) = \bigcap_{\substack{A \in \Psi^0(M) \\ Au \in C^\infty(M)}} \text{Char}(A).$$

Proof. If $Au \in C^\infty$ then, by the one-operator lemma, $\text{WF}(u) \cap \text{Ell}(A) = \emptyset$, i.e. $\text{WF}(u) \subset \text{Char}(A)$; so $\text{WF}(u)$ is contained in the intersection. Conversely if $\alpha \notin \text{WF}(u)$, the same lemma provides A elliptic at α with Au smooth, and then $\alpha \notin \text{Char}(A)$, so α is not in the intersection. $\square$

Fourier characterization and first properties

Lemma 8.23 (Fourier characterization). *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and $(x_0, \xi_0) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus 0)$. Then*

$$(x_0, \xi_0) \notin \text{WF}(u) \iff \exists \varepsilon > 0, \forall \phi \in C_c^\infty(B_\varepsilon(x_0)), \forall N, \exists C :$$

$$\boxed{|\widehat{\phi u}(\xi)| \leq C \langle \xi \rangle^{-N}} \quad \text{whenever} \quad \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \varepsilon.$$

Proof of " $\Leftarrow$ ". We produce an operator elliptic at (x_0, ξ_0) that smooths u , and invoke the one-operator lemma.

Take $\chi \in C^\infty(S^{n-1})$ supported in the ε -cap around $\xi_0/|\xi_0|$ with

$$\chi\left(\frac{\xi_0}{|\xi_0|}\right) \neq 0.$$

Extend it to a symbol $\tilde{\chi} \in S^0(\mathbb{R}_\xi^n)$ with

$$\tilde{\chi}(\xi) = \chi\left(\frac{\xi}{|\xi|}\right) \quad \text{for } |\xi| \geq \frac{1}{2}.$$

Let $\phi \in C_c^\infty(B_\varepsilon(x_0))$ with $\phi(x_0) \neq 0$, and define (right quantization)

$$A = \text{Op}_R(\tilde{\chi}(\xi)\phi(x)) \in \Psi^0(\mathbb{R}^n), \quad \text{so} \quad Au = \mathcal{F}^{-1}(\tilde{\chi}(\xi)\widehat{\phi u}(\xi)).$$

Then:

- A is **elliptic at** (x_0, ξ_0) , since its symbol equals $\tilde{\chi}(\xi)\phi(x) \neq 0$ there;
- by hypothesis,

$$|\widehat{Au}(\xi)| = |\tilde{\chi}(\xi)\widehat{\phi u}(\xi)| = O(\langle \xi \rangle^{-\infty}),$$

because $\tilde{\chi}$ is supported exactly in the cone where the decay hypothesis applies;

- hence $Au \in C^\infty(\mathbb{R}^n)$.

By the one-operator lemma, $(x_0, \xi_0) \notin \text{WF}(u)$.

The direction " $\Rightarrow$ " is obtained by running the same computation backwards: if $\alpha \notin \text{WF}(u)$, apply an operator of the above form which is $\equiv 1$ microlocally on a small cone, and read off the decay of $\widehat{\phi u}$ from the smoothness of Au . $\square$

Remarks.

- (1) $\text{WF}(u) \subset S^*M$ is **closed** (conic in $T^*M \setminus 0$).
- (2) $\pi(\text{WF}(u)) = \text{sing supp}(u)$, where $\pi : S^*M \rightarrow M$.
- (3) $\text{WF}(\bar{u}) = -\text{WF}(u)$ (use the Fourier characterization: $\widehat{\phi \bar{u}}(\xi) = \overline{\widehat{\phi u}(-\xi)}$).
- (4) **All definitions are coordinate invariant.** If $f : M \xrightarrow{\sim} N$ and $u \in \mathcal{D}'(N)$, then

$$f_{\#}^{-1}(\text{WF}(u)) = \text{WF}(f^*u), \quad f_{\#}(x, \xi) = (f(x), Df(x)^{-T}\xi).$$

Hence $\text{WF}(u)$ is a well-defined closed conic subset of $T^*M \setminus 0$ for $u \in \mathcal{D}'(M)$.

Example 8.24 (The delta function). For $\delta_0 \in \mathcal{D}'(\mathbb{R})$,

$$\text{WF}(\delta_0) = \{0\} \times (\mathbb{R} \setminus \{0\}),$$

since $\widehat{\phi\delta_0}(\xi) = \phi(0)$ does not decay in any direction.

Example 8.25 (A half-space). Let $\Omega = \{x_1 > 0\} \subset \mathbb{R}^n$. Then

$$\text{WF}(\mathcal{K}_\Omega) = \{(x, \xi) : x_1 = 0, \xi_1 \neq 0, \xi_i = 0 \ \forall i = 2, \dots, n\}.$$

Example 8.26 (A smooth domain). If $\Omega \subset M$ has smooth boundary $\partial\Omega$, then

$$\text{WF}(\mathcal{K}_\Omega) = N^*\partial\Omega \setminus 0,$$

the conormal bundle of the boundary, punctured at the zero section. Both conormal directions (inward and outward) occur.

Example 8.27 (A corner). Let $\Omega = \{x \in \mathbb{R}^2 : x_1 x_2 > 0\}$. Away from the origin the smooth-boundary calculation applies; **at the origin all directions appear** in $\text{WF}(\mathcal{K}_\Omega)$ —the corner “radiates” in every codirection.

Figure 8.6 compares these three boundary geometries.

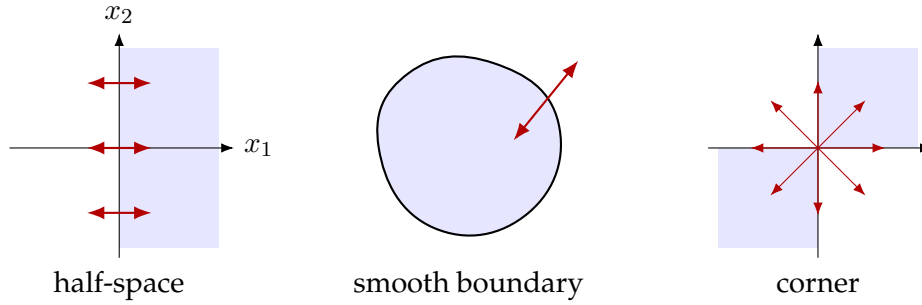


Figure 8.6: Wave front directions for a half-space, a smooth boundary, and a corner.

Example 8.28 (A one-sided boundary value). Let

$$(x + i0)^{-1} = \lim_{\varepsilon \downarrow 0} (x + i\varepsilon)^{-1} \in \mathcal{S}'(\mathbb{R}).$$

Its Fourier transform is supported in a half-line,

$$\mathcal{F}((x + i0)^{-1}) = c \mathcal{K}_{[0, \infty)} \quad (c = 2\pi i \text{ with the convention } \widehat{u}(\xi) = \int e^{-ix\xi} u \, dx),$$

so decay holds in the direction $\xi < 0$ but not $\xi > 0$:

$$\text{WF}((x + i0)^{-1}) = \{0\} \times \mathbb{R}_{>0}.$$

Figure 8.7 displays the resulting one-sided frequency direction.

This one-sidedness is what makes such distributions multipliable, as explained in the multiplication subsection below.

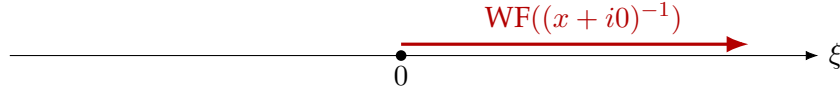


Figure 8.7: The boundary value $(x + i0)^{-1}$ has only positive frequency directions.

Microlocality and microlocal elliptic regularity

Proposition 8.29 (Microlocality). *Let $u \in \mathcal{D}'(M)$ and $A \in \Psi^m(M)$. Then*

$$\boxed{\text{WF}(Au) \subseteq \text{WF}'(A) \cap \text{WF}(u).}$$

Proof. Two separate inclusions.

(a) $\text{WF}(Au) \subseteq \text{WF}'(A)$. Let $\alpha \in S^*M$ with $\alpha \notin \text{WF}'(A)$. Take $B \in \Psi^0(M)$ elliptic at α with

$$\text{WF}'(B) \cap \text{WF}'(A) = \emptyset$$

(possible since $\text{WF}'(A)$ is closed and α lies in its open complement). Then

$$\text{WF}'(BA) \subseteq \text{WF}'(B) \cap \text{WF}'(A) = \emptyset \implies BA \in \Psi^{-\infty} \implies BAu \in C^\infty.$$

Since B is elliptic at α , the one-operator lemma gives $\alpha \notin \text{WF}(Au)$.

(b) $\text{WF}(Au) \subseteq \text{WF}(u)$. Now let $\alpha \notin \text{WF}(u)$; we want $\alpha \notin \text{WF}(Au)$.

By the one-operator lemma there is $B \in \Psi^0$ elliptic at α with $Bu \in C^\infty$. Choose $\tilde{B} \in \Psi^0$ elliptic at α with

$$\text{WF}'(\tilde{B}) \cap \text{WF}(u) = \emptyset, \quad \text{WF}'(\tilde{B}) \subset \text{Ell}(B).$$

Take a microlocal parametrix Q of B over $\text{WF}'(\tilde{B}) \subset \text{Ell}(B)$, so that with $R := I - QB$,

$$\text{WF}'(R) \cap \text{WF}'(\tilde{B}) = \emptyset.$$

Then

$$\tilde{B}Au = \tilde{B}A(QB + R)u = \underbrace{\tilde{B}AQ(Bu)}_{\text{smooth}} + \underbrace{\tilde{B}ARu}_{\text{smooth}}.$$

- The first term is smooth because $Bu \in C^\infty$ and $\tilde{B}AQ$ is a Ψ DO.
- For the second, $\text{WF}'(\tilde{B}AR) \subseteq \text{WF}'(\tilde{B}) \cap \text{WF}'(R) = \emptyset$, so $\tilde{B}AR \in \Psi^{-\infty}$.

Thus $\tilde{B}Au \in C^\infty$ with $\tilde{B}$ elliptic at α , so $\alpha \notin \text{WF}(Au)$ by the one-operator lemma. $\square$

Proposition 8.30 (Microlocal elliptic regularity). *Let $u \in \mathcal{D}'(M)$ and $A \in \Psi^m(M)$ properly supported. Then*

$$\boxed{\text{WF}(u) \subseteq \text{WF}(Au) \cup \text{Char}(A).}$$

Proof. Let $\alpha \in S^*M$ with

$$\alpha \in \text{Ell}(A) \quad \text{and} \quad \alpha \notin \text{WF}(Au);$$

we must show $\alpha \notin \text{WF}(u)$.

Both $\text{Ell}(A)$ and $\text{WF}(Au)^c$ are open neighborhoods of α , so by the microlocalizer lemma we may take $B \in \Psi^0(M)$ elliptic at α with

$$\begin{cases} \text{WF}'(B) \cap \text{WF}(Au) = \emptyset, \\ \text{WF}'(B) \subset \text{Ell}(A). \end{cases}$$

Take $Q \in \Psi^{-m}(M)$ a microlocal parametrix of A over the closed set $\text{WF}'(B) \subset \text{Ell}(A)$:

$$\text{WF}'(\underbrace{I - QA}_{=:R}) \cap \text{WF}'(B) = \emptyset.$$

Then

$$Bu = BQAu + BRu.$$

- BRu is smooth: $\text{WF}'(BR) \subseteq \text{WF}'(B) \cap \text{WF}'(R) = \emptyset$, so $BR \in \Psi^{-\infty}$.
- $BQAu$ is smooth: by microlocality,

$$\text{WF}(BQAu) \subseteq \text{WF}'(BQ) \cap \text{WF}(Au) \subseteq \text{WF}'(B) \cap \text{WF}(Au) = \emptyset.$$

Hence $Bu \in C^\infty$ with B elliptic at α , so $\alpha \notin \text{WF}(u)$. □

Example 8.31. Let $u = \mathcal{K}_{[0,+\infty)}$, so $u' = \delta_0$, i.e. $Au = \delta_0$ with $A = \frac{d}{dx}$. Since $\sigma^1(\frac{d}{dx}) = i\xi \neq 0$ for $\xi \neq 0$,

$$\text{Char}(\frac{d}{dx}) = \emptyset,$$

and microlocal elliptic regularity gives

$$\text{WF}(u) \subseteq \underbrace{\text{Char}(\frac{d}{dx})}_{=\emptyset} \cup \text{WF}(\delta_0) = \text{WF}(\delta_0) = \{0\} \times (\mathbb{R} \setminus 0).$$

(In fact equality holds.)

Operations on distributions with wave front constraints

Motivation The product $\delta_0 \cdot \delta_0$ is **not** well defined. But $\delta_0 \cdot \delta_1$ is (it equals 0: the supports are disjoint). More refined: two distributions can be multiplied as soon as their singularities do not "collide in phase space with opposite directions". The wave front set is exactly the bookkeeping device for this.

Definition 8.32 (Distributions with constrained wave front set). Let $\Lambda \subset S^*M$ be **closed**. Define

$$\mathcal{D}'_\Lambda(M) = \{u \in \mathcal{D}'(M) : \text{WF}(u) \subset \Lambda\}.$$

The topology is specified by declaring that $u_j \rightarrow u_*$ in $\mathcal{D}'_\Lambda(M)$ iff

- (1) $u_j \rightarrow u_*$ in $\mathcal{D}'(M)$; **and**
- (2) for all $A \in \Psi^0(M)$ properly supported with $\text{WF}'(A) \cap \Lambda = \emptyset$,

$$Au_j \longrightarrow Au_* \quad \text{in the } C^\infty\text{-topology.}$$

Extreme cases.

$$\Lambda = \emptyset \implies \mathcal{D}'_\Lambda(M) = C^\infty(M), \quad \Lambda = S^*M \implies \mathcal{D}'_\Lambda(M) = \mathcal{D}'(M).$$

Example 8.33. $C^\infty(M) \hookrightarrow \mathcal{D}'_\Lambda(M)$ has dense range.

Pairing

Proposition 8.34 (Pairing under disjoint wave front conditions). *Let μ be a smooth nonvanishing density and let $\Lambda, \Sigma \subset S^*M$ be **disjoint closed** sets. Then the $L^2(\mu)$ -pairing*

$$\langle \cdot, \cdot \rangle_{L^2(\mu)} : C^\infty(M) \times C_c^\infty(M) \rightarrow \mathbb{C}$$

extends (separately continuously) to

$$\mathcal{D}'_\Lambda(M) \times \mathcal{E}'_\Sigma(M) \rightarrow \mathbb{C}, \quad \mathcal{E}'_\Sigma := \mathcal{E}' \cap \mathcal{D}'_\Sigma.$$

Proof. Since Λ, Σ are disjoint and closed, by the microlocalizer lemma choose $A \in \Psi^0(M)$ with

$$\text{WF}'(I - A) \cap \Lambda = \emptyset, \quad \text{WF}'(A) \cap \Sigma = \emptyset.$$

For $u \in C^\infty, v \in C_c^\infty$ we have the identity

$$\langle u, v \rangle_{L^2} \doteq \langle u, Av \rangle_{L^2_\mu} + \langle (I - A)^*u, v \rangle_{L^2_\mu},$$

which is a tautology because $\langle u, Av \rangle + \langle u, (I - A)v \rangle = \langle u, v \rangle$.

We now check that **each term on the right extends** to $u \in \mathcal{D}'_\Lambda, v \in \mathcal{E}'_\Sigma$:

- $v \in \mathcal{E}'_\Sigma \implies$ by microlocality

$$\text{WF}(Av) \subseteq \text{WF}'(A) \cap \text{WF}(v) \subseteq \text{WF}'(A) \cap \Sigma = \emptyset,$$

so $Av \in C_c^\infty(M)$. Hence $\langle u, Av \rangle$ makes sense for any $u \in \mathcal{D}'$.

- $u \in \mathcal{D}'_\Lambda \implies$ using $\text{WF}'((I - A)^*) = \text{WF}'(I - A)$ from the operator wave front set calculus,

$$\text{WF}((I - A)^*u) \subseteq \text{WF}'(I - A) \cap \text{WF}(u) \subseteq \text{WF}'(I - A) \cap \Lambda = \emptyset,$$

so $(I - A)^*u \in C^\infty(M)$. Hence $\langle (I - A)^*u, v \rangle$ makes sense for any compactly supported v .

The result is independent of the choice of A (two such choices differ by an operator that is smoothing on both factors), and reduces to the usual pairing on $C^\infty \times C_c^\infty$. $\square$

Multiplication

Proposition 8.35 (Multiplication under the Hörmander condition). *Let $\Lambda, \Sigma \subset S^*M$ with*

$$\boxed{\Sigma \cap (-\Lambda) = \emptyset.}$$

Then multiplication

$$C^\infty \times C^\infty \rightarrow C^\infty, \quad (u, v) \mapsto u \cdot v$$

extends to

$$\mathcal{D}'_\Lambda(M) \times \mathcal{D}'_\Sigma(M) \longrightarrow \mathcal{D}'(M).$$

Proof. Fix a density μ . For $\phi \in C_c^\infty(M)$ define

$$\langle u \cdot v, \phi \rangle_{L^2_\mu} \doteq \langle u, \phi \bar{v} \rangle,$$

which is the correct formula for smooth u, v . Now $\phi \bar{v} \in \mathcal{E}'$ and, by the remarks following the Fourier characterization,

$$\text{WF}(\phi \bar{v}) \subseteq \text{WF}(\bar{v}) = -\text{WF}(v) \subseteq -\Sigma.$$

The hypothesis $\Sigma \cap (-\Lambda) = \emptyset$ is exactly $\Lambda \cap (-\Sigma) = \emptyset$, so Λ and $-\Sigma$ are disjoint closed sets and the pairing proposition applies: $\langle u, \phi \bar{v} \rangle$ is well defined for $u \in \mathcal{D}'_\Lambda$. Continuity in ϕ gives $u \cdot v \in \mathcal{D}'(M)$. $\square$

Example 8.36. Let $u = (x + i0)^{-1}$ on $\mathbb{R}$. By the one-sided boundary-value example,

$$\text{WF}(u) = \{0\} \times \mathbb{R}_{>0}, \quad -\text{WF}(u) = \{0\} \times \mathbb{R}_{<0},$$

so

$$\text{WF}(u) \cap (-\text{WF}(u)) = \emptyset \implies u^2 \in \mathcal{D}'(\mathbb{R}).$$

But $\text{WF}(\bar{u}) = -\text{WF}(u)$, so $\text{WF}(u) \cap (-\text{WF}(\bar{u})) = \text{WF}(u) \neq \emptyset$: the product

$$u \cdot \bar{u} \quad \text{is **not** defined!}$$

(Indeed $|x + i0|^{-2} = x^{-2}$ is genuinely non-integrable at 0.)

*Next: **Propagation of singularities.***

Chapter 9

Propagation of Singularities and Local Solvability

Throughout, M is a **closed** (compact, boundaryless) smooth manifold, $\Psi^m(M)$ denotes the classical pseudodifferential operators of order m , and $T^*M \setminus 0$ is the cotangent bundle with the zero section removed. All symbols are classical (polyhomogeneous) unless stated otherwise.

For $A \in \Psi^m(M)$ we have the three basic conic subsets of $T^*M \setminus 0$:

- $\text{WF}'(A)$ — the **microsupport** (essential support) of A : the complement of the largest open cone on which the full symbol is $O(\langle \xi \rangle^{-\infty})$;
- $\text{Ell}(A)$ — the **elliptic set**: where $\sigma^m(A)$ is invertible with the expected size $|\sigma^m(A)| \gtrsim |\xi|^m$;
- $\text{Char}(A) = (T^*M \setminus 0) \setminus \text{Ell}(A)$ — the **characteristic set**.

Definition 9.1 (Wave front and Sobolev wave front sets). For $u \in \mathcal{D}'(M)$, set

$$\text{WF}(u) = \bigcap_{\substack{A \in \Psi^0(M) \\ Au \in C^\infty(M)}} \text{Char}(A), \quad \text{WF}^s(u) = \bigcap_{\substack{A \in \Psi^0(M) \\ Au \in H^s(M)}} \text{Char}(A).$$

The **Sobolev wave front set** $\text{WF}^s(u)$ measures microlocal H^s -regularity:

$$(x_0, \xi_0) \notin \text{WF}^s(u) \iff u \text{ is microlocally } H^s \text{ near } (x_0, \xi_0),$$

$$\text{WF}^s(u) = \emptyset \iff u \in H^s(M), \quad \text{WF}(u) = \overline{\bigcup_{s \in \mathbb{R}} \text{WF}^s(u)}.$$

The families are nested: $\text{WF}^s(u) \subseteq \text{WF}^{s'}(u)$ for $s \leq s'$.

Microlocal elliptic regularity. If $P \in \Psi^m(M)$ and $Pu = f$, then

$$\text{WF}^s(u) \subseteq \text{WF}^{s-m}(f) \cup \text{Char}(P), \quad \text{in particular} \quad \text{WF}(u) \subseteq \text{WF}(f) \cup \text{Char}(P).$$

Remark 9.2. The question of this chapter. Elliptic regularity says nothing at all *inside* $\text{Char}(P)$. Propagation of singularities (POS) is exactly the statement that on $\text{Char}(P)$ the singularities are not arbitrary: they are **invariant under a canonically defined flow**. This is strictly more information.

9.1 The model case of real vector fields

Let X be a smooth real vector field on M with flow $\varphi_t : M \rightarrow M$ (complete, since M is closed). Its **cotangent lift** is

$$\Phi_t : T^*M \setminus 0 \hookrightarrow, \quad \Phi_t(x, \xi) = (\varphi_t(x), (d\varphi_t|_x)^{-T}\xi) = (\varphi_t)_\#(x, \xi).$$

This is the unique lift making Φ_t a homogeneous canonical (symplectic) transformation; it is a one-parameter group.

Theorem 9.3 (POS for flows). *Let $u \in \mathcal{D}'(M)$ satisfy $\frac{1}{i}Xu = 0$. Then $\text{WF}(u)$ is invariant under Φ_t for all t .*

Proof. Since $\frac{d}{dt}\varphi_t^*u = \varphi_t^*(Xu) = 0$ and $\varphi_0^*u = u$, we get $\varphi_t^*u = u$ for all $t \in \mathbb{R}$. The wave front set transforms functorially under a diffeomorphism φ :

$$\text{WF}(\varphi^*u) = \{(x, (d\varphi|_x)^T\eta) : (\varphi(x), \eta) \in \text{WF}(u)\} = \varphi_\#^{-1}(\text{WF}(u)).$$

Hence $\text{WF}(u) = \text{WF}(\varphi_t^*u) = \Phi_t^{-1}(\text{WF}(u))$, i.e. $\text{WF}(u)$ is Φ_t -invariant. $\square$

Two remarks that motivate everything below:

- (1) The operator $P = \frac{1}{i}X$ lies in $\Psi^1(M)$ (it is a differential operator of order 1) and its principal symbol is

$$p(x, \xi) = \xi(X(x)) = \langle \xi, X(x) \rangle,$$

which is **real** and **homogeneous of degree 1** in ξ .

- (2) Φ_t is exactly the **Hamiltonian flow of p** , as shown below. Thus the theorem says that $\text{WF}(u)$ is invariant under the Hamiltonian flow of the principal symbol.

Generalisation programme.

baby version	general version
$\frac{1}{i}X$	$P \in \Psi^m(M)$, $m \in \mathbb{R}$, with $\sigma^m(P)$ having a representative $p \in S_{\text{hom}}^m(T^*M \setminus 0; \mathbb{R})$ — real and homogeneous
lifted flow Φ_t	Hamiltonian flow of p
$Pu = 0$	$Pu = f$, with f controlling where propagation stops

9.2 Symplectic geometry on $T^*M \setminus 0$

Definition 9.4 (Tautological and symplectic forms). Let $\pi : T^*M \rightarrow M$ be the projection. The **tautological (Liouville) 1-form** is

$$\alpha_{(x, \xi)}(\theta) = \xi(d\pi(\theta)), \quad \theta \in T_{(x, \xi)}(T^*M),$$

and the symplectic form is $\omega = -d\alpha$. In local canonical coordinates $\alpha = \sum_j \xi_j dx^j$ and

$$\omega = \sum_j dx^j \wedge d\xi_j.$$

Definition 9.5 (Hamilton vector field). Given $p : T^*M \setminus 0 \rightarrow \mathbb{R}$, the **Hamilton vector field** X_p (also written H_p) is defined by

$$\omega(X_p, \cdot) = dp.$$

Local computation. Write $X_p = \sum_j a_j \partial_{x_j} + \sum_j b_j \partial_{\xi_j}$. Then

$$\omega(X_p, \cdot) = \sum_j a_j d\xi_j - b_j dx^j, \quad dp = \sum_j \partial_{x_j} p dx^j + \partial_{\xi_j} p d\xi_j,$$

so that

$$\boxed{a_j = \partial_{\xi_j} p, \quad b_j = -\partial_{x_j} p} \iff X_p = \sum_j \partial_{\xi_j} p \partial_{x_j} - \partial_{x_j} p \partial_{\xi_j}.$$

Its flow $\Phi_t(x, \xi) = (x(t), \xi(t))$ solves **Hamilton's equations**

$$\dot{x} = \partial_{\xi} p, \quad \dot{\xi} = -\partial_x p.$$

Two structural facts.

- $X_p p = \{p, p\} = 0$, so Φ_t **preserves the level sets of p** . In particular

$$\text{Char}(P) = \{p = 0\} \text{ is } \Phi_t\text{-invariant.}$$

- If p is homogeneous of degree m in ξ , then X_p is homogeneous of degree $m - 1$, so Φ_t commutes with the dilations up to reparametrization. Consequently, after fixing a metric (or a nonvanishing density) the flow **descends to the cosphere bundle** $S^*M = (T^*M \setminus 0)/\mathbb{R}_{>0}$, at least up to a positive reparametrization of time. For $m = 1$ the flow descends literally.

Figure 9.1 illustrates an orbit contained in a level set of p .

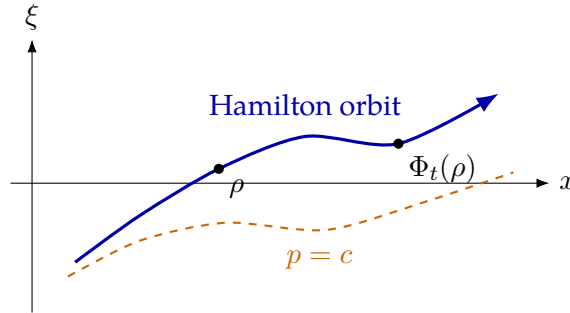


Figure 9.1: Hamiltonian flow preserves each level set of the principal symbol. Null bicharacteristics are the orbits in $p^{-1}(0)$.

Consistency with cotangent lifts. For $p(x, \xi) = \langle \xi, X(x) \rangle$, Hamilton's equations read $\dot{x}^j = X^j(x)$ and $\dot{\xi}_k = -\xi_j \partial_{x_k} X^j$, which is precisely the cotangent lift $\Phi_t = (\varphi_t)_\#$. ✓

Null bicharacteristics

Definition 9.6. The orbits of Φ_t contained in $\text{Char}(P) = \{p = 0\}$ are called the **null bicharacteristics** (or *bicharacteristic strips*) of P . Orbits in $\{p = c\}$, $c \neq 0$, are *bicharacteristics*; only the null ones matter for regularity, because only $\text{Char}(P)$ can carry singularities.

Definition 9.7 (real principal type). $P \in \Psi^m(M)$ is of **real principal type** if $p = \sigma^m(P)$ is real-valued, homogeneous of degree m , and $dp \neq 0$ on $\text{Char}(P)$ (often one also requires X_p to be nowhere radial on $\text{Char}(P)$, so that null bicharacteristics project to genuine curves in M). This is the standing hypothesis for the theorems below.

9.3 The wave operator

Let $M = \mathbb{R}_t \times N$ with (N, g) Riemannian, and

$$P = \partial_t^2 - \Delta_g, \quad p(t, \tau, x, \xi) = -\tau^2 + |\xi|_g^2, \quad |\xi|_g^2 = g^{ij}(x)\xi_i\xi_j.$$

(Sign convention: $\sigma(\partial_j) = i\xi_j$, so $\sigma(\partial_t^2) = -\tau^2$ and $\sigma(-\Delta_g) = |\xi|_g^2$.) The characteristic set is the **light cone** $\{\tau^2 = |\xi|_g^2\}$. Hamilton's equations split:

$$\begin{cases} \dot{t} = \partial_\tau p = -2\tau, & \dot{x}^j = \partial_{\xi_j} p = 2g^{ij}\xi_i, \\ \dot{\tau} = -\partial_t p = 0, & \dot{\xi}_k = -\partial_{x_k} p = -(\partial_{x_k} g^{ij})\xi_i\xi_j. \end{cases}$$

The first pair integrates to $\tau \equiv \tau_0$, $t(s) = t_0 - 2\tau_0 s$. The second pair is exactly the **(co-)geodesic flow on T^*N** , run at twice the usual speed. Hence

$$\Phi_s(t, \tau, x, \xi) = (t - 2s\tau, \tau, G_{2s}(x, \xi)),$$

where G denotes the geodesic flow.

Interpretation. On $\text{Char}(P)$ one has $\tau = \pm|\xi|_g$, so $|dx/dt|_g = 1$: the projections of null bicharacteristics to M are **light rays**, i.e. curves $t \mapsto (t, \gamma(t))$ with γ a unit-speed geodesic in N . Thus POS says:

Remark 9.8. Singularities of solutions of the wave equation travel along null geodesics at unit speed and do not disappear.

Figure 9.2 shows the projected null bicharacteristics.

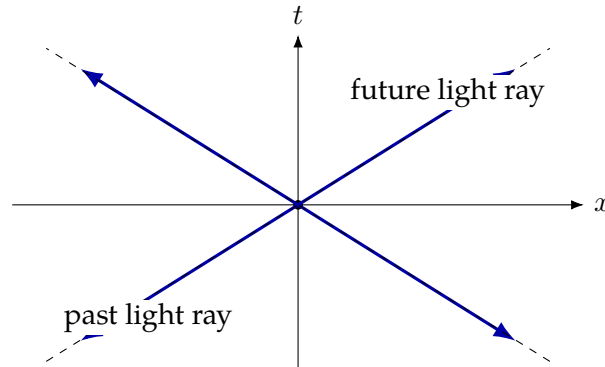


Figure 9.2: Projection of the wave operator's null bicharacteristics: singularities travel along light rays at unit speed.

This is the rigorous version of geometric optics, and the reason POS is the workhorse behind e.g. Weyl-law remainder estimates, inverse problems (boundary/lens rigidity), and control theory (the geometric control condition).

9.4 Hörmander's propagation theorem

Theorem 9.9 (POS, Hörmander). *Let M be closed, $P \in \Psi^m(M)$ with $\sigma^m(P) = [p]$, where $p \in S_{\text{hom}}^m(T^*M \setminus 0; \mathbb{R})$ is real and homogeneous, and let $u, f \in \mathcal{D}'(M)$ satisfy*

$$Pu = f.$$

Then for every $s \in \mathbb{R}$,

$$\boxed{\text{WF}^s(u) \setminus \text{WF}^{s-m+1}(f) \subseteq \text{Char}(P) \setminus \text{WF}^{s-m+1}(f)}$$

and it is a union of maximally extended null bicharacteristics of p contained in $\text{Char}(P) \setminus \text{WF}^{s-m+1}(f)$.

Equivalently, in "moving" form: if $\gamma : [0, T] \rightarrow \text{Char}(P)$ is a null bicharacteristic with

$$\gamma(0) \notin \text{WF}^s(u) \quad \text{and} \quad \gamma([0, T]) \cap \text{WF}^{s-m+1}(f) = \emptyset,$$

then $\gamma(T) \notin \text{WF}^s(u)$. (And symmetrically backwards in t , since the hypothesis on p is time-symmetric.)

Regularity indices and the source term

- **Where the containment in $\text{Char}(P)$ comes from.** Off $\text{Char}(P)$ this is just elliptic regularity, and elliptic regularity gains m derivatives ($\text{WF}^s(u) \subseteq \text{WF}^{s-m}(f) \cup \text{Char}(P)$). POS is a genuinely different mechanism and gains only $m - 1$ derivatives, which is why the index $s - m + 1$ appears. This loss of one derivative relative to the elliptic case is sharp.
- **Why $\text{WF}(f)$ blocks propagation.** The theorem only propagates regularity where f is microlocally H^{s-m+1} . If a null bicharacteristic meets $\text{WF}^{s-m+1}(f)$, the source may inject a new singularity and the argument stops. A curve avoiding $\text{WF}(f) \cap \text{Char}(P)$ transports the implication

$$(x_0, \xi_0) \notin \text{WF}(u) \implies (y_0, \xi_1) \notin \text{WF}(u),$$

whereas a curve through the source wave front set gives no conclusion.

- **Applies equally to P^* .** Since $\sigma^m(P^*) = \bar{p} = p$, the operator P^* has the same characteristic set and the same null bicharacteristics. This is what will be used for solvability.
- **$f = 0$ case.** If $Pu = 0$ then $\text{WF}^s(u)$ is, for every s , a Φ_t -invariant subset of $\text{Char}(P)$ —the exact analogue of the baby version.

Figure 9.3 distinguishes bicharacteristics that meet the source wave front set from those that do not.

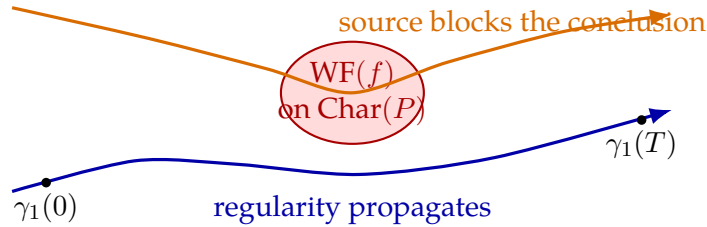


Figure 9.3: Propagation along bicharacteristics. The lower orbit avoids the source wave front set; the upper orbit meets it.

Sketches of the standard proofs (a) **Positive commutator (Hörmander's energy argument).** Assume for simplicity $P = P^*$ modulo Ψ^{m-1} (possible since p is real). For $A \in \Psi^{s-(m-1)/2}$ set $Q = A^*A \geq 0$ and compute

$$\langle i[P, Q]u, u \rangle = i(\langle Qu, Pu \rangle - \langle Qf, u \rangle) + (\text{lower order})$$

so the left-hand side is controlled by f and by lower-order terms. On the other hand $i[P, Q] \in \Psi^{2s}$ has principal symbol

$$\sigma(i[P, Q]) = H_p(|a|^2) = 2a H_p a.$$

One now chooses the symbol a , supported in a small conic neighborhood of the bicharacteristic segment, so that

$$H_p(a^2) = -b^2 + e^2,$$

with b elliptic at the *target* point $\gamma(T)$ and e supported near the *source* point $\gamma(0)$ where regularity is assumed. (Such an a exists because $H_p \neq 0$ there: one solves a transport ODE along the flow.) Rearranging gives the a priori estimate

$$\|Bu\|^2 \lesssim \|Eu\|^2 + |\langle Qf, u \rangle| + \|u\|_{\text{lower order}}^2,$$

i.e. microlocal H^s control at $\gamma(T)$ in terms of control at $\gamma(0)$ plus the source term. A regularisation argument (inserting a family of cutoffs a_ε with uniform bounds) upgrades this from an a priori estimate to a genuine regularity statement.

(b) Normal form / FIO conjugation. Near a point of $\text{Char}(P)$ with $dp \neq 0$ and X_p non-radial, Darboux's theorem produces a homogeneous canonical transformation taking p to the symbol ξ_1 . Quantising it by an elliptic Fourier integral operator conjugates P microlocally to D_{x_1} modulo smoothing. For D_{x_1} the statement is an explicit ODE computation: $D_{x_1}u = f$ integrates along the x_1 -lines, which are exactly the null bicharacteristics of ξ_1 . Egorov's theorem guarantees that the wave front-set statement is transported correctly.

9.5 Local solvability

References: Folland, *Introduction to PDE*; Hörmander, *ALPDO* vol. IV (ch. 26).

Background: what can go wrong

Theorem 9.10 (Malgrange–Ehrenpreis). *If $P \in \text{Diff}(\mathbb{R}^n)$ has **constant coefficients** and $P \neq 0$, then for every $f \in \mathcal{E}'(\mathbb{R}^n)$ there exists $u \in \mathcal{D}'(\mathbb{R}^n)$ with $Pu = f$. (Indeed P has a fundamental solution.)*

Example 9.11 (Lewy's nonsolvable operator). For variable coefficients, local solvability can fail dramatically. The operator

$$L = \partial_{x_1} + i\partial_{x_2} - 2i(x_1 + ix_2)\partial_{x_3}$$

admits $F \in C^\infty(\mathbb{R}^3)$ for which $Lu = F$ has **no** distributional solution — not even locally, on any open set. Note $\sigma^1(L)$ is *not* real, so L is not of real principal type; the obstruction is genuinely microlocal.

The general picture. Nirenberg–Treves formulated **condition** (Ψ) on the sign changes of $\text{Im } p$ along the bicharacteristics of $\text{Re } p$, and conjectured it characterises local solvability for principal-type operators. Necessity was known (Hörmander, Moyer); sufficiency was proved by **Dencker (2006)**, settling the Treves–Nirenberg conjecture. The real-principal-type case treated below is the easy end of this story: $\text{Im } p \equiv 0$, so (Ψ) holds trivially, and one can give a self-contained proof.

Setting and the non-trapping condition M closed, $P \in \Psi^m(M)$ of real principal type, $p : T^*M \setminus 0 \rightarrow \mathbb{R}$ real homogeneous, Φ_t the Hamiltonian flow. Fix once and for all a smooth nonvanishing density μ , so that adjoints P^* and the pairing $\langle \cdot, \cdot \rangle_{L^2(\mu)}$ are defined; since Φ_t is homogeneous it descends to S^*M .

Definition 9.12 (non-trapping). A compact set $K \subseteq M$ is **non-trapping** for Φ_t if

$$\forall (x, \xi) \in T^*M \setminus 0 \text{ with } x \in K, \quad \exists T \in \mathbb{R} : \quad \pi \circ \Phi_T(x, \xi) \notin K.$$

In words: no bicharacteristic stays over K for all time — every one of them eventually escapes, forward or backward. By homogeneity it is enough to test $(x, \xi) \in S^*M$.

A bicharacteristic may cross the strip $\pi^{-1}(K)$ and leave it, but it may not remain over K for all time; see figure 9.4.

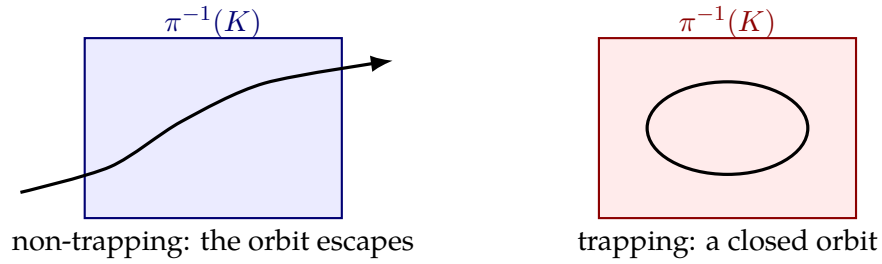


Figure 9.4: The non-trapping condition excludes bicharacteristics that remain over K for all time.

Special case: $K = \{x_0\}$ is non-trapping iff no null bicharacteristic has projection identically equal to x_0 ; a convenient sufficient condition is $\partial_\xi p(x_0, \xi) \neq 0$ for all $\xi \neq 0$, since $\dot{x} = \partial_\xi p$.

The goal

Theorem 9.13 (local solvability). Suppose $x_0 \in M$ is such that $\{x_0\}$ is non-trapping. Then there is an open $U \ni x_0$ such that

$$\forall f \in C^\infty(M) \quad \exists u \in \mathcal{D}'(M) : \quad Pu = f \text{ in } U.$$

The proof has three ingredients: (i) POS turns non-trapping into a *smoothness* statement for the “dual kernel”; (ii) functional analysis turns this into an a priori estimate; (iii) Hahn–Banach turns the estimate into solvability.

The obstruction space $N(K)$ For $K \subseteq M$ compact non-trapping put

$$N(K) = \{u \in \mathcal{D}'(M) : P^*u = 0 \text{ on } M, \text{ supp}(u) \subseteq K\}.$$

(i) Smoothness. $N(K) \subseteq C^\infty(M)$. Let $u \in N(K)$. By elliptic regularity $\text{WF}(u) \subseteq \text{Char}(P^*) = \text{Char}(P)$, and by POS (with $f = 0$) $\text{WF}(u)$ is a union of maximally extended null bicharacteristics. But $\pi(\text{WF}(u)) \subseteq \text{supp}(u) \subseteq K$, so every such bicharacteristic stays over K for all time—contradicting non-trapping unless $\text{WF}(u) = \emptyset$. Hence $u \in C^\infty(M)$.

(ii) **Finite dimensionality.** $\dim N(K) < \infty$. By (i) and the closed-graph estimate established below, applied with $P^*u = 0$, one has $\|u\|_{H^1} \leq C\|u\|_{H^{-N}}$ for all $u \in N(K)$; that is, the identity map

$$(N(K), \|\cdot\|_{H^{-N}}) \longrightarrow (N(K), \|\cdot\|_{H^1})$$

is bounded. Composing with the compact embedding $H^1(M) \hookrightarrow H^{-N}(M)$ from Rellich's theorem, the identity on $N(K)$ is compact. Hence $N(K)$ is finite dimensional.

(iii) **Point support.** $N(\{x_0\}) = 0$. A distribution supported at a single point is a finite linear combination of derivatives of δ_{x_0} ; by (i) it is also smooth; a smooth function supported at a point is 0.

Continuity of $K \mapsto N(K)$

Lemma 9.14. *Let K be compact and non-trapping, and let $U_1 \supseteq U_2 \supseteq \dots \supseteq K$ be open with $\bigcap_j U_j = K$. Put $K_j = \overline{U_j}$. Then:*

- (1) *for $j \gg 1$, K_j is non-trapping;*
- (2) $\lim_{j \rightarrow \infty} \dim N(K_j) = \dim N(K)$.

Proof of 1. Suppose not; then for infinitely many j there is $\alpha_j \in S^*M \cap \pi^{-1}(K_j)$ that is trapped, i.e. $\pi\Phi_t(\alpha_j) \in K_j$ for all t . Fix $T \in \mathbb{R}$. Since $\bigcap_j K_j = K$ and the K_j decrease, $\Phi_T(\alpha_j) \in \pi^{-1}(K_j)$ and, passing to a subsequence (compactness of S^*M), $\alpha_j \rightarrow \alpha_* \in \pi^{-1}(K)$. By continuity of the flow $\Phi_T(\alpha_j) \rightarrow \Phi_T(\alpha_*)$, and since the sets $\pi^{-1}(K_j)$ shrink to $\pi^{-1}(K)$ we get $\pi\Phi_T(\alpha_*) \in K$. As T was arbitrary (diagonalise over a countable dense set of times and use continuity), α_* is trapped in $\pi^{-1}(K)$ — contradicting that K is non-trapping. $\square$

Proof of 2. Clearly $N(K) \subseteq N(K_j)$, so $\dim N(K_j) \geq \dim N(K)$; suppose the limit is strictly larger. Then for infinitely many j there exists

$$u_j \in N(K_j) \cap N(K)^\perp, \quad \|u_j\|_{L^2} = 1.$$

The uniform a priori estimates below (valid with constants uniform in j , since $K_j \subseteq K_1$ and all K_j are non-trapping for $j \gg 1$) bound u_j in H^1 ; by Rellich a subsequence converges in L^2 to some u with $\|u\|_{L^2} = 1$, $P^*u = 0$, $\text{supp}(u) \subseteq \bigcap_j K_j = K$, and $u \perp N(K)$. So $u \in N(K) \cap N(K)^\perp = \{0\}$ — a contradiction. $\square$

Combining with (iii): **given x_0 with $\{x_0\}$ non-trapping, there is an open $U \ni x_0$ with $K := \overline{U}$ non-trapping and $N(K) = N(\{x_0\}) = 0$.**

A priori estimates Fix $K \subseteq M$ compact non-trapping. Let $u \in \mathcal{D}'(M)$ with $\text{supp}(u) \subseteq K$.

(a) **(regularity).** $P^*u \in H^{s-m+1}(M) \implies u \in H^s(M)$.

Proof. Put $g = P^*u \in H^{s-m+1}$, so $\text{WF}^{s-m+1}(g) = \emptyset$. By POS, $\text{WF}^s(u)$ is a union of maximally extended null bicharacteristics inside $\text{Char}(P)$. Since $\pi(\text{WF}^s(u)) \subseteq \text{supp}(u) \subseteq K$ and K is non-trapping, $\text{WF}^s(u) = \emptyset$, i.e. $u \in H^s$. $\square$

(b) **(estimate, via the closed graph theorem).** There are N and $C = C(s, N, K)$ with

$$\|u\|_{H^s} \leq C(\|P^*u\|_{H^{s-m+1}} + \|u\|_{H^{-N}}) \quad \text{for all } u \in \mathcal{D}'(M), \text{ supp } u \subseteq K.$$

Proof. Let

$$X = \{u \in H^{-N}(M) : \text{supp } u \subseteq K, P^*u \in H^{s-m+1}\}, \quad \|u\|_X = \|u\|_{H^{-N}} + \|P^*u\|_{H^{s-m+1}}.$$

X is a Banach space: if (u_j) is $\|\cdot\|_X$ -Cauchy then $u_j \rightarrow u$ in H^{-N} and $P^*u_j \rightarrow g$ in H^{s-m+1} ; continuity of P^* on distributions forces $P^*u = g$, and the support condition is closed. The inclusion $\iota : X \rightarrow H^s$ is well defined by (a), and has closed graph (both convergences imply convergence in $\mathcal{D}'$). By the closed graph theorem ι is bounded, which is the estimate. $\square$

(c) (**improved estimate**). If moreover $N(K) = 0$, then for every s there is C with

$$\boxed{\|u\|_{H^s} \leq C \|P^*u\|_{H^{s-m+1}}} \quad \text{supp } u \subseteq K.$$

Proof. Suppose not: there are u_j with $\text{supp } u_j \subseteq K$, $\|u_j\|_{H^s} = 1$ and $\|P^*u_j\|_{H^{s-m+1}} \rightarrow 0$. By (b), $\|u_j\|_{H^{-N}} \geq \frac{1}{2C}$ for j large. Since $H^s \hookrightarrow H^{-N}$ is compact, after passing to a subsequence $u_j \rightarrow u$ in H^{-N} ; applying (b) to the differences,

$$\|u_j - u_k\|_{H^s} \leq C(\|P^*(u_j - u_k)\|_{H^{s-m+1}} + \|u_j - u_k\|_{H^{-N}}) \rightarrow 0,$$

so $u_j \rightarrow u$ in H^s as well. Then $\|u\|_{H^s} = 1$, $\text{supp } u \subseteq K$ and $P^*u = 0$, i.e. $0 \neq u \in N(K) = 0$ — contradiction. $\square$

From the estimate to solvability (duality) This is the standard Hahn–Banach argument completing the solvability proof.

Proof of local solvability. Let $U \ni x_0$ be as produced by the continuity argument above, so that $K = \overline{U}$ is non-trapping and $N(K) = 0$; thus (c) holds for every s . Fix $f \in C^\infty(M)$ and fix $s \in \mathbb{R}$.

Consider the linear subspace $E = \{P^*\phi : \phi \in C_c^\infty(U)\} \subseteq H^{s-m+1}(M)$ and define

$$L : E \rightarrow \mathbb{C}, \quad L(P^*\phi) = \langle f, \phi \rangle.$$

- **Well defined:** if $P^*\phi_1 = P^*\phi_2$ then $\|\phi_1 - \phi_2\|_{H^s} \leq C\|P^*(\phi_1 - \phi_2)\|_{H^{s-m+1}} = 0$ by (c), so $\phi_1 = \phi_2$. (In particular P^* is injective on $C_c^\infty(U)$.)
- **Bounded:** $|L(P^*\phi)| = |\langle f, \phi \rangle| \leq \|f\|_{H^{-s}} \|\phi\|_{H^s} \leq C\|f\|_{H^{-s}} \|P^*\phi\|_{H^{s-m+1}}$.

By Hahn–Banach, L extends to a bounded functional on $H^{s-m+1}(M)$, hence is represented by some

$$u \in (H^{s-m+1})^* = H^{m-1-s}(M), \quad \langle u, P^*\phi \rangle = \langle f, \phi \rangle \quad \forall \phi \in C_c^\infty(U).$$

By definition of the distributional action this says exactly

$$Pu = f \quad \text{in } U.$$

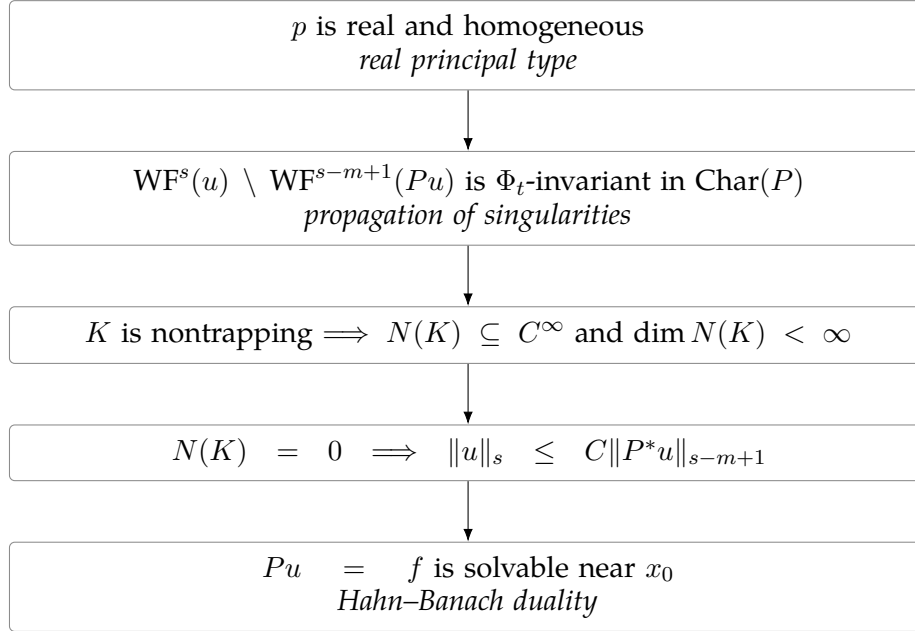
$\square$

Remarks.

- Since (c) holds for every s , choosing $s = m - 1 - k$ produces a solution $u \in H^k(U)$ for any prescribed k ; the solution depends on k and one does not directly obtain a single smooth solution this way (a Baire-category / iteration argument is needed for that).
- Nothing in the argument used $f \in C^\infty$ beyond $f \in H^{-s}$, so the same proof gives solvability for f in a Sobolev space.

- The obstruction space $N(K)$ is the exact source of the "solvability is only local and only after shrinking U " phenomenon: $N(K)$ is the cokernel of P acting on distributions supported in K , and it vanishes only for small enough K .

9.6 Summary of the Logical Chain



And in the other direction, the failure of any of the hypotheses (real principal symbol, non-trapping) is not an artefact: Lewy's operator shows that complex principal symbols can destroy solvability entirely, and the sharp criterion in that generality is Nirenberg–Treves' condition (Ψ) , proved sufficient by Dencker in 2006.

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